8 Cleaning and Shaping of the Root Canal System

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Introduction

Clinical endodontics encompasses a number of treatments that share the common goal of preventing and treating microbial contamination of pulps and root canal systems. Treatment of traumatic dental injuries and vital pulp treatment are fundamentally different from pulpectomies and root canal instrumentation of teeth with infected pulps (see Chapter 1 for more details).

Endodontic therapy is directed toward one specific set of aims: to cure or prevent periradicular periodontitis.[309,459] The ultimate goal is to aid in the long-term retention in patients' natural teeth in function and esthetics.

To date, available pertinent clinical trials[89,317,326,341] and in vitro studies suggest that certain practices in canal preparation and disinfection are more appropriate than others. This chapter provides a summary of this information.

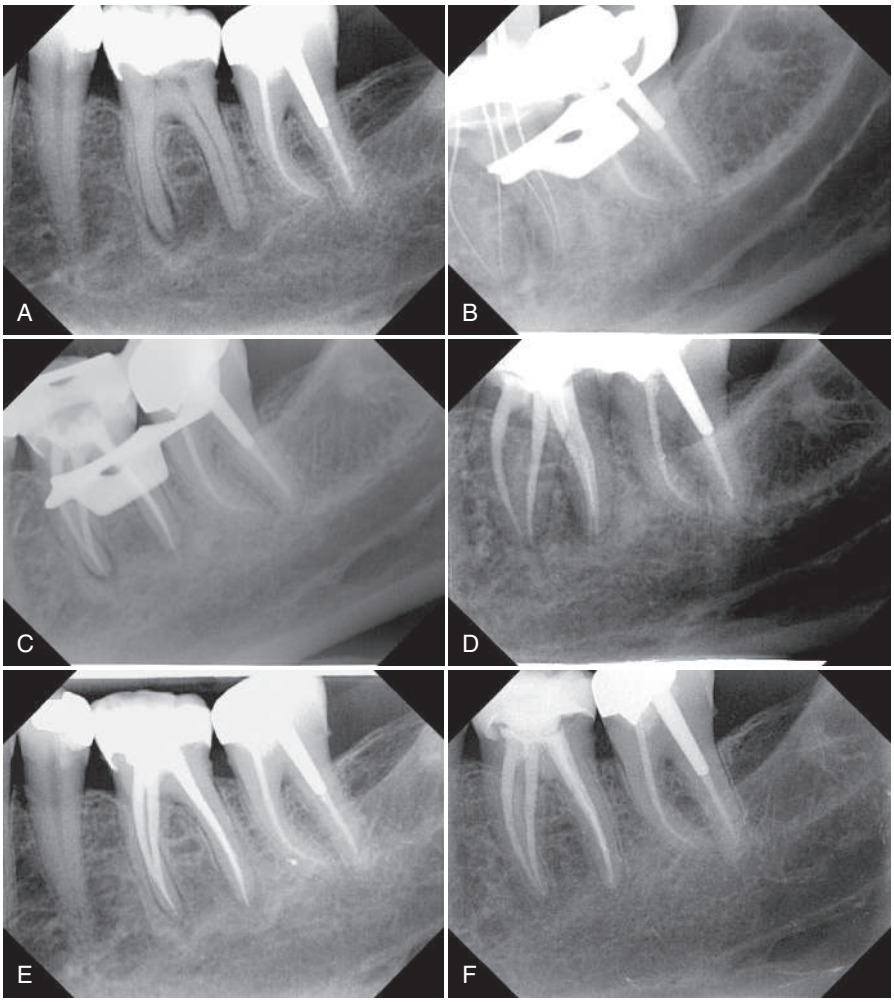


Fig. 8.1 Effect of routine root canal treatment of a mandibular molar. **A,** Pretreatment radiograph of tooth #19 shows radiolucent lesions adjacent to both mesial and distal root apices. **B,** Working length radiograph shows two separate root canals in the mesial root and two merging canals in the distal root. **C,** Posttreatment radiograph after shaping of root canal systems with nickel-titanium rotary files and obturation with thermoplasticized gutta-percha. **D,** Six-month recall radiograph after restoration of tooth #19 with an adhesively inserted full ceramic crown; some periradicular bone fill can be seen. **E,** One-year recall radiograph displays evidence of additional periradicular healing. **F,** Five-year recall radiograph; tooth not only is periapically sound but also clinically asymptomatic and fully functional.

Orthograde root canal treatment is a predictable and usually highly successful procedure, both in relatively straightforward (Fig. 8.1) and more complex cases (Fig. 8.2). In recent studies and reviews, favorable outcome rates of up to 95% were reported for the treatment of teeth diagnosed with irreversible pulpitis;[37,93,139] favorable outcome rates of up to 85% were reported for teeth with infected, necrotic pulps.[92,140,301,326,358]

Box 8.1 Basic Objectives in Cleaning and Shaping

The primary objectives in cleaning and shaping the root canal system are to:

- Remove infected soft and hard tissue
- Give disinfecting irrigants access to the apical canal space
- Create space for the delivery of medicaments and subsequent obturation
- Retain the integrity of radicular structures

The ultimate cause of pulpal pathosis are microorganisms that can breach dental hard-tissue barriers through several avenues—the most common being dental caries (Fig. 8.3). Shaping and cleaning procedures (Box 8.1) as part of root canal treatment are directed against microbial challenges to the root canal system. However, proper disinfection per se does not guarantee long-term retention of root canal–treated teeth; there is good evidence that this outcome is closely related to placement of an adequate coronal restoration.[20,300,352,369] Moreover, the impact of preservation of radicular structural strength on tooth longevity has been emphasized.[154]

Principles of Cleaning and Shaping

Endodontists agree that a major biologic aim of root canal therapy is to address apical periodontitis by disinfection and subsequent sealing of root canal systems. However, considerable disagreement exists over how this goal should be achieved. Although the terms “cleaning and shaping” are

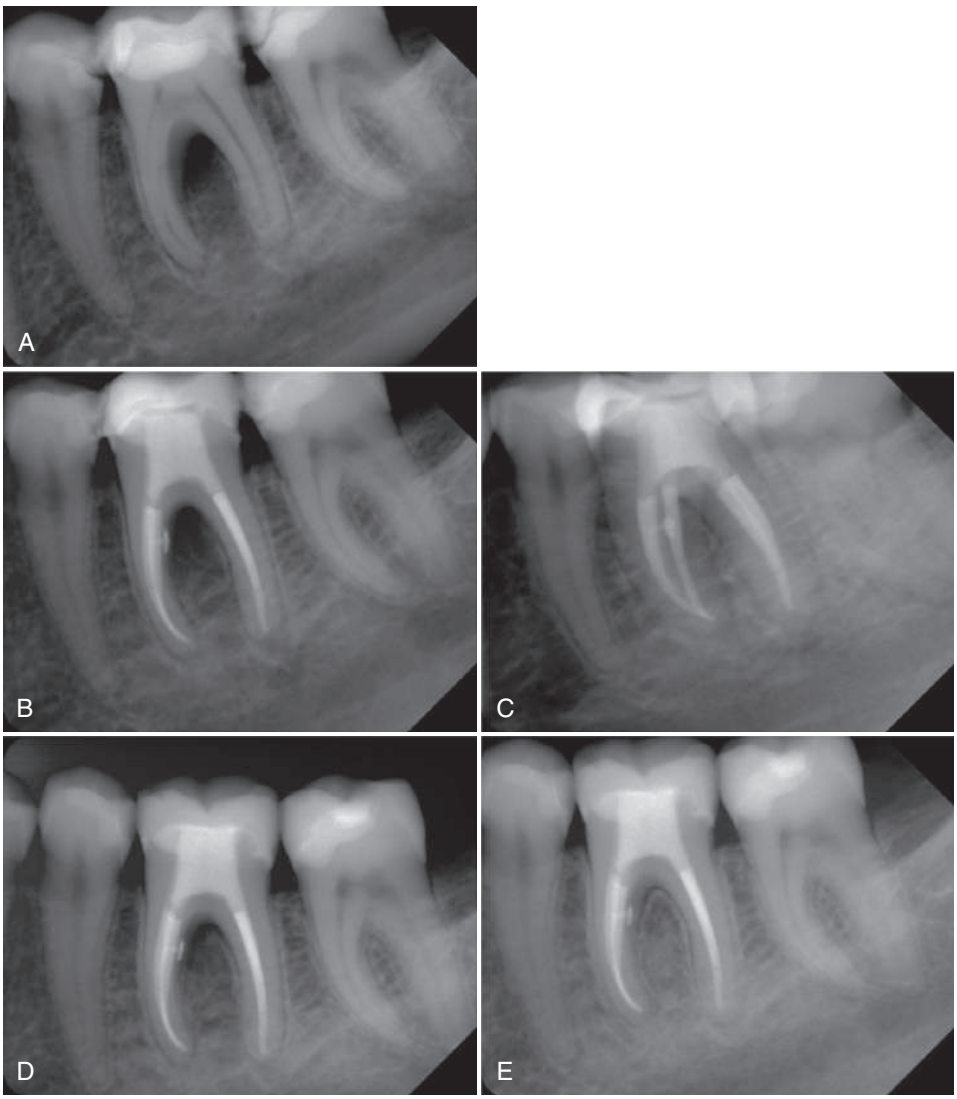


Fig. 8.2 Root canal treatment in a case of apical and interradicular pathosis. **A,** Pretreatment radiograph of tooth #19 shows an interradicular lesion. **B–C,** Posttreatment radiographs after root canal preparation and obturation. Note the lateral canal in the coronal third of the root canal. **D–E,** Two-month recall radiograph suggests rapid healing. (Courtesy Dr. H. Walsch.)

commonly used to describe root canal treatment procedures, reversing the order to "shaping and cleaning" may more correctly reflect the fact that enlarged canals direct and facilitate the cleaning action of irrigants.

Microorganisms in the pulp cavity and coronal root canal may be readily killed by irrigants early in a procedure, but bacteria and their toxins in less accessible canal areas can survive and elicit or maintain apical periodontitis. In current everyday practice, these persisting bacteria can be targeted only after root canal preparation.

MECHANICAL OBJECTIVE

An ideal mechanical objective of root canal instrumentation is complete and centered incorporation of the original canals into the prepared shape, meaning that all root canal surfaces are mechanically prepared (*green areas* in Fig. 8.4, *A* and *B*); however, this goal is unlikely to be met with current techniques.[312,340]

Preparation errors such as deviations, zipping, and perforations should be absent. Although these negative effects of canal shaping and other procedural mishaps (see later) may not directly affect the probability of a favorable outcome,[254] they leave parts of the root canal system inaccessible for disinfection and are undesirable for that reason alone.

Another important mechanical objective is to retain as much cervical and radicular dentin as possible so as not to weaken the root structure, thereby preventing root fractures. Before root canal shaping, dentin wall thickness dimensions of 1 mm and below have been demonstrated in anatomical studies.[113] Straightening of canal paths can lead to precarious thinning of curved root walls (Fig. 8.5). Although no definitive minimal radicular wall thickness has been established, 0.3 mm is considered critical by some authors.[252] To avoid overpreparation and outright perforations, adequate access cavity preparation and optimal enlargement of the coronal

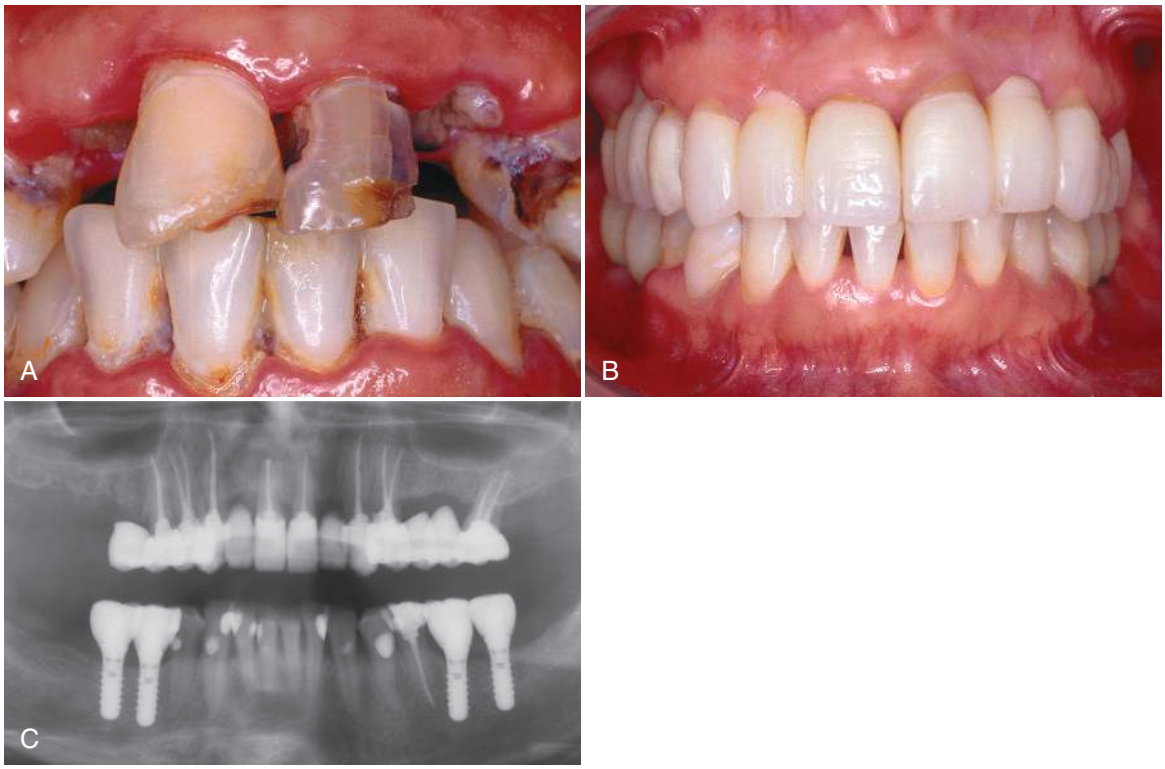


Fig. 8.3 Root canal therapy as part of a comprehensive treatment plan. The patient, who was recovering from intravenous drug addiction, requested restorative dental treatment. Because of extensive decay, several teeth had to be extracted, and nine teeth were treated endodontically. Root canal treatment was aided by nickel-titanium rotary instruments, and obturation was done with lateral compaction of gutta-percha and AH26 as the sealer. Microsurgical retrograde therapy was performed on tooth #8, and the distobuccal root of #14 had to be resected. Metal-free adhesively luted restorations were placed, and missing mandibular teeth were replaced by implants. **A,** Pretreatment intraoral status, showing oral neglect. **B,** Posttreatment intraoral status at 4-year follow-up, showing fully functional, metal-free, tooth-colored reconstructions. **C,** Panoramic radiograph at 4-year recall shows sound periradicular tissues in relation to endodontically treated teeth. (Restorations done by Dr. Till N. Göhring.)

third of the root canal has to be ascertained (see upcoming discussion).

BIOLOGIC OBJECTIVE

Schilder suggested that canals should be prepared to a uniform and continuous taper;[386] however, this guideline was aimed at facilitating obturation rather than targeting antimicrobial efficacy. For optimal disinfection, preparation shape and antimicrobial efficacy are intimately related through efficient removal of infected pulp and dentin (Fig. 8.6) and creation of space for delivery of irrigants.

Traditionally, fluids have been dispensed passively into root canals by syringe and needle (Fig. 8.7). When delivered with passive needle irrigation, solutions have been shown to progress only about 1 mm farther than the tip of the needle.[169,350,370] Enlarged apical canals and finer more flexible needles allow increasingly deeper needle placement, and this improves débridement and disinfection of canals.[14,464] Irrigation methods and solution are discussed in detail elsewhere in this chapter.

TECHNICAL OBJECTIVE

While a continuous taper that encompasses the original shape and curvature of a given root canal is an accepted goal, final apical preparation size remains a much-disputed entity in root canal therapy, as does final taper of the preparation.[43] Arguments were made for better disinfection with larger sizes (i.e., #50 or greater)[79,363] in combination with smaller tapers of .02 to .05. Others found no difference whether the selected final size was small or large.[95,500]

CLINICAL ISSUES

A wide spectrum of possible strategies exists for attaining the goal of removing the canal contents and eliminating infection. Lussi and colleagues[264] introduced an approach to removing canal contents and accomplishing disinfection that did not involve the use of a file: the so-called noninstrumentation technique. This system consisted of a pump, a hose, and a special valve that was cemented into the access cavity (Fig. 8.8, *A*) to provide oscillation of irrigation solutions (1% to 3% sodium hypochlorite) at a reduced pressure. Although in vitro data[263] suggested that canals could be cleaned and filled using this noninvasive system (see Fig. 8.8, *B* and *C*), clinical results were not as convincing (see Fig. 8.8, *D*).[25] Recently, a strategically similar technique was introduced, the GentleWave (Sonendo, Orange, CA, USA) unit, described in detail later.

At the opposite end of the spectrum of débridement strategies is to remove all intraradicular infection through extraction of the tooth in question. Almost invariably, periradicular lesions heal after extraction of the involved tooth.

Clinical endodontic therapy takes place somewhere along this spectrum of treatment strategies. This is reflected in

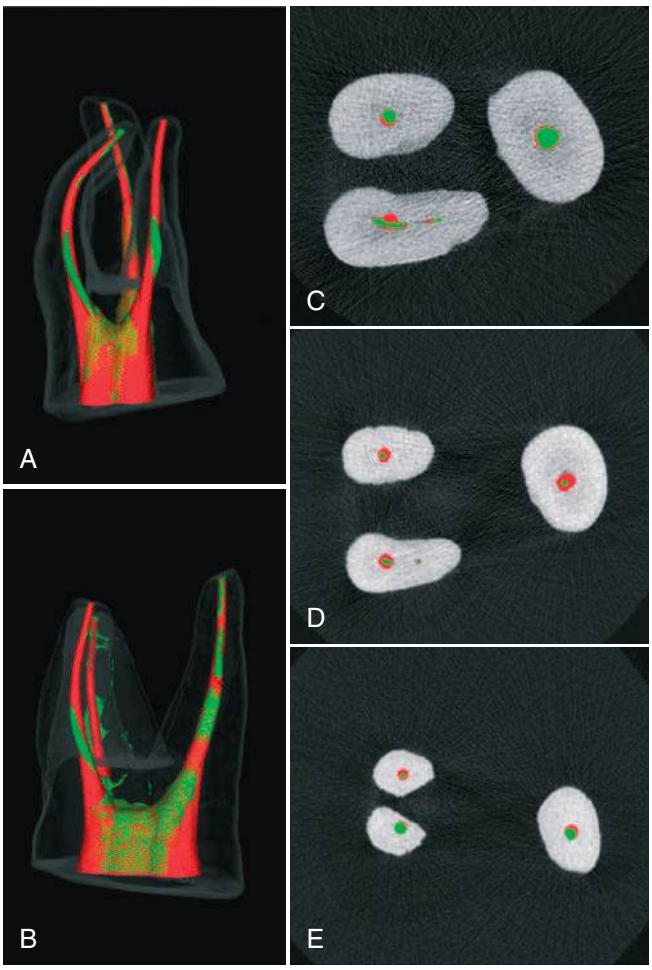


Fig. 8.4 Example of a desired shape, with the original root canal fully incorporated into the prepared outline. **A–B,** Micro–computed tomography reconstructions in clinical and mesiodistal views of a maxillary molar prepared with a NiTi rotary system. The *green area* indicates the pretreatment shape, and the *red area* indicates the posttreatment shape. Areas of *mixed red and green* indicate no change (i.e., no removal of radicular dentin). **C–E**, Cross sections of the coronal, middle, and apical thirds; the pretreatment cross sections *(green)* are encircled by the posttreatment outlines *(red)* in most areas. (***A–B***, From Hübscher W, Barbakow F, Peters OA: Root-canal preparation with FlexMaster: canal shapes analysed by micro-computed tomography, *Int Endod J* 36:740–747, 2003.)

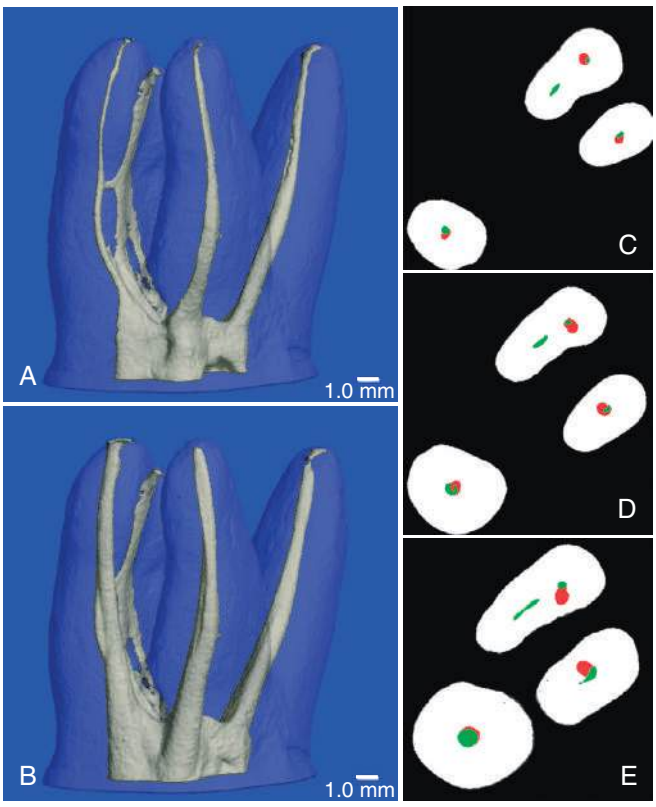


Fig. 8.5 Example of excessive thinning of dental structure during root canal treatment. **A–B**, Micro–computed tomography reconstructions show pretreatment and posttreatment root canal geometry of a maxillary molar. **C–E**, Cross sections of the coronal, middle, and apical thirds with pretreatment canal cross sections. Note the transportation and thinning, in particular, in the main mesiobuccal canal.

some of the controversies that surround the cleaning and shaping process. Once the decision has been made to initiate endodontic treatment, a clinician must integrate their knowledge of dental anatomy, immunology, and bioengineering science with clinical information.

Endodontic therapy has been compared with a chain of events, wherein the chain is only as strong as each individual link. For the purpose of this chapter, shaping and cleaning of the root canal system is considered a decisive link, because shaping determines the efficacy of subsequent procedures. It includes mechanical débridement, the creation of space for the delivery of medicaments, and optimized canal geometries for adequate obturation.[326] These tasks are attempted within a complex anatomic framework, as recognized in the early 20th century by Walter Hess (Fig. 8.9; see also Chapter 7 for a complete description of root canal anatomy).[196]

A clinician must choose appropriate strategies, instruments, and devices to overcome challenges and accomplish precise preparation in shape, length, and width. This allows endodontic therapy to address various forms of the disease processes described previously (Fig. 8.10). Recall radiographs taken at appropriate intervals predictably demonstrate longevity and favorable outcomes (see Figs. 8.1, 8.2, and 8.11) if a systematic approach to root canal shaping is adhered to (see Box 8.1).

Endodontic files traditionally have been manufactured according to empiric designs, and most instruments still are conceived based on individual clinicians' philosophies rather than developed through an evidence-based approach. Similar to the development of composite resins in restorative dentistry, the development of new files is a fast and market-driven process. With new instruments becoming available, the clinician may find it difficult to pick the file and technique most suitable for an individual case. Clinicians must always bear in mind that all file systems have specific benefits and weaknesses. Ultimately, clinical experience, handling properties, usage safety, and case outcomes, rather than marketing or the inventor's name, should drive decisions toward a particular design. The following section describes typical instruments used in root canal shaping.

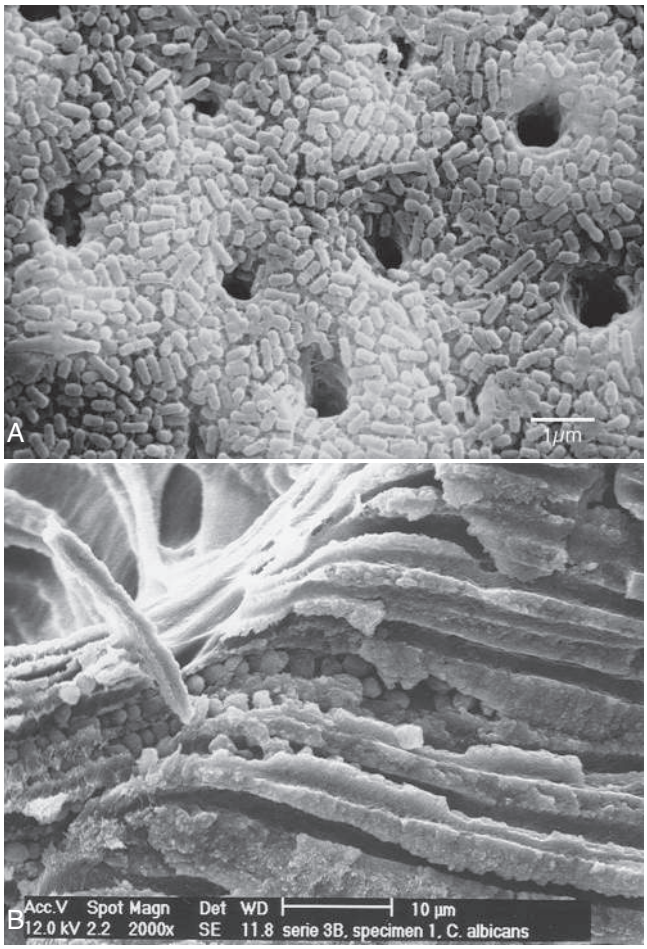


Fig. 8.6 Presence of microorganisms inside the main root canal and dentinal tubules. **A,** Scanning electron micrograph of a root canal surface shows a confluent layer of rod-shaped microbes (×3000). **B,** Scanning electron micrograph of a fractured root with a thick smear layer and fungi in the main root canal and dentinal tubules. (***A***, Courtesy Professor C. Koçkapan; ***B***, Courtesy Professor T. Waltimo.)

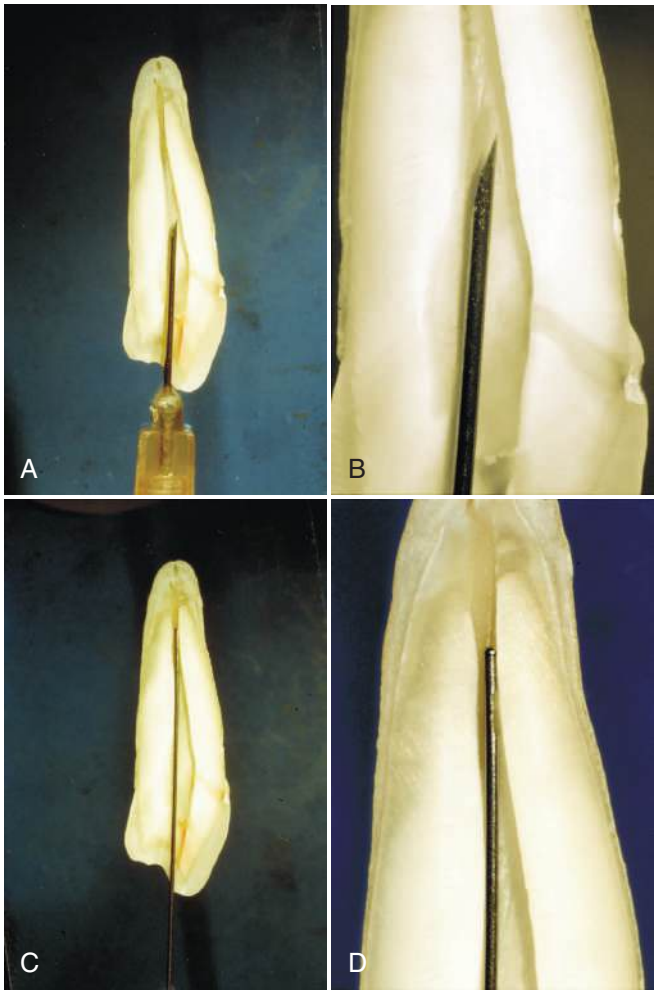


Fig. 8.7 Irrigation needles inserted into prepared root canals. **A**–**B**, A 27-gauge needle barely reaches the middle third. **C**–**D**, A 30-gauge, side-venting needle reaches the apical third in adequately enlarged canals.

Endodontic Instruments

GENERAL CHARACTERISTICS

Design Elements

Root canal preparation instruments such as K-files and nickel-titanium rotary instruments follow certain design principles that relate to drills and reamers used for work in wood and metal, respectively, while other instruments such as broaches and Hedström files do not find a direct technological correlate. Design elements such as the tip, flutes, and cross sections are considered relevant for files and reamers used in rotary motion. These pertinent aspects are briefly described later; for a more detailed review, the reader is referred to the literature.[162,365,371,423]

Tip Design. In root canal preparation, an instrument tip has two main functions: to guide the file through the canal and to aid the file in penetrating deeper into the canal. A clinician unfamiliar with the tip design, in particular of a rotary instrument, may do either of the following: (i) transport the canal (if the tip is capable of enlarging the canal and is used too long in one position in a curved canal) or (ii) encounter excessive torsion and break the file (if a noncutting tip is forced into a canal with a smaller diameter than the tip).

The angle and radius of its leading edge and the proximity of the flute to its actual tip end determine the cutting ability of a file tip. Cutting ability and file rigidity determine the propensity to transport the canal. The clinician must keep in mind that as long as a flexible file with a noncutting tip is engaged, 360-degree canal transportation is unlikely to occur.[360]

Studies have indeed shown that tip design affects file control, efficiency, and outcome in the shaping of root canal systems.[283,284] The tip of the original K-file resembled a pyramid; instrument tips have been described as *cutting*, *noncutting*, and *partially cutting*, although no clear distinction exists among the three types (Fig. 8.12).

Noncutting tips, also called *Batt tips*, are created by grinding and smoothing the apical end of the instrument (see Fig. 8.12, *A*). A tip modification was introduced with the Flex-R file, which was manufactured fully by grinding

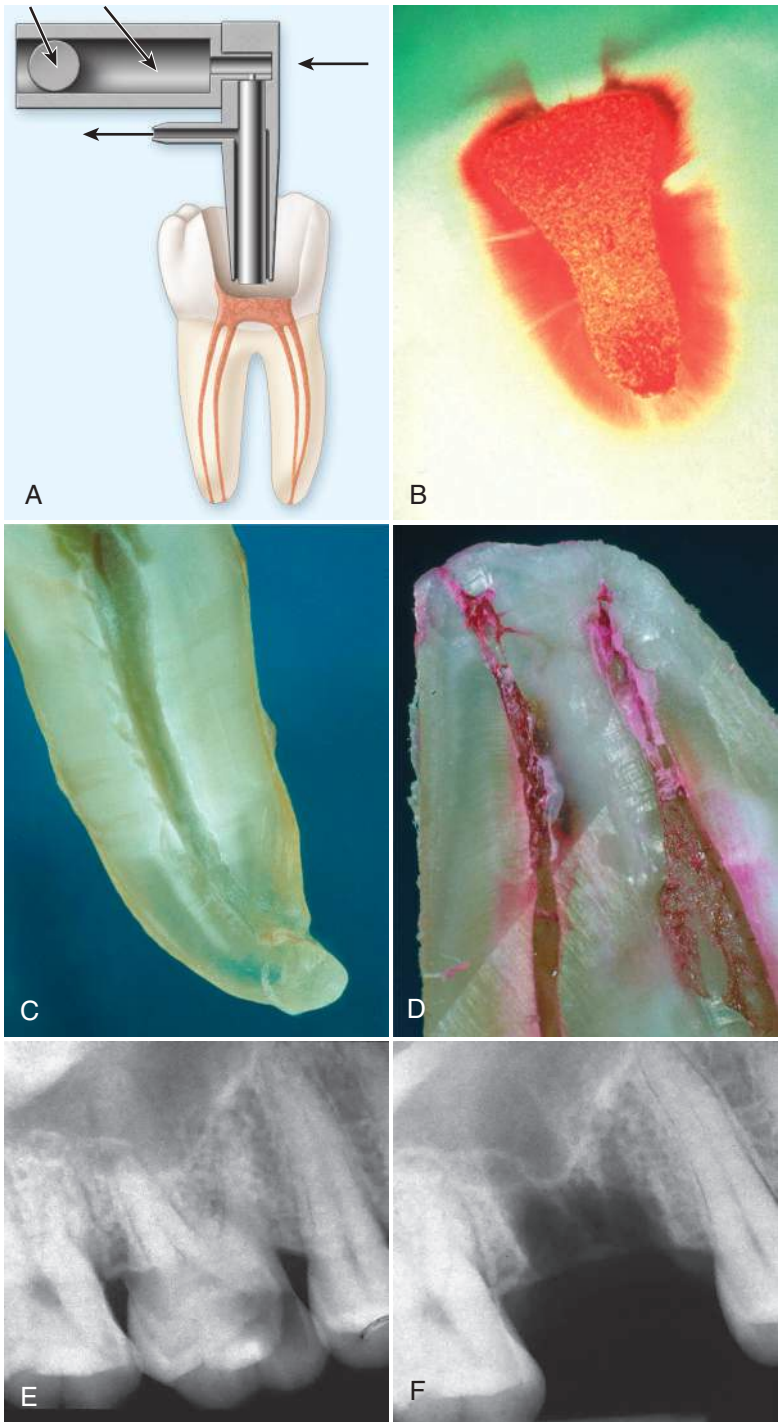


Fig. 8.8 Spectrum of strategies for accomplishing the primary aim of root canal treatment: elimination of infection. **A,** Schematic diagram of minimally invasive therapy using the noninstrumentation technique (NIT). **B,** Example of teeth cleaned in vitro using NIT. Note the clean intracanal surface, which is free of adhering tissue remnants. **C**–**D**, Examples of teeth cleaned in vivo and later extracted to investigate the clinical effects of NIT. Note the relatively clean, tissue-free canal space in **C** and the significant tissue revealed by rhodamine B staining in **D. E**–**F**, Course of maximally invasive therapy; apically involved tooth #30 was extracted, effectively removing the source of periradicular inflammation. (***A***–***B***, Courtesy Professor A. Lussi. ***C***–***D***, Courtesy Professor T. Attin. ***E***–***F***, Courtesy Dr. T. Kaya.)

so that the transitional angles between the tip and the instrument's working parts blend smoothly.[361] Similar techniques are required to manufacture NiTi K-files.[449]

For NiTi rotary files, typically rounded noncutting tips are used (see Fig. 8.12, *B*), which are effective in preventing preparation errors that were found with earlier so-called safe cutting tips.[210] One exception to this rule are rotaries specifically designed as retreatment instruments; cutting tips in that case facilitate removal of the existing root canal filling material and are sufficiently safe.

Longitudinal and Cross-Sectional Design. The *flute* of the file is the groove in the working surface used to collect soft tissue and dentin chips from the wall of the canal. The

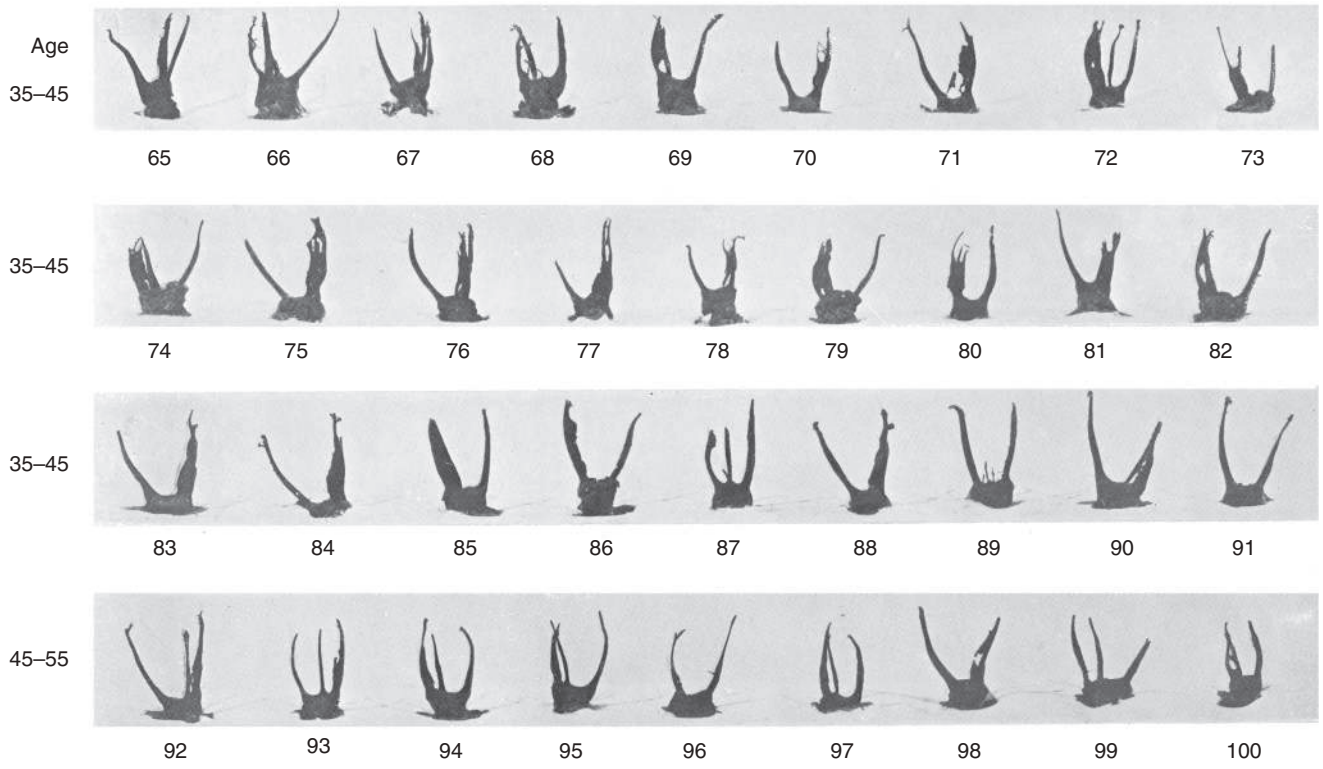


Fig. 8.9 Panel of 36 anatomic preparations of maxillary molars from the classic work by Professor Walter Hess of Zurich. Note the overall variability of root canal systems and the decrease of canal dimensions with age. (From Hess W: *The anatomy of the root canals of teeth of the permanent dentition*, London, 1925, John Bale, Sons & Danielsson).

effectiveness of a flute depends on its depth, width, configuration, and surface finish. The surface with the greatest diameter that follows the groove (where the flute and land intersect) as it rotates forms the *leading (cutting) edge* or the *blade* of the file. The cutting edge creates and deflects dentin chips along the wall of the canal and severs or snags soft tissue. Its effectiveness depends on its angle of incidence and sharpness.

Some instruments have a feature between trailing and cutting edge that forms a larger contact area with the radicular wall; this surface is called a radial *land* (Fig. 8.13). This design is believed to reduce the tendency of the file to thread into the canal. It also supports the cutting edge and limits the depth of cut; its position relative to the opposing cutting edge and its width determine its effectiveness. On the other hand, landed files are typically less cutting efficient compared with triangular cross sections.[335]

To reduce frictional resistance, some of the surface area of the land that rotates against the canal wall may be reduced to form the *relief*. The angle the cutting edge forms with the long axis of the file is called the *helical angle* (see Fig. 8.13).

If a file is sectioned perpendicular to its long axis, the *rake angle* is the angle formed by the leading edge and the radius of the file through the point of contact with the radicular wall. The rake angle may be *positive* or *cutting* (Fig. 8.14, *A*). If the angle formed by the leading edge and the surface to be cut is 90 degrees, the rake angle is said to be *neutral*. It may also be *negative* or *scraping* (see Fig. 8.14, *C*). The *cutting angle* is considered a better indication of a file's cutting ability and is determined by measuring the angle formed by the leading edge of the file and a tangent to the radicular wall in the point of contact. The clearance angle corresponds to the cutting angle at the trailing edge of the file and, in case of reciprocating action, becomes the cutting angle. The sum of cutting angle and rake angle is 90 degrees.

The *pitch* of the file is the distance between a point on the leading edge and the corresponding point on the adjacent leading edge (the distance from one "spiral twist" to the next; see Fig. 8.13). The smaller the pitch or the shorter the distance between corresponding points, the more spirals the file has and the greater the helix angle. While K-files have a constant pitch typically in the range of 1 mm, many NiTi rotaries have a variable pitch—one that changes along the working surface. When variable pitch is used, for example to modify the helical angle, usually tighter spirals are located close to the tip of the file and more space between the flutes is located toward the coronal part of the file. A longitudinal section through an instrument reveals the *core* (see Fig. 8.13). The outer diameter of a tapered instrument increases from the file tip toward the handle. Depending on core dimensions, flutes may become proportionately deeper, resulting in a *core taper* that may be different from the *external taper*.

Cutting angles, helix angles, and external and core tapers may vary along the working surface of the file, and the ratios of these quantities can vary among instruments of the same series.

Taper. The *taper* usually is expressed as the amount that the file diameter increases each millimeter along its working surface from the tip toward the file handle. For example,

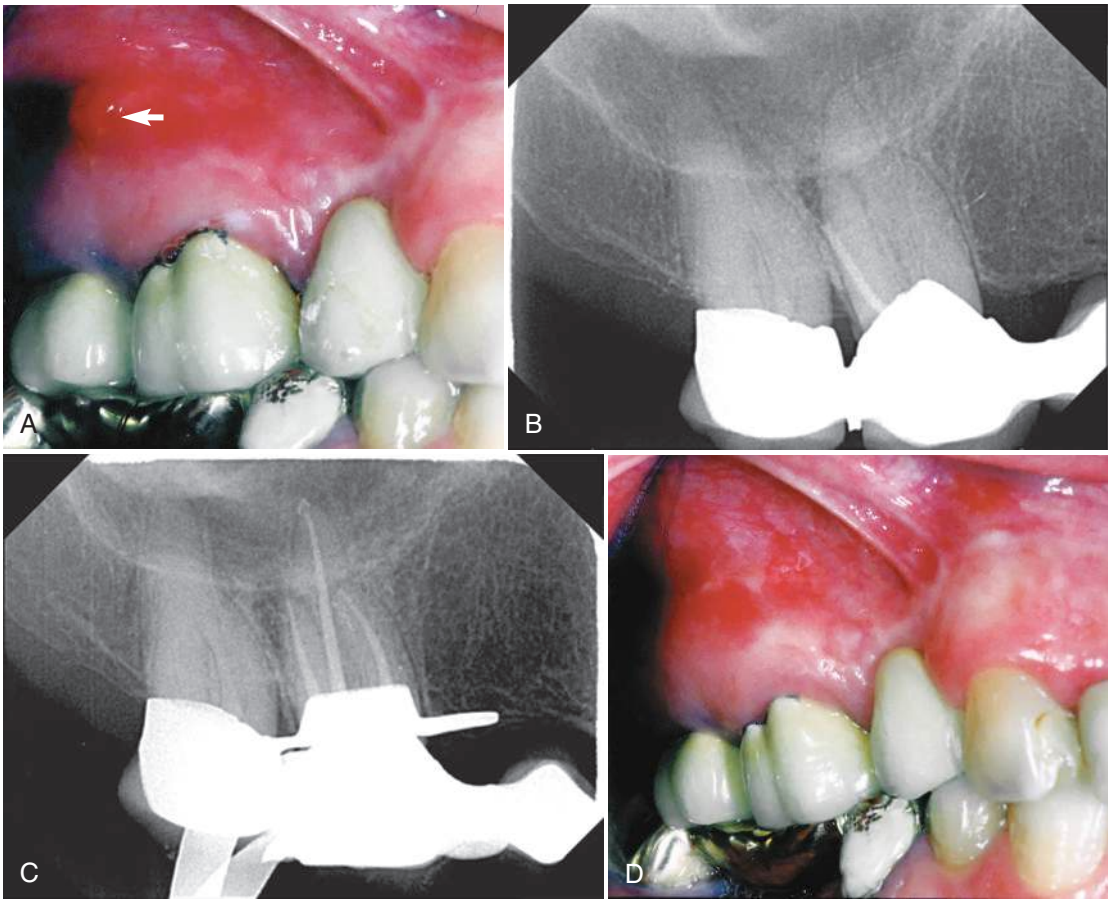


Fig. 8.10 Sinus tract as a sign of a chronic apical abscess and effect of routine root canal treatment. **A,** Intraoral photograph of left maxillary region with draining sinus tract *(arrow)* periapical to tooth #14. **B,** Pretreatment radiograph with gutta-percha point positioned in the sinus tract, pointing toward the distobuccal root of #14. **C,** Finished root canal fillings after 2 weeks of calcium hydroxide dressing. **D,** Intraoral photograph of the same region as in **A,** showing that the sinus tract had closed by the time obturation was performed.

a size #25 file with a .02 taper would have a 0.27 mm diameter 1 mm from the tip, a 0.29 mm diameter 2 mm from the tip, a 0.31 mm diameter 3 mm from the tip, and so forth. Instruments can have constant or variable taper. Current instrument developments include variations in helical angle, pitch, and taper along the cutting portion, which along with variations in alloy and rotational speed (rpm) all affect cutting behavior.[335] The ability to determine cross-sectional diameter at a given point on a file can help the clinician determine the file size in the point of curvature and the relative stress being placed on the instrument. Instruments with greater tapers are designed so that the tip of the instrument functions as a guide, and the middle and coronal part of the instrument's working part is the one engaging the canals walls.

International Standards Organization Norms

Standardized specifications have been established to improve endodontic instrument quality.[212] For example, the International Standards Organization (ISO) has worked with the Fédération Dentaire Internationale (FDI) and other stakeholders to define specifications. These standards are designated with an ISO number. The American Dental Association (ADA) also has been involved in this effort, as has the American National Standards Institute (ANSI); these standards are designated with an ANSI number. However, new instrument designs have resulted in a need for reconsideration of the standards.

Two ISO standards pertain to endodontic instruments: ISO No. 3630-1 deals with K-type files (as does ANSI No. 28), Hedström files (ANSI No. 58), and barbed broaches and rasps (ANSI No. 63). ISO No. 3630-3 deals with condensers, pluggers, and spreaders (ANSI No. 71); however, the term *ISO-normed instruments* is often synonymous for K-files (Fig. 8.15).

One important feature of ISO-normed hand instruments is a defined increase in tip diameter of 0.05 or 0.1 mm, depending on the instrument size (Fig. 8.16). ISO-normed K- and Hedström files (Fig. 8.17) are available in different lengths (21, 25, and 31 mm), but all have a 16-mm-long section of cutting flutes (see Figs. 8.12 and 8.15). The cross-sectional diameter at the first rake angle of any file is labeled D_0. The point 1 mm coronal to D_0 is D_1, the point 2 mm coronal to D_0 is D_2, and so on up to D_{16}. The D_{16} point is the largest diameter of an ISO-normed instrument. Each file derives its numeric name from the diameter at D_0 and is assigned a specific color code (see Fig. 8.15). Another aspect of ISO files is the standard taper of 0.32 mm over 16 mm of cutting blades, or 0.02 mm increase in diameter per millimeter of flute length (.02 taper).

The ISO-normed design is a simplification that has specific disadvantages, and it may explain the clinical observation

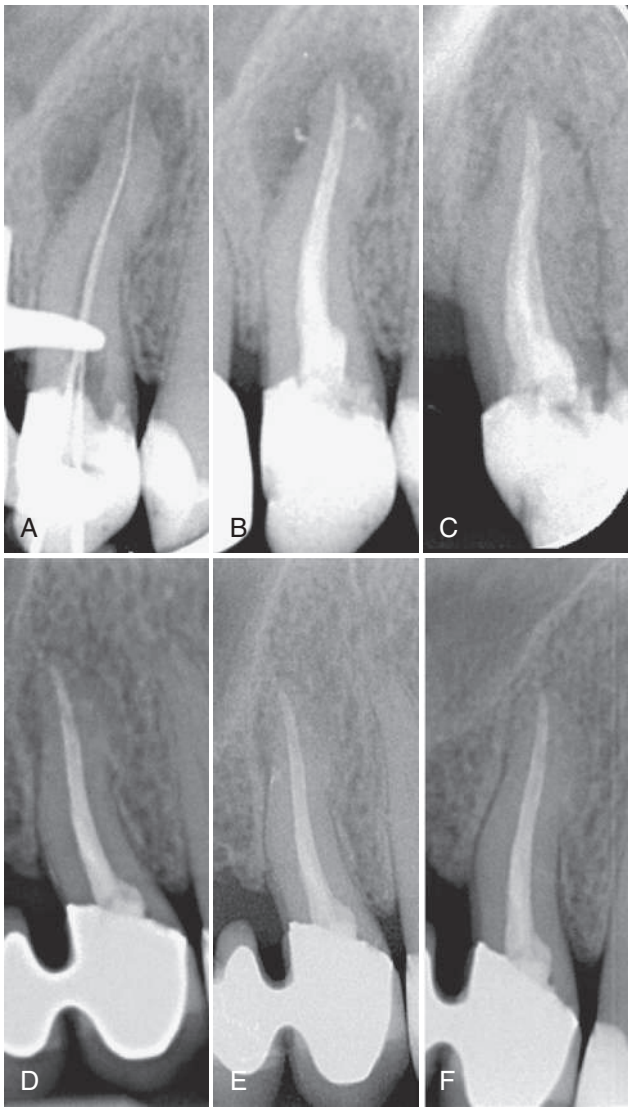


Fig. 8.11 Relationship of radicular anatomy and endodontic disease as shown by filled accessory canals. **A,** Working length radiograph of tooth #13 shows lesions mesially and distally but not apically. **B,** Posttreatment radiograph shows the accessory anatomy. **C,** Six-month recall radiograph before placement of the restoration. **D,** Two-year recall radiograph after resection of the mesiobuccal root of tooth #14 and placement of a fixed partial denture. Excess sealer appears to have been resorbed, forming a distal residual lesion. **E,** Four-year recall radiograph shows almost complete bone fill. **F,** Seven-year recall radiograph; tooth #14 is radiologically sound and clinically within normal limits.

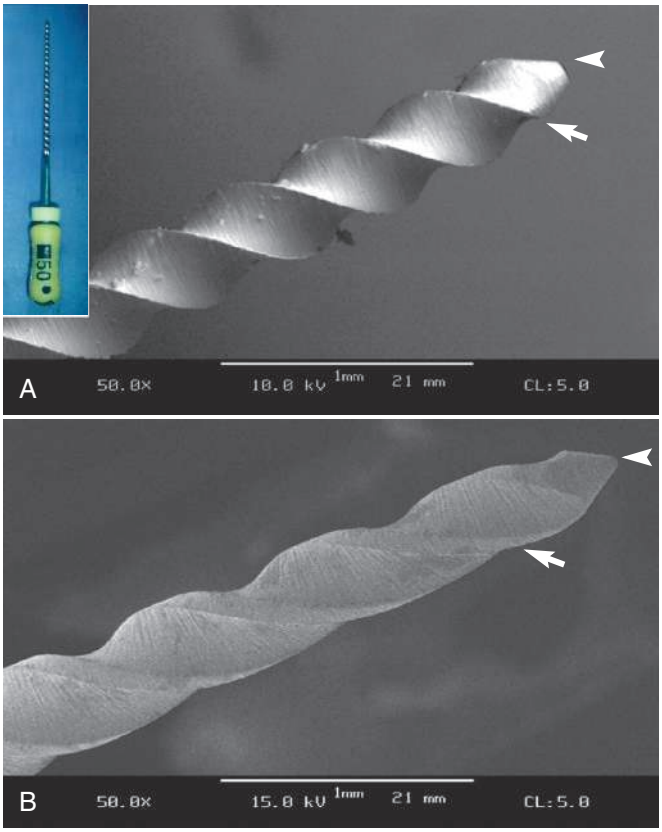


Fig. 8.12 Comparison of the flute geometry and tip configuration of a hand file *(insert)* and a NiTi rotary instrument. **A,** K-file with sharp cutting edges *(arrow)* and Batt tip *(arrowhead)*. **B,** GT rotary file with rounded, noncutting tip *(arrowhead),* smooth transition, and guiding radial lands *(arrow)*.

that enlarging a root canal from size #10 to #15 is more difficult than the step from size #55 to #60. The introduction of K-type files with tip sizes between the ISO-stipulated diameters seemed to solve the problem. However, the use of such file is not universally recommended, perhaps because the approved machining tolerance of 0.02 mm could negate any intended advantages. Moreover, although 0.02 mm tolerance is stipulated by the ISO norm, most manufacturers do not adhere to it.[223,389,437,511] Another suggested modification relates to tips with a constant 29% of diameter increments.

Further changes to the numbering system for files with different sizes have been implemented by several manufacturers. One system has introduced "half" sizes in the range of #15 through #60, resulting in instruments in sizes #15, #17.5, #20, #22.5, and so on.

Alloys

There are currently two principally different types of alloys used for endodontic instruments: stainless steel and nickel-titanium. Most manually operated endodontic instruments are fabricated from stainless steel and have considerable resistance to fracture. A clinician who is careful in applying force and adheres to a strict program of discarding instruments after use should have few instrument fractures. Stainless steel files are comparably inexpensive so that adequate cleaning and sterilization for reuse of files in sizes up to #60 may not be cost effective. Indeed, files in the range up to #60 have been considered disposable instruments.[429]

Several burs and instruments designed for slow-speed handpiece operation such as Gates Glidden drills, Peeso burs, and pilot drills for intraradicular posts are also manufactured from stainless steel. Instruments designed for rotary root canal instrumentation, however, are typically made of nickel-titanium.[393] This alloy offers unique properties, specifically flexibility, and corrosion resistance.

Physical and Chemical Properties of Steel and Nickel Titanium Alloys. Basic engineering terms relate to metals and their behavior when used to manufacture

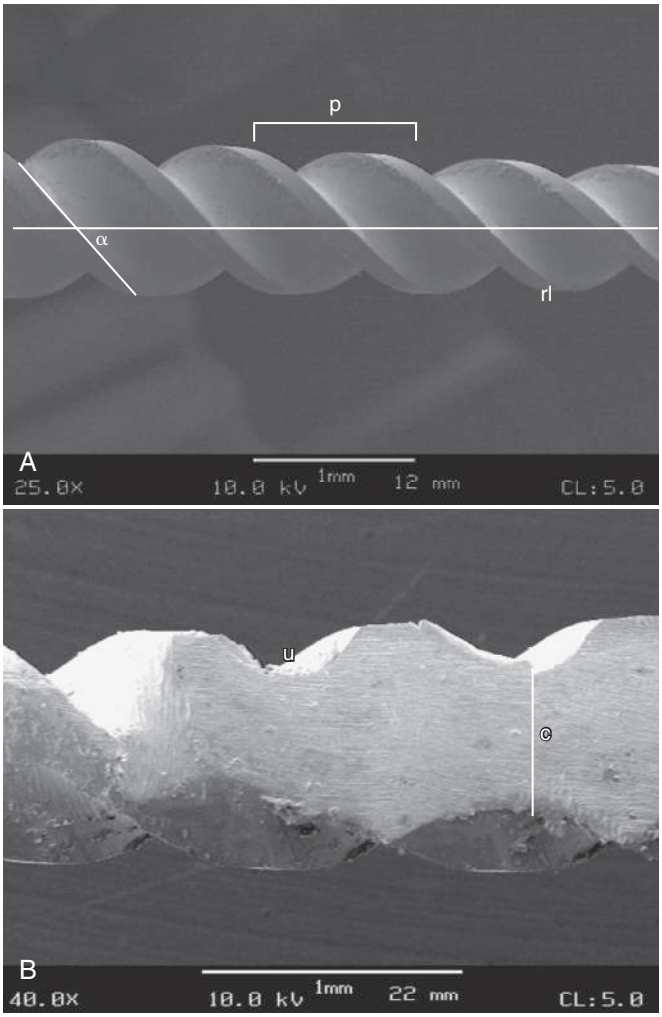


Fig. 8.13 Design characteristics of nickel-titanium rotary instruments. **A,** Lateral view showing the details of the helical angle, pitch *(p),* and the presence of guiding areas, or radial lands *(rl)* (scanning electron micrograph, ×25). **B,** Ground-working part of the instrument in **A,** showing U-shaped excavations and the dimension of the instrument core *(c).*

endodontic instruments. Stress-strain diagrams describe the response of metal wires under loading depending on their crystal configuration (Fig. 8.18).

During the development of the equiatomic *nitinol* (this acronym is derived from **ni**ckel-**ti**tanium investigated at the **N**aval **O**rdinance **L**aboratory) alloy (55% [by weight] nickel and 45% [by weight] titanium), several effects were noted that relate to its specific crystal arrangement with two stable main phases, *austenite* and *martensite* (Fig. 8.19): a shape memory effect as temperature- and strain-dependent pseudoelasticity, all attributable to specific thermodynamic properties of the new alloy.[69,119,310,331,449]

Walia and others[474] thought that the pseudoelastic properties of 55-nitinol might prove advantageous in endodontics and initially tested hand instruments. They found that size #15 NiTi instruments were two to three times more flexible than stainless steel instruments; moreover, the instruments showed superior resistance to angular deflection.[474]

Furthermore, hardly any plastic deformation of cutting flutes was recorded when an instrument was bent up to 90 degrees, and forces required to bend endodontic files to 45 degrees were reduced by 50% with nickel titanium, also known as NiTi.[393,474] Serene and others[393] speculated that heat, probably during sterilization cycles, could even restore the molecular structure of used NiTi files, resulting in an increased resistance to fracture. Such a behavior is claimed to occur for current martensitic instruments.[332]

These unusual properties are the result of a molecular crystalline phase transformation in specific crystal structures of the austenitic and martensitic phases of the alloy.[449] External stresses transform the austenitic crystalline form of NiTi into a martensitic crystalline structure that can accommodate greater stress without increasing the strain. As a result, a NiTi file has transformational elasticity, also known as *pseudoelasticity*, or the ability to return to its original shape after being deformed (see Fig. 8.19, *B*). This property dictates that typical NiTi instruments are manufactured by milling rather than twisting. Twisting incorporates plastic deformation and is used, for example, to produce stainless steel K-files. Another recently introduced manufacturing technique is electric discharge machining, which apparently does not introduce defects.[344]

Similar to the application of deforming forces, temperature changes also result in phase transformation (see Fig. 8.19, *A*) from austenite to martensite and vice versa.[190,282] Moreover, thermal heat applied either during the production of the raw wire or after manufacturing of the file can be used to modify its properties, most importantly, its flexibility.[172,399] For austenitic endodontic instruments a recoverable elastic response of up to 7% is expected (Fig. 8.20). However, more martensitic instruments will

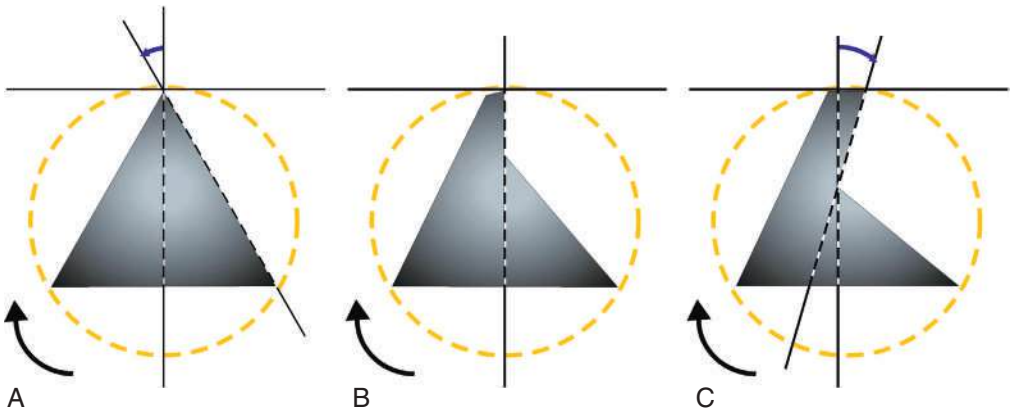


Fig. 8.14 The cutting angle of an endodontic file can be negative (**A**), neutral (**B**), or positive (**C**).

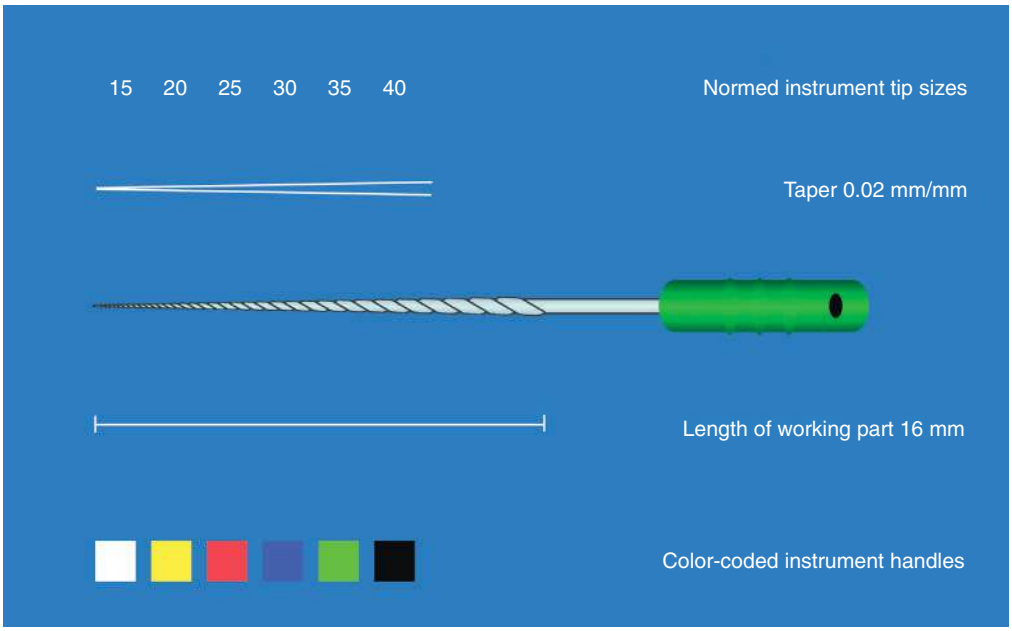


Fig. 8.15 Schematic drawing of an ISO-normed hand instrument size #35. Instrument tip sizing, taper, and handle colors are regulated by the ISO/ANSI/ADA norm.

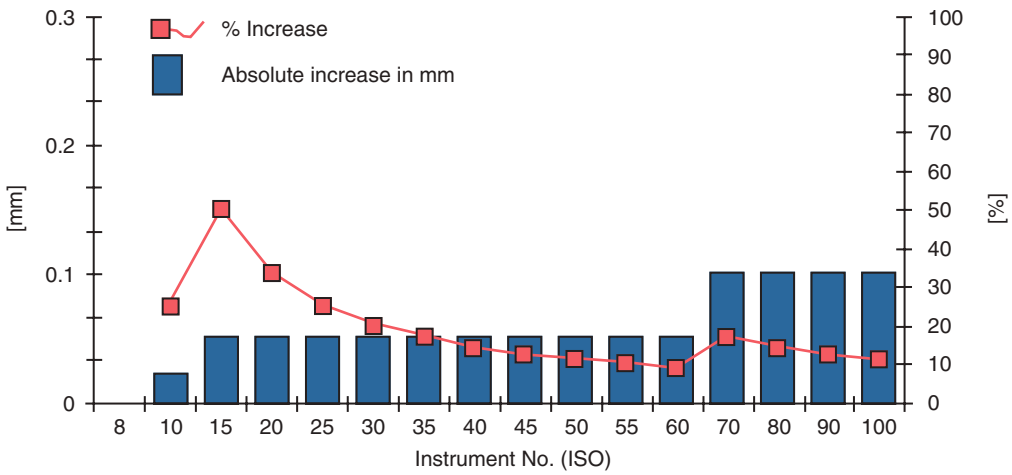


Fig. 8.16 Increase in tip diameter in absolute figures and in relation to the smaller file size. Note the particularly large increase from size #10 to size #15.

have less of an elastic range and are more likely to plastically deform during use.[332,399]

NiTi instruments may have imperfections such as milling marks, metal flash, or rollover.[474] Some researchers have speculated that fractures in NiTi instruments originate at such surface imperfections.[13]

Surface irregularities may provide reservoirs of corrosive substances—most notably sodium hypochlorite (NaOCl). Chloride corrosion may lead to micropitting[372] and possibly subsequent fracture in NiTi instruments.[184] Immersion in various disinfecting solutions for extended periods (e.g., overnight soaking) produced corrosion of NiTi instruments and subsequent decreased torsional resistance.[304,424] For ProTaper,[46] RaCe, and ProFile[339] instruments, also 2-hour immersion damaged the integrity of the alloy.

In the majority of studies sterilization procedures did not appear to negatively impact torsional strength[199,407] or fatigue resistance[71,198] of most NiTi instruments: austenitic[470] and martensitic[81] alloy behaved grossly similarly in this aspect. Some NiTi instruments are available in presterilized form, which eliminates the need to process and sterilize before use and provides a simpler workflow overall. There is an ongoing discussion over the impact of other aspects of clinical usage on the mechanical properties of NiTi rotaries. Most likely, clinical usage leads to some changes in the alloy, potentially through work hardening,[12,224]

Another strategy to improve file characteristics is electropolishing. Surface coatings and ion implantation have been tried. Electropolishing is a process that removes surface irregularities such as flash and bur marks. It is believed to improve material properties, specifically fatigue and corrosion resistance; however, the evidence for both these claims is contradictory. One study[19] found an extension of fatigue life for electropolished instruments, while others

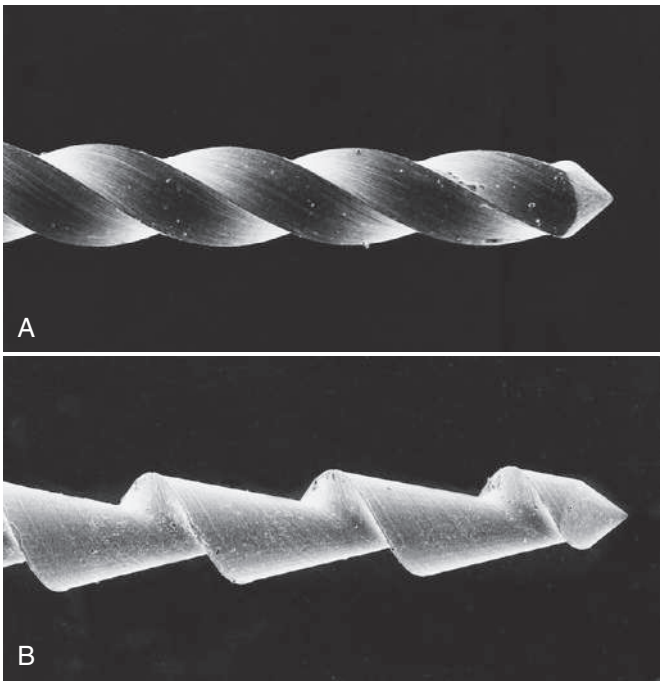


Fig. 8.17 Scanning electron micrographs of endodontic hand files fabricated by twisting (K-file size #40, **A**) and grinding (Hedström file #50, **B**). (***A–B***, Courtesy Dentsply Maillefer, Ballaigues, Switzerland.)

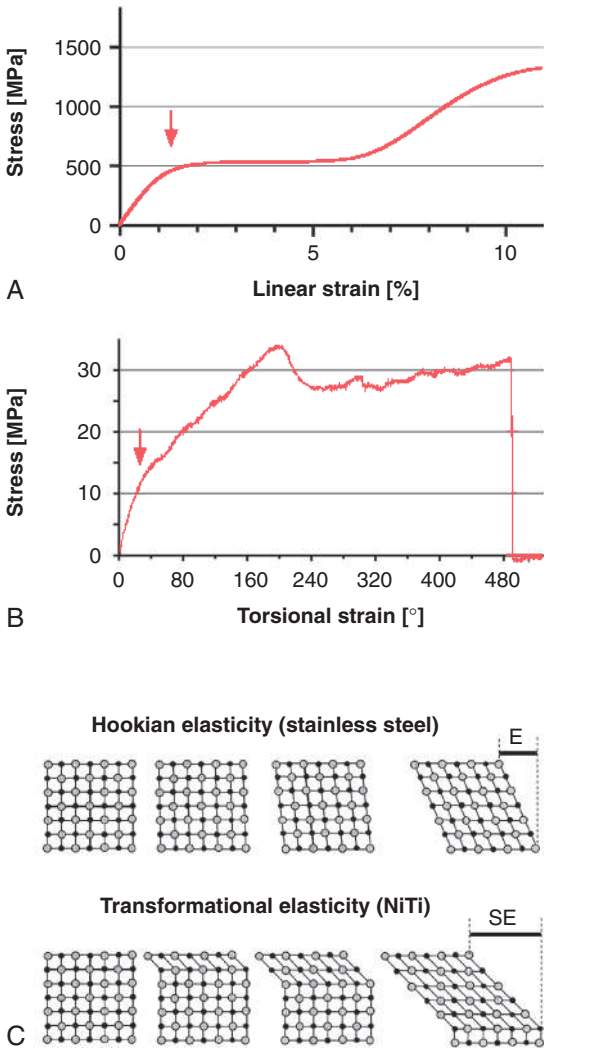


Fig. 8.18 Stress-strain behavior of nickel-titanium alloy. **A,** Schematic diagram of linear extension of a NiTi wire. **B,** Torque to failure test of a size #60, .04 taper ProFile NiTi instrument. Note the biphasic deformation, indicated by arrows in **A–B. C,** Comparison of stainless steel and nickel-titanium crystal lattices under load. Hookian elasticity accounts for the elastic behavior *(E)* of steel, whereas transformation from martensite to austenite and back occurs during the pseudoelastic *(PE)* behavior of NiTi alloy. (**C,** Modified from Thompson SA: An overview of nickel-titanium alloys used in dentistry, *Int Endod J* 33:297–310, 2000.)

found no improvement of fatigue resistance of electropolished instruments.[70,195] Boessler et al.[54] suggested a change in cutting behavior with an increase of torsional load after electropolishing.

Perhaps more relevant than surface treatment are modifications of the base alloy that significantly alter material properties within the atomic ratio.[323] The first commercialized alloy to exploit this was M-Wire (SportsWire, Langley, OK, USA), which was shown to have higher fatigue resistance with similar torsional strength.[220]

More recently, most instrument are manufactured from drawn standard NiTi wires, which are then worked by micro milling and finally submitted to specific annealing and cooling steps.[331,399] Instruments manufactured with this workflow tend to be relatively more martensitic at room and possible at body temperature during root canal treatment, as defined by their M_s temperature (see Fig. 8.19, A). Typically, such heat-treated alloys are more flexible[447] and present higher fatigue resistance.[332] Examples are the so-called gold and blue alloys types (Dentsply Sirona, York, PA, USA) or so-called controlled memory alloy used in Hyflex instruments (Coltene Endo, Cuyahoga Falls, OH, USA).

MANUALLY OPERATED INSTRUMENTS

Endodontic trays contain many items familiar from general dentistry, but certain hand instruments are designed specifically for endodontic procedures. This includes instruments employed for procedures inside the pulp space—for example, hand- and engine-driven instruments for root canal preparation and energized instruments for root canal shaping. Special instruments and devices for root canal obturation are selected for filling prepared canal spaces.

K-Type Instruments

Manually operated instruments are generically called *files.* Defined by function, files are instruments that enlarge canals with apico-coronal insertion and withdrawal motions.

Files were first mass produced by the Kerr Manufacturing Co. of Romulus, Michigan, in the early 1900s—hence the name K-type file (or K-file). K-files and K-reamers were manufactured by the same process (i.e., by twisting square or triangular metal blanks along their long axis; see Fig. 8.17, *A*). Three or four equilateral, flat surfaces were ground at increasing depths on the sides of a piece of wire, producing a tapered pyramidal shape. The wire then was stabilized on one end, and the distal end was rotated to form the spiral instrument. The number of sides and the number of spirals determine whether the instrument is best suited for filing or reaming. In general, a three-sided configuration with fewer

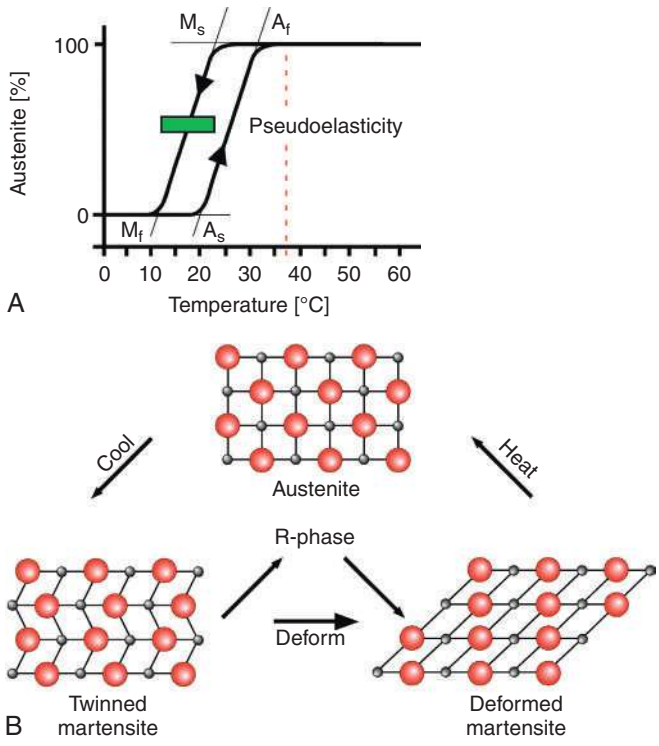


Fig. 8.19 Pseudoelastic behavior of nickel titanium is based on the two main crystal configurations, martensite and austenite, which depend on temperature (**A**) and applied strain (**B**). Formation of the respective configuration initiates at the start temperatures, M_s and A_s.

spirals (e.g., 16 per 16 mm working portion) is used for reaming (i.e., cutting and enlarging canals with rotational motions). A file has more flutes per length unit (e.g., 20) than a reamer, whereas a three-sided shape is generally more flexible than a four-sided one.[381]

K-type instruments are useful for penetrating and enlarging root canals. In general, a reaming motion (i.e., constant file rotation) causes less transportation than a filing motion (translational or "in-and-out" motion).[153] *Transportation* in root canal preparation may be defined in general as the movement away from the initial canal axis and is described in more detail later in this chapter.

Stainless steel K-files may be precurved by overbending; this procedure subjects the file to substantial strain and should therefore be done carefully. Permanent deformation occurs when the flutes become wound more tightly or opened more widely (Fig. 8.21). When such deformation occurs, an instrument should no longer be used; file fracture is likely to occur during clockwise motion after plastic deformation.[395]

Interestingly, although the force required for failure is the same in both directions of rotation,[244] failure occurs in the counterclockwise direction at half the number of rotations required for failure in the clockwise direction. Therefore K-type instruments should be operated more carefully when pressure is applied in a counterclockwise direction.

Cross-sectional analysis of a K-file reveals why this design allows careful application of clockwise and counterclockwise rotational and translational working strokes: The

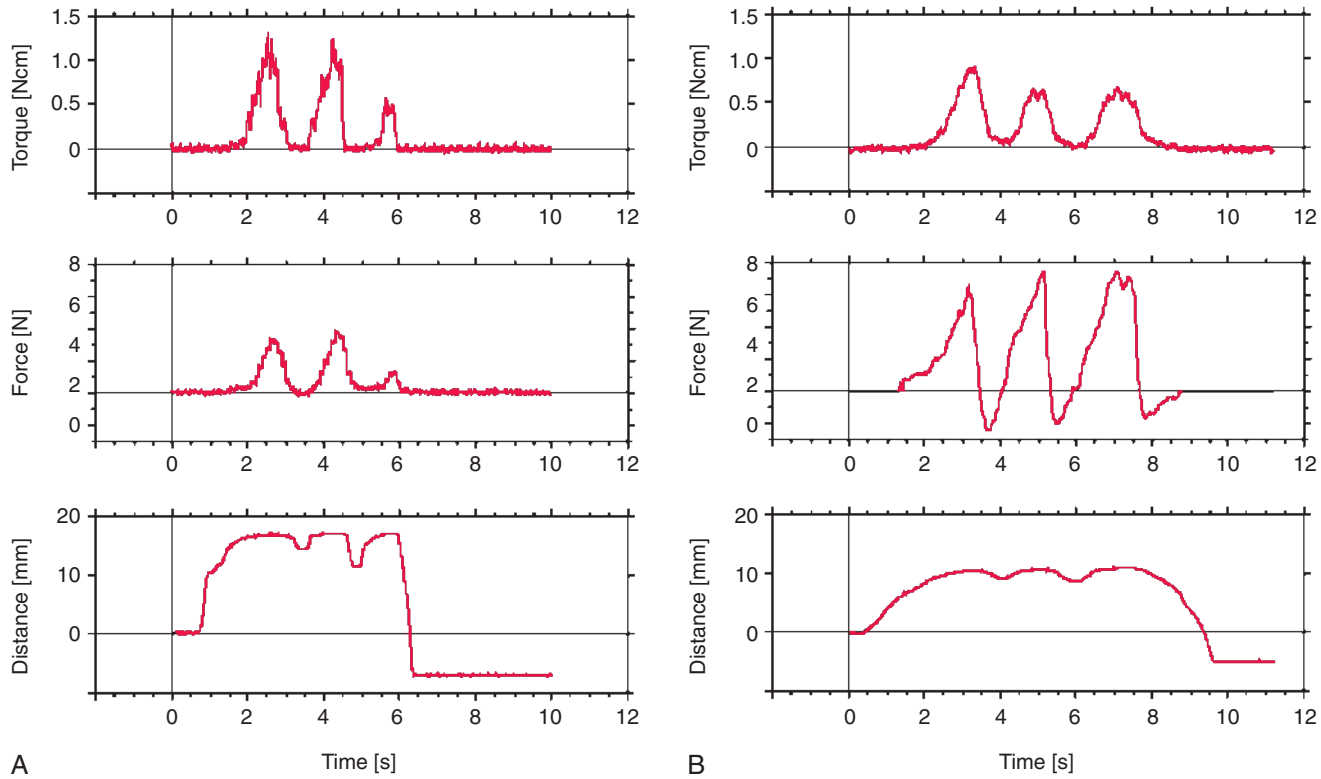


Fig. 8.20 Physical factors (torque, axial force, and insertion depth) that affect root canal instrumentation documented with a torque-testing platform. **A,** ProFile size #45, .04 taper used in a mildly curved canal of a single-rooted tooth, step-back after apical preparation to size #40. **B,** FlexMaster size #35, .06 taper used in a curved distobuccal canal of a maxillary first molar, crown-down during the initial phase of canal preparation.

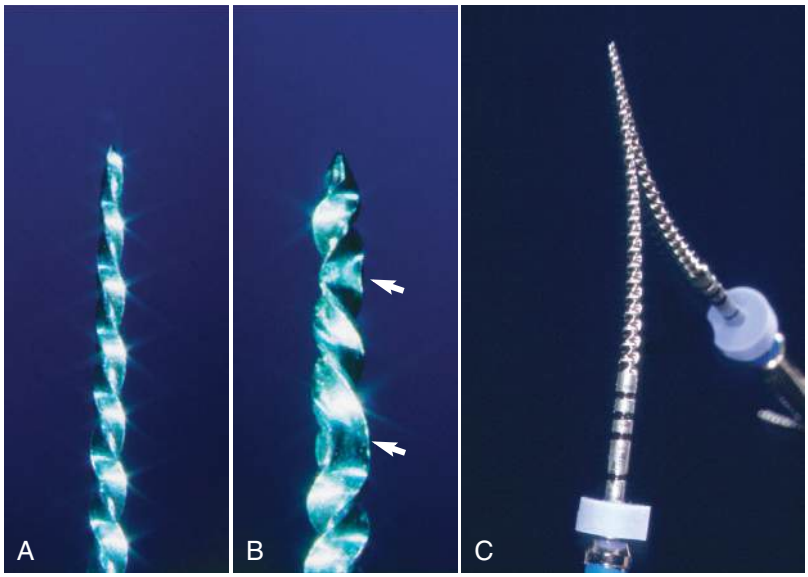


Fig. 8.21 Deformation of endodontic instruments manufactured from nickel-titanium alloy. **A** and **B,** Intact and plastically deformed ProFile instruments (*arrow* indicates areas of permanent deformation). **C,** ProFile instrument placed on a mirror to illustrate elastic behavior.

cross section is symmetrical with negative rake angles, allowing dentin to be adequately cut in both clockwise and counterclockwise direction.

H-Type Instruments

H-type instruments, also known as Hedström files (see Fig. 8.17, *B*), are milled from round, stainless steel blanks. These files are very efficient for translational strokes[381] because of a positive rake angle and a blade with a cutting rather than a scraping angle. Rotational working movements are discouraged because of the possibility of fracture. Hedström files up to size #25 can be efficiently used to relocate canal orifices and, with adequate filing strokes, to remove overhangs. Similarly, wide oval canals can be instrumented with Hedström files, as well as with rotary instruments. On the other hand, overzealous filing can lead to considerable thinning of the radicular wall and strip perforations (Fig. 8.22).

Precurving Hedström files results in points of greater stress concentration than in K-type instruments. Such prestressed areas may lead to the propagation of cracks and ultimately fatigue failure.[182] Note that clinically, fatigue fractures may occur without visible signs of deformation.

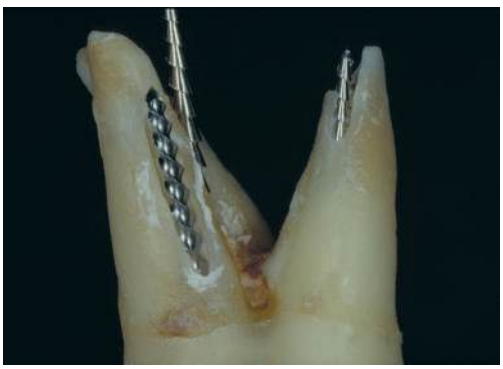


Fig. 8.22 Result of an overenthusiastic attempt at root canal treatment of a maxillary second molar with large stainless steel files. Multiple strip perforations occurred, so consequently the tooth had to be extracted.

Hedström files are produced by grinding a single continuous flute into a tapered blank. Computer-assisted machining technology has allowed the development of H-type instruments with very complex forms. Because an H-file generally has sharper edges than a K-file, it has a tendency to thread into the canal during rotation, particularly if the instrument's blades are nearly parallel. Awareness of threading-in forces is important to avoid instrument failure, also with NiTi rotary instruments.

Effectiveness and Wear of Instruments. The ability of an endodontic hand instrument to cut and machine dentin is essential; however, no standard exists for either the cutting or machining effectiveness of endodontic files, nor have clear requirements been established for resistance to wear. In any study of the effectiveness of an instrument, two factors must be investigated: (1) effectiveness in cutting or breaking loose dentin and (2) effectiveness in machining dentin.

Attempts have been made to evaluate the effectiveness of an instrument when used with a linear movement,[383] and these studies showed that instruments might differ significantly, not only when comparing brands and types but also within one brand and type. For K-files, effectiveness varies 2 to 12 times between files of the same brand. This variation for Hedström files is greater, ranging from 2.5 to more than 50 times.[436]

During canal preparation, a file's rake edge shaves off dentin that accumulates in the grooves between the rake edges. The deeper and larger this space, the longer the stroke can be before the instrument is riding on its own debris, making it ineffective. These design variations and the rake angle of the edges determine the effectiveness of a Hedström file. Of the hybrid files, the K-Flex (Kavo Kerr Endo, Orange, CA, USA) file has properties similar to those of K-files. The Flex-R file (Integra Miltex, Plainsboro, NJ, USA), which is a ground instrument with a triangular cross section similar to

a K-file, more closely resembles a Hedström file in its variations in cutting behavior. It also is more effective at substrate removal than the K-files but cannot measure up to the H-files' ability to machine radicular dentin.[436]

Barbed Broaches

Barbed broaches (Fig. 8.23) are produced in a variety of sizes and color codes. They are manufactured by cutting sharp, coronally angulated barbs into metal wire blanks. Broaches are intended to remove vital pulp from root canals, and in cases of mild inflammation and a large canal space, they work well for severing pulp at the constriction level in toto. The use of broaches has declined since the advent of NiTi rotary instruments, but broaching occasionally may be useful for expediting emergency procedures (see Chapter 19) and removing materials (e.g., cotton pellets or absorbent points) from root canals.

LOW-SPEED ENGINE-DRIVEN INSTRUMENTS

Burs

Specialized burs are available for endodontic access cavities. These burs are used in both high-speed and slow-speed handpieces and are manufactured from stainless steel. (Access cavity preparation and used materials are described in detail in Chapter 7.)

Gates-Glidden Drills

Gates-Glidden (GG) drills (Fig. 8.24) have been used for almost 150 years without noteworthy design changes. GG drills are typically used to enlarge coronal canal areas.[107] When misused, GG drills can dramatically reduce radicular wall thickness.[155,213]

GG instruments are available in six sizes and various lengths, by several manufacturers. Each instrument has a long, thin shaft with parallel walls and a short, oval cutting head with safety tips (see Fig. 8.24, *B*); these drills with tip diameters of 0.5 to 1.1 mm are produced in stainless steel and NiTi varieties. Because of their design and physical properties,[62] GG drills are side-cutting instruments; they can be used to cut dentin as they are withdrawn from the canal (i.e., on the outstroke) and should be used only in the straight portions of a canal.[456]

Two procedural sequences have been proposed: With the step-down technique, the clinician starts with a large drill and progresses to smaller ones; conversely, with the step-back technique, the clinician starts with a small drill and progresses to larger ones. Either technique efficiently opens root canal orifices and works best when canals exit the access cavity without severe angulations.

When used adequately, GG instruments are inexpensive, safe, and clinically beneficial tools. However, high speeds

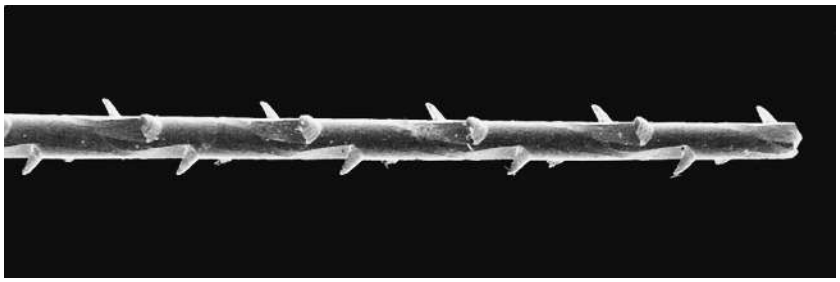


Fig. 8.23 Scanning electron micrograph of a barbed broach. (Courtesy Moyco Union Broach, York, PA, USA.)

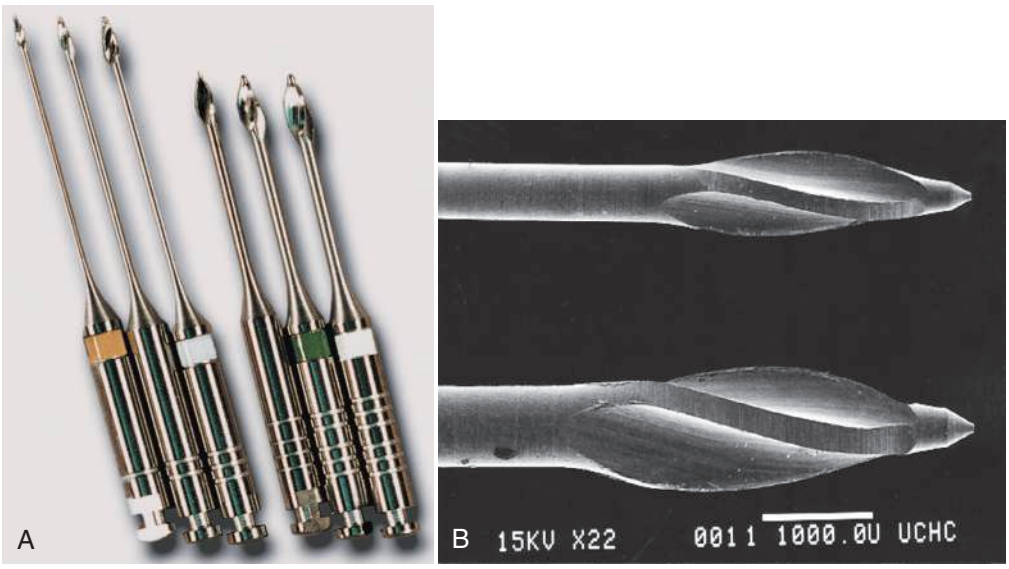


Fig. 8.24 Various Gates-Glidden (GG) burs made of stainless steel (**A**) and scanning electron micrograph (**B**, working tip). (***A***, From Johnson WT: *Color atlas of endodontics*, St Louis, 2002, Saunders.)

excessive pressure, an incorrect angle of insertion, and the use of GG instruments to aggressively drill into canals have resulted in mishaps such as strip perforation.

Peeso Reamers and Similar Drills

Peeso reamers and other drills are occasionally used in root canal preparation for coronal flaring or during postpreparation. At this point, these drills are manufactured mainly from stainless steel and commonly by milling blanks. Peeso drills are available with cutting and noncutting tips but should be used with caution to avoid excessive preparation and thinning of radicular dentin walls.[8] Materials and clinical procedures for post space generation are described in Chapter 23.

ENGINE-DRIVEN INSTRUMENTS FOR CANAL PREPARATION

Instrument Types

Engine-driven instruments for root canal preparation made of stainless steel have been in use for more than half a century—the first decades mainly in handpieces that permitted reciprocation (alternating clockwise-counterclockwise motion). The major two problems with this type of instrument were canal transportation and file fracture. This changed with the advent of NiTi rotaries in the early 1990s; the much more flexible alloy allowed continuous rotation and reduced both canal preparation errors and instrument fracture compared with earlier engine-driven techniques.

Currently more than 100 types of rotary instrument systems are marketed, and more continue to be developed.[1,48] The instruments vary greatly in terms of design, alloy used, and recommended cutting movement (Table 8.1). Various built-in features may help prevent procedural errors, increase efficiency, and improve the quality of canal shaping. For example, a longer pilot tip may guide the instrument and help to stay centered in the canal long axis.

Another direction of instrument development is the prevention of instrument fractures (Fig. 8.25). There are several ways to modify an instrument to make it less likely to fracture—for example, increasing the core diameter will increase torsional resistance. Another approach is to use a torque-limiting motor (see later). Alternatively, a zero taper or nearly parallel and fluted working portion of the file may be provided for curved canals so that the apical portion of the canal can be enlarged without undue file stress and compression of debris. Changing continuous rotation to reciprocation motion when applied to NiTi rotary instruments has been successful in preventing threading-in and instrument fractures in general. Another direction of instrument development is to improve shaping as it relates to circumferential root canal wall contact, also described as conforming shaping. One example for this strategy is a file manufactured from an expandable and flexible hollow NiTi tube, the so-called Self-adjusting File (SAF; ReDent-Nova, Raanana, Israel) (Fig. 8.26).

More recently, a substantial S-shape was imposed onto a flexible NiTi rotary. This strategy provides an adaptable, larger envelope of motion while maintaining a limited

Table 8.1 Grouping of Instruments According to Their Mode of Cutting and Manufacturer Details

Group	Enlargement Potential	Preparation Errors	Fracture Resistance	Clinical Performance
I: ProFile,[1] ProSystem GT, GTX,[1] Quantec,[2] Pow-R,[3] Guidance,[4] K3[2] LightSpeed var.*,[2]	+, Depending in sizes, often time consuming	++, Low incidence, usually <150 μm canal transportation	±, Fatigue +, Torsional load, depending on system	++, Good, depending on treatment conditions. No difference between instruments shown so far, except for inexperienced clinicians, who perform better with landed instruments
II: ProTaper var.,[1] RaCe,[5] Hero 642,[6] FlexMaster,[7] Mtwo,*,[7] Sequence,**,[8] Alpha[9]... ProFile Vortex[1] Twisted File[1]	±, Good with use of hybrid techniques	±, Overall more demanding in clinicians' ability	+, Fatigue ±, Torsional load, depending on taper, handling	
III: EndoEZE AET[10], WaveOne,[1] Reciproc[7] OneShape[6] SAF, TRUShape,[1] TruNatomy,[1] XP-Shaper,[8] XP-Finisher[8]	Limited	Varies EndoEZE AET—WaveOne, Reciproc+	Varies +, With WaveOne, Reciproc	Varies

Group I consists of radial-landed instruments with reaming action. Group II instruments are such with triangular cross section and cutting action, while Group III is made up of atypical instruments with unusual geometry, movement, or sequence

Manufacturers:

[1]Dentsply TSirona, York, PA, USA/Dentsply Maillefer, Ballaigues Switzerland.
[2]Analytic Endodontics, Orange, CA, USA/SKavo Kerr, Orange, CA, USA.
[3]Moyco Union Broach, York, PA, USA.
[4]Guidance Endo, Albuquerque, NM, USA.
[5]FKG, La Chaux-De-Fonds, Switzerland.
[6]MicroMega, Besançon, France.
[7]VDW, Munich, Germany ** originally by Sweden & Martina, Padova, Italy.
[8]BrasselerUSA, Savannah, GA * instruments made by FKG.
[9]Brasseler, Lemgo, Germany.
[10]Ultradent, South Jordan, UT, USA.

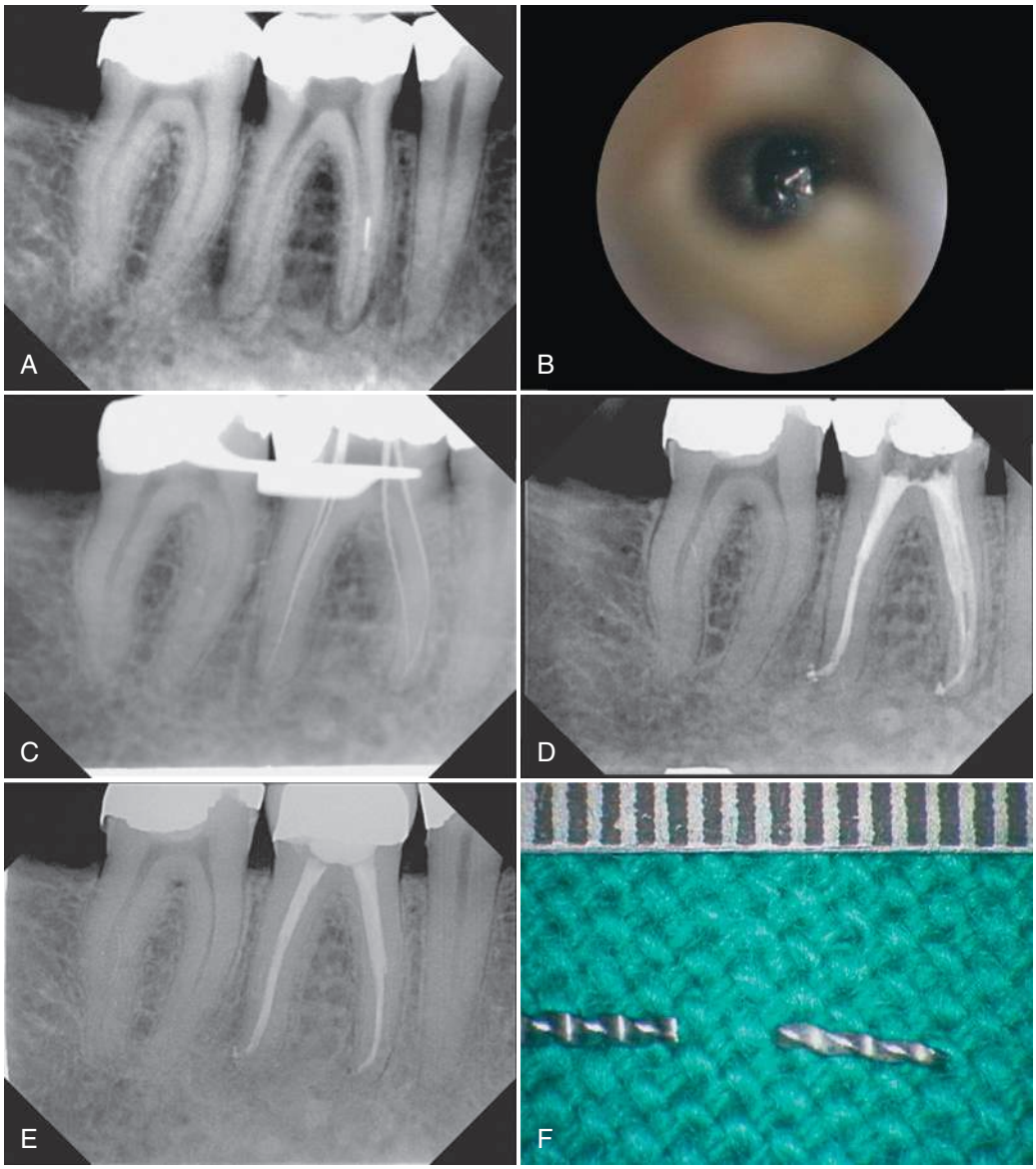


Fig. 8.25 Removal of a fractured NiTi instrument from a mesiolingual canal of a mandibular molar. **A,** Fragment located in the middle third of the root. **B,** Clinical aspect of the fragment after enlargement of the coronal third of the root canal with modified Gates-Glidden drills, visualized with an operating microscope (×25). **C,** Radiograph taken after removal of the fragment; four hand files have been inserted into the canals. **D,** Final radiograph shows slight widening of the coronal third of the mesiolingual canal and fully sealed canal systems. A full crown was placed immediately after obturation. **E,** Recall radiograph 5 years after obturation shows sound periradicular tissues. **F,** Removed fragment and separated file (gradation of ruler is 0.5 mm).

maximum flute diameter (TRUShape, Dentsply Sirona; XP Shaper, Brasseler, Savannah, GA, USA). Marketed files vary greatly in terms of their specific design characteristics, such as tip sizing, taper, cross section, helix angle, and pitch (see Fig. 8.13). Some of the early systems have been removed from the market or relegated to minor roles.

Most instruments described in the following section are manufactured by multiaxial grinding or micromilling processes, although some are produced by laser etching and yet others by plastic deformation under specific heating and cooling processes or electric discharge machining.

Many variables and physical properties influence the clinical performance of NiTi rotaries.[239,329,393,449] Clinical practice has generated much of what is known about NiTi instruments, including reasons for instrument fracture and the application of specific instrument sequences; NiTi rotary instruments have substantially reduced the incidence of relevant canal shaping errors.[89]

Table 8.1 and the following sections describe the instrument groups most widely used for root canal preparation at this point in time. Typical clinical strategies apply to all NiTi rotary instruments, regardless of the specific design or brand. However, three design groups will be analyzed separately next: group I, instruments designed for passive preparation; group II, rotary instruments designed for active cutting; and group III, unique designs that do not fit in either group I and II.

Group I: Passive Preparation; Presence of Radial Lands. The first commercially successful rotary instruments were ProFile (Dentsply Sirona), Lightspeed (marketed in its current form by Kavo Kerr, Orange, CA, USA),



Fig. 8.26 The SAF instrument. The instrument is made as a hollow, thin NiTi lattice cylinder that is compressed when inserted into the root canal and adapts to the canal's cross section. It is attached to a vibrating handpiece. Continuous irrigation is applied through a special hub on the side of its shank. (Courtesy ReDent-Nova, Raanana, Israel.) Insert shows the abrasive surface of the instrument.

and GT rotaries (Dentsply Sirona), and have in common a cross section with so-called radial lands (Fig. 8.27, red arrow). These are created by three round excavations, also known as the U-shape. The design of the instrument tip and also the lateral file surface (radial land) guide the file as it progresses apically. This makes rotaries listed in group I fairly safe regarding preparation errors. On the other hand, it results in a reaming action rather than cutting of dentin, and this makes them less efficient. Moreover, the smear layer produced with radial-landed rotaries is different in consistency and amount compared with the debris and smear created by cutting files.[318]

LIGHTSPEED. The LightSpeed file, developed by Drs. Steve Senia and William Wildey in the early 1990s and later known as *LS1*, was introduced as an instrument with a thin, noncutting shaft and short anterior cutting head. The same design principles apply to the subsequently marketed LSX instrument (Kavo Kerr) that is manufactured not by milling but by a stamping process. A full set consists of 25 LightSpeed LS1 instruments in sizes #20 to #100, including half sizes (e.g., 22.5, 27.5); LSX does not have half sizes, and a set includes sizes #20 to #80.

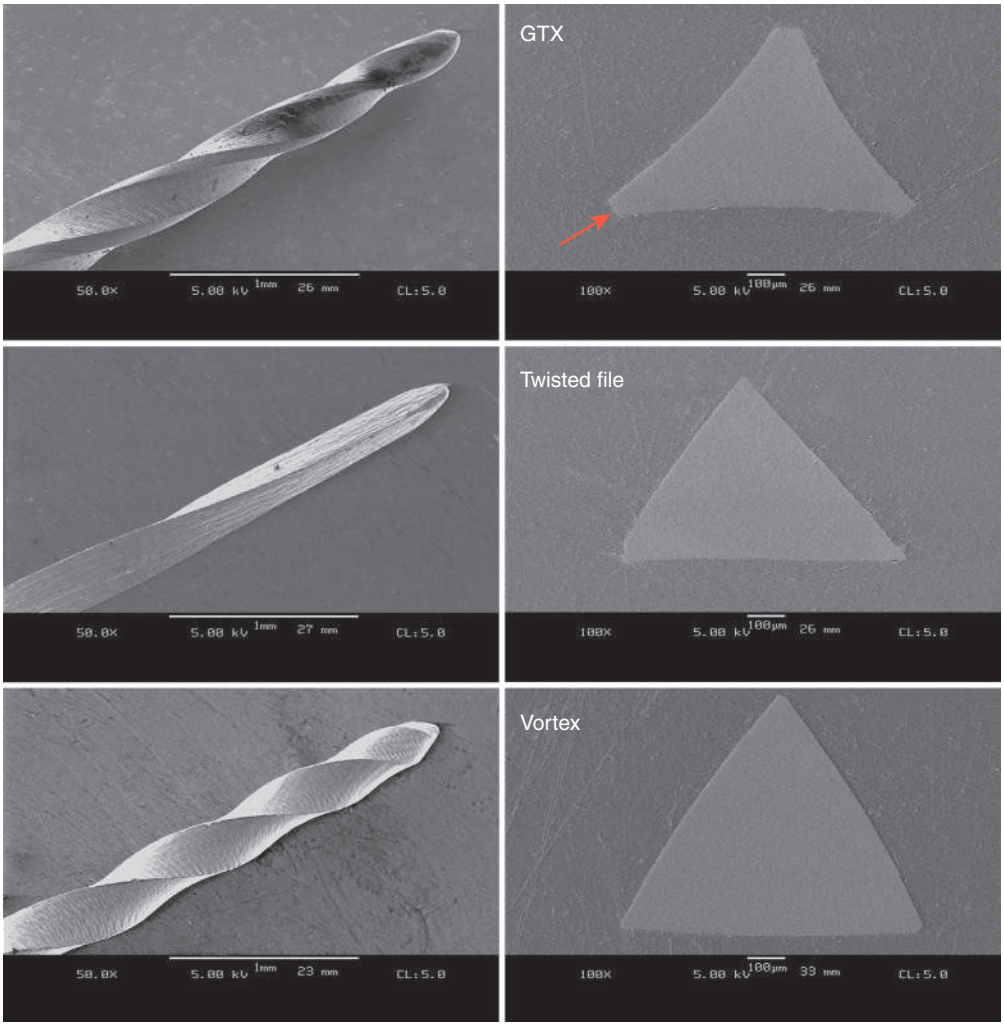


Fig. 8.27 Scanning electron micrographs (SEM) of current nickel-titanium rotary instruments, seen laterally (*left panel*, ×50) and in cross sections (*right panel*, ×100). Note the radial landed cross section of the *GTX* instrument (*red arrow*).

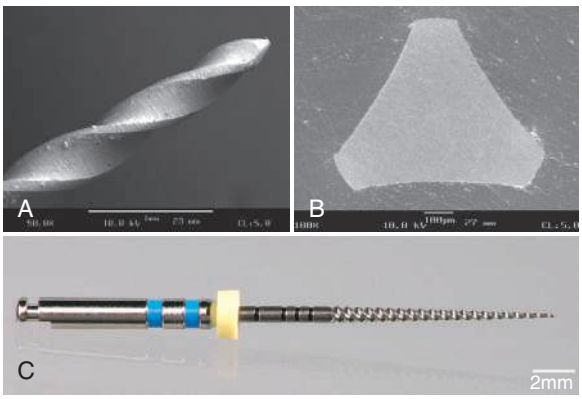
PROFILE. The ProFile system (Dentsply Sirona) was introduced by Dr. Ben Johnson in 1994. ProFile instruments have increased tapers compared with conventional hand instruments. The tips of the original ProFile Series 29 rotary instruments had a constant proportion of diameter increments (29%). Later, a ProFile series with ISO-sized tips (Dentsply Maillefer, Ballaigues, Switzerland) was developed and marketed in Europe.

Cross sections of a ProFile instrument show a U-shape design with radial lands (Fig. 8.28) and a parallel central core. Lateral views show a 20-degree helix angle, a constant pitch, and bullet-shaped noncutting tips (see Fig. 8.12). Together with a slightly negative rake angle, this configuration facilitates a reaming action on dentin rather than cutting. ProFile instruments shaped canals without major preparation errors in a number of in vitro investigations.[66,67,451,452]

GT AND GTX FILES. The Greater Taper, or GT file, was introduced by Dr. Steve Buchanan in 1994. This instrument incorporates a radial landed cross-sectional design and was marketed as ProFile GT (Dentsply Sirona). The instruments were available in four tapers (.06, .08, .10, and .12), and the maximum diameter of the working part was 1 mm. The design limited the maximum shank diameter and decreased the length of the cutting flutes with increasing taper. The instruments had a variable pitch and a growing number of flutes in progression to the tip; the apical instrument diameter was 0.2 mm.

K3. In a sequence of design developments by their inventor, Dr. McSpadden, the Quantec 2000 files were followed by the Quantec SC, the Quantec LX, and the current K3 system (all by Kavo Kerr Endo). K3, similar to ProFile instruments, includes instruments with .02, .04, and .06 tapers. The most obvious difference between the Quantec and K3 models is the K3's unique cross-sectional design: a slightly positive rake angle for greater cutting efficiency, wide radial lands, and a peripheral blade relief for reduced friction. Unlike the Quantec, a two-flute file, the K3 features a third radial land to help prevent threading-in. K3 features a round safety tip, but the file is about 4 mm shorter than other files (although it has the same length of cutting flutes) because of the so-called Axxess handle. The instruments are coded by ring color and number.

In the lateral aspect, the K3 has a variable pitch and variable core diameter, which provide apical strength. Manufacturing this intricate design can result in some metal flash.



D

No. of instruments/ set	Tip sizes	Size increments	r.p.m. (recommended)	Lengths
Orifice Shapers: 6	20–80	10; from 60: 20	150–350, low apical force, torque to fracture and working torque dependent on instrument size	19 mm
ProFile .06: 6	15–40	5		21 mm, 25 mm, some 31 mm
ProFile .04: 9	15–90	5; from #45: 15; from #60: 30		
ProFile .02: 6	15–45	5		
Profile Series 29	13–100	Varies, 29%		21 mm, 25 mm

Fig. 8.28 Design features of a ProFile instrument. **A,** Lateral view (scanning electron micrograph [SEM], ×50). **B,** Cross section (SEM, ×200). **C,** Lateral view. **D,** Design specifications.

Summary. Radial landed rotary instruments are considered very safe, even when accidentally taken beyond the confines of the root canal. Fracture resistance to torsional loading and cyclic varies depending on specific instrument design. The limited cutting efficacy of these files was perceived as a downside and is a reason that the market share has diminished.[50] However, their track record in clinic and research continue to support the use of rotaries listed in group I.

Group II: Active Cutting; Triangular Cross Section. Rotary instruments in group II all have a more active cutting flute design in common and have the largest market share. Radial lands are absent (see Fig. 8.13), and this fact results in a higher cutting efficacy. This translates to a higher potential for preparation errors, in particular when the instrument is taken through the apical foramen, thus eliminating the guide derived from the noncutting tip.

PROTAPER UNIVERSAL, GOLD. The ProTaper system originally comprised six instruments: three shaping files and three finishing files. This set is now complemented by two larger finishing files and a separate set of three retreatment rotaries. ProTaper instruments were designed by Drs. Cliff Ruddle, John West, and Pierre Machtou. In cross sections, ProTaper shows a convex triangle with sharp cutting edges and no radial lands. The cross section of finishing files F3, F4, and F5 is slightly relieved for increased flexibility. The three shaping files have tapers that increase coronally, and the reverse pattern is seen in the five finishing files.

Shaping files Nos. 1 and 2 have tip diameters of 0.185 mm and 0.2 mm, respectively, 14-mm-long cutting blades, and partially active tips. The finishing files (F1 to F5) have tip diameters of 0.2, 0.25, 0.3, 0.4, and 0.5 mm, respectively, between D_0 and D_3, and the apical tapers are .07, .08, .09, .05, and .04, respectively. The finishing files have rounded noncutting tips.

Two aspects of handling have been emphasized for ProTaper. The first is the preparation of a glide path, either manually[321] or with special rotary instruments.[47] An enlargement to a size approaching the subsequent rotaries' tips, at least larger than the file's core diameter, prevents breakage and allows assessment of the canal size.[321] This means that a glide path should correspond to a size #15 or 20. Glide path preparation is currently recommended for most NiTi instruments.

The second specific recommendation for ProTaper use is the use of a more laterally directed "brushing" working stroke. Such a stroke allows the clinician to coronally direct larger files away from danger zones and counteract any "threading-in" effect.[53] Brushing strokes should be applied judiciously and away from the furcation in molars to avoid thinning of radicular structure.

In a study using plastic blocks, ProTaper created acceptable shapes more quickly than GT rotary, ProFile, and Quantec instruments[502] but also created somewhat more aberrations. A study using micro-CT showed that the original ProTaper file created consistent shapes in constricted canals, without obvious preparation errors, although wide canals may be insufficiently prepared with this system.[338]

A newer version, called ProTaper Next, was introduced in 2013. Current research suggests that mechanical properties of these instruments, manufactured from M-Wire, are better than ProTaper Universal.[23,121,322]

HERO 642, HERO SHAPER. Several systems in group II (see Table 8.1) were designed with positive rake angles, which provide greater cutting efficiency. HERO instruments (MicroMega, Besançon, France) are an example, the original version was known as *HERO 642* (the name *HERO* is an acronym for *h*igh *e*lasticity in *ro*tation); the name has now changed to HERO Shaper, with few apparent differences in the instrument design.

Cross sections of HERO instruments show geometries similar to those of an H-file without radial lands. Tapers of .02, .04, and .06 are available in sizes ranging from #20 to #45. The instruments are relatively flexible but maintain an even distribution of force into the cutting areas.[460,461] HERO instruments have a progressive flute pitch and a noncutting passive tip, similar to other NiTi rotary systems. The instruments are coded by handle color.

RACE, BIORACE, BT RACE. The RaCe file has been manufactured since 1999 by FKG and was later distributed in the United States by Brasseler. The name, which stands for r*eamer with* a*lternating cutting e*dges, describes just one design feature of this instrument. Light microscopic imaging shows flutes and reverse flutes alternating with straight areas; this design is aimed at reducing the tendency of files to thread into a root canal. Cross sections are triangular or square for .02 instruments with size #15 and #20 tips. The lengths of cutting parts vary from 9 to 16 mm.

The surface quality of RaCe instruments has been modified by electropolishing, and the two largest files (size #35, .08 taper and size #40, .10 taper) are also available in stainless steel. The tips are round and noncutting, and the instrument handles are color-coded and marked by milled rings. RaCe instruments have been marketed in various packages to address small and large canals; recently they are sold as BioRaCe, purportedly to allow larger preparation sizes, with an emphasis on the use of .02 tapered instruments. A new variant, BT RaCe was recently introduced and incorporates different tip designs, as well as different sequences.

Few results of in vitro experiments comparing RaCe to other contemporary rotary systems are available;[384,385] canals in plastic blocks and in extracted teeth were prepared by the RaCe system with less transportation than with ProTaper files.[384] In another study, ProTaper and RaCe performed similarly when canals were prepared to an apical size #30.[313] As with any rotary system, this file may be included in a hybrid hand-rotary technique. BioRace instruments prepared S-shaped canals in plastic blocks (to size #40) similarly to ProTaper and MTwo but were superior when combined with S-Apex.[56]

ENDOSEQUENCE. The EndoSequence rotary instrument is produced by FKG in Switzerland and marketed in the United States by Brasseler. This instrument adheres to the conventional length of the cutting flutes, 16 mm, and to larger tapers, .04 and .06, to be used in a crown-down approach. The overall design, including the available tapers and cross sections, is thus similar to many other files; however, the manufacturer claims that a unique longitudinal design called *alternating wall contact points* (ACP) reduces torque requirements and keeps the file centered in the canal. It also has varying comparatively small helical angles. Another feature of the Sequence design is an electrochemical treatment after manufacturing, similar to that of RaCe files, resulting in a smooth, polished surface. This is believed to promote better fatigue resistance; hence a rotational speed of 600 rpm is recommended for EndoSequence.[231] Most in vitro results, however, suggest that EndoSequence is not superior to other files in cyclic fatigue resistance.[195,243,353]

TWISTED FILE. In 2008, SybronEndo, now Kavo Kerr Endo, presented the first fluted NiTi file manufactured by plastic deformation, a process similar to the twisting process that is used to produce stainless steel K-files: the Twisted File (TF). According to the manufacturer, a thermal process allows twisting during a phase transformation into the so-called R-phase of nickel-titanium. The instrument is currently available with size tip from 25 to 50 and in tapers from .04 to .12.

The unique production process is believed to result in superior physical properties; indeed, early studies suggested significantly better fatigue resistance when size #25 .06 taper TFs were compared with K3 instruments of the same size GTX instrument.[144] Moreover, as determined by bending tests according to the ISO norm for hand instruments, 3630-1, TFs size #25 .06 taper were more flexible than ProFiles of the same size.[143] Others found similar levels for fatigue resistance for TF and Profile of similar size.[243] A more recent development for TF is the use of an electric motor that allows different file movement, both continuous rotation and reciprocation, depending on the clinical situation (TF Adaptive, Kavo Kerr).

PROFILE VORTEX. Profile Vortex instruments are manufactured from NiTi. Two versions are on the market—one made from M-Wire and another from so-called blue wire (Vortex Blue, which showed greater resistance to cyclic fatigue and increased torque resistance) and have varying helical angles to counteract the tendency of nonlanded files to thread into the root canal. Vortex instruments are recommended to be used at 500 rpm; higher rotational speed results in less torque generated.[35] Canal preparation with ProFile Vortex in vitro was similar to other rotary instruments.[75,493] Vortex instruments are available in sizes 15 to 50 and in .04 and .06 tapers.

MTWO. This instrument entered the European market in 2004. It possesses a two-fluted S-shaped cross section. The original strategy allowed for three distinct shaping approaches after the use of a basic sequence with tip sizes

from #10 to #25, and tapers ranging from .04 to .06. Subsequent enlargement was meant to either create apical sizes up to #40 .04, alternatively to #25 .07 or to larger apical sizes with so-called apical files. MTwo is a well-researched and cutting-efficient instrument; clinically it is an example for the so-called single length technique. Canal shapes with MTwo were similar to other contemporary root canal instruments, either in rotation or reciprocation.[72]

EDGE FILES. A selection of instruments in designs that are similar to other designs, made primarily from heat-treated NiTi alloys, are available from Edge Endo (Albuquerque, NM, USA). There is comparatively limited research available for the individual files but extended fatigue life has been reported for them, at least at room temperature.[22]

SUMMARY. The market share for rotary files without radial lands continues to expand because of perceived higher efficacy. The overall incidence of clinically relevant preparation errors (see later for details) appears to be low, in spite of more aggressive cutting by files without radial lands. Instrument fractures remain a concern as does the tendency of continuously rotating instruments to thread or pull into the canal, specifically as working length (WL) is approached.

Group III, Atypical Instruments

WAVEONE, RECIPROC, GOLD, BLUE. A way to mitigate problems with continuous rotation (e.g., taper lock, fatigue fracture, threading-in) is to replace continuous rotation by reciprocation action. A case report[496] described this approach using ProTaper instruments. Based on experiments evaluating the maximal rotational angle before plastic deformation for the selected instrument, a forward angle of 144 degrees, followed by a reverse rotation of 72 degrees, was recommended.[496]

Subsequently, two instruments specifically designed for reciprocation were brought to the market (WaveOne, Dentsply Sirona) and Reciproc (VDW, Munich, Germany, currently not available in the United States). WaveOne instruments are available in three tip sizes, #21, #25, and #40, with tapers of .06 and .08, respectively. Corresponding sizes for Reciproc are tips of #25, #40, and #50, with tapers of .08, .06, and .05, respectively. Both instruments feature variable tapers that are largest toward the tips. The main WaveOne cross section is triangular, similar to ProTaper, while Reciproc is a two-fluted file with a design similar to MTwo (VDW).

Special motors are used for both systems to provide reciprocation action with alternating counterclockwise and clockwise rotations of about 150 to 170 and 30 to 50 degrees, respectively.[134]

Both files are machined with left-leaning flutes, and therefore the cutting direction for both is clockwise. One problem that may occur with this design is the transportation of dentin debris into the apical area, rather than moving debris coronally. There is mixed evidence for this phenomenon[73,110] in vitro. Clinically, frequent careful cleaning of the cutting blades with a moist gauze is recommended.

Shaping ability of these systems seems, according to current in vitro data, similar to established systems with continuous rotation.[72,278,469]

SELF-ADJUSTING FILE. The SAF (ReDent-Nova) represented a completely different approach, both in file design and mode of operation.[280] It is a cylindrical, hollow design with a thin-walled NiTi lattice displaying a lightly abrasive surface (see Fig. 8.26). An initial glide path to a #20 K-file allows insertion of the SAF file. The file is operated with a handpiece that generates in-and-out movements (4000 per minute) and 0.4 mm amplitudes.

In vitro data for this system suggest that indeed more wall contact is made compared with rotary files,[336] resulting in better débridement and superior antimicrobial efficacy.[415] Shaping quality is also on par with rotary instruments.[336]

TRUSHAPE. Two strategies have recently emerged in file design, which are both present in the TRUShape (Dentsply Sirona) instrument: so-called conforming shaping and a limited maximum flute diameter. The strong S-shaped longitudinal design allows TRUShape to expand and contract during rotation, similar to what had been described for the SAF, in an attempt to increase wall contact. TRUShape is marketed in sizes #20, #25, #30, and #40, all with a maximum fluted diameter of 0.8 mm, thus limiting coronal dentin removal. Used in continuous rotation at 300 rpm, TRUShape has been shown to create adequate shapes in vitro during initial canal preparation in molars[328] and premolars.[216] Specifically in root canal retreatment, the ability of TRUShape files to create more canal wall contact has attracted attention.

Further design developments led to the TruNatomy system (Dentsply Sirona), which includes a dedicated Orifice Modifier, Glide Path file, and three shaping instruments, as well as a special irrigation needle and conforming gutta percha. TruNatomy is operated at 500 rpm and 1.5 Ncm and with gentle stroking motion.

XP-SHAPER, XP-FINISHER. A similar design is incorporated in the XP instruments. Both have strong curves in the longitudinal aspect and a small taper, so that the maximum fluted diameter is limited. The files are made of a NiTi solution that changes its conformation in the transition from room to body temperature, which in turn leads to some stiffening and specific preparation outcomes. The finisher is thought of as an irrigation enhancement device and is run at about 800 rpm. Initial reports suggest that using the XP-Shaper resulted in acceptable canal preparation[28] and that the use of the finisher had débridement potential that is similar or better than with other shaping and cleaning strategies.[122]

SONIC AND ULTRASONIC SHAPING INSTRUMENTS. An alternative way of instrumenting root canals was introduced when clinicians became able to activate files by electromagnetic ultrasonic energy. Piezoelectric ultrasonic units are also available for this purpose (Fig. 8.29). These units activate an oscillating sinusoidal wave in the file with a frequency of about 30 kHz.

Two types of units, ultrasonic and sonic, are marketed. Ultrasonic devices, which operate at 25 to 30 kHz, include the magnetostrictive Cavi-Endo (Dentsply Caulk, Milford, DE, USA), the piezoelectric ENAC (Osada, Tokyo), the EMS Piezon Master 400 (Electro Medical Systems, Vallée de Joux, Switzerland), and the P5 Neutron (Satelec, Merignac Cedex, France).

Sonic devices, which operate at 2 to 3 kHz, include the Sonic AirMM 1500 (Micro Mega, Prodonta, Geneva, Switzerland), the Megasonic 1400 (Megasonic Corp, House Springs, MO, USA), and the Endostar (Syntex Dental Products, Valley Forge, PA, USA).



Fig. 8.29 Example of an ultrasonic unit (Neutron P5 piezoelectric). (Courtesy Satelec, Merignac Cedex, France.)

The piezoelectric transducer transfers more energy to the file than does the magnetostrictive system, making it more cutting efficient. The file in an ultrasonic device vibrates in a sinus wavelike fashion. A standing wave has areas with maximal displacement (i.e., antinodes) and areas with no displacement (i.e., nodes). The tip of the instrument exhibits an antinode. If powered too high, especially with no contact with the canal wall, the instrument may break because of the intense vibration. Therefore, files must be used only for a short time, must remain passive within the canal, and the power must be controlled carefully. The frequency of breakage in files used for longer than 10 minutes may be as high as 10%, and the breakage normally occurs at the nodes of vibrations.[9] Ultrasonic devices have been linked to a higher incidence of preparation errors and to reduced radicular wall thickness.[260,277,512]

SUMMARY. An abundance of NiTi systems is currently on the market, Table 8.1 lists these instruments systematically and illustrates some of the most relevant properties. Most systems included in Table 8.1 have files with tapers greater than the .02 stipulated by the ISO norm, while some have variable tapers and yet others have canal-conforming shapes and limited maximum flute diameters. Differences exist in tip designs, cross sections, and manufacturing processes. In vitro tests continue to identify the effect of specific designs on shaping capabilities and fracture resistance. An attractive goal is the instrumentation of more canal surface, in an effort to make biofilm more susceptible to subsequent chemical disinfection. At the same time, instrumentation paradigms that conserve more dentin appear desirable to promote long-term function.[154] Meaningful differences in clinical outcomes in regard to various specific design variations have yet to be ascertained.

Motors

Motors for rotary instruments have become more sophisticated since the simple electric motors of the first generation in the early 1990s (Fig. 8.30, *A*). Electric motors with gear reduction are most suitable for rotary NiTi systems, because they ensure a constant rpm level and constant torque. They can also be programmed to provide alternative rotation patterns—for example, reciprocation with freely selectable angles of rotation.[134] Electric motors often have presets for rpm and torque and are capable of delivering torques much higher than those required to break tips. Some authors believe that torque-controlled motors (see Fig. 8.30, *B–D*) increase operational safety.[142] Others have suggested that such motors may mainly benefit inexperienced



Fig. 8.30 Examples of motors used with rotary nickel-titanium endodontic instruments. **A,** First-generation motor without torque control. **B,** Fully electronically controlled second-generation motor with sensitive torque limiter. **C,** Frequently used simple torque-controlled motor. **D,** Newer-generation motor with built-in apex locator and torque control.

clinicians.[499] These motors probably do not reduce the risk of fracture caused by cyclic fatigue, and even if the torque is below the fracture load at D_3, a fracture at a smaller diameter (D_2) is still possible.

To complicate matters further, a differential exists between torque at failure at D_3 and the working torque needed to operate an instrument effectively (Fig. 8.31 and Box 8.2).[52,329,376] In many cases, the working torque is greater than the torque required to fracture the instrument's tip.

This differential between working torque and tip fracture load is especially large with files with a taper equal to or greater than .06; therefore, these files are rather ineffective in most torque-controlled motors. Some contemporary motors offer microprocessor-controlled features, for example, information about the specific instrument being used, such as the presets and the usage history, as well additional connectivity for example to an apex locator.

Some motors have built-in apex locators (see Fig. 8.30, *D*). This type of motor may be programmed to stop a rotary instrument when WL is reached or go into reverse gear. Similar changes in movement can occur in an adaptive fashion, depending on the torsional load that the instrument experiences in the root canal. The TF Adaptive motor (Kerr Endo) is an example for this technology that appeared only recently.

Some of the factors besides the motor itself that may influence the incidence of fracture in engine-driven NiTi rotary instruments are lubrication, specific instrument motion, and speed of rotation. It cannot be overemphasized that NiTi rotary instruments should only be used in canals that have been flooded with irrigants. Although lubricants such as RC-Prep (Premier, Norristown, PA) and Glyde (Dentsply Maillefer) have been recommended in the past, their benefit appears to be restricted to plastic blocks,[18] and could not be demonstrated when rotary instruments engage dentin surfaces. To the contrary, experimental data suggests that the use of gel-type lubricants fails to reduce torque values during simulated canal preparation.[55,330] Finally, because of chemical interactions between NaOCl and ethylenediamine tetra-acetic acid (EDTA),[164] alternating irrigants and using lubricants that contain EDTA may even be counterproductive. Details regarding irrigation solution interactions are described later in this chapter.

For instrument motion, some manufacturers recommend a pecking, up-and-down motion. This not only prevents threading in of the file; it is also believed to distribute stresses away from the instrument's point of maximum flexure where fatigue failure would likely occur.[250,348] However, such in-and-out movements did not significantly enhance the lifespan of ProFile size .04 or GT rotary instruments rotated around a 5-mm radius cylinder with a 90-degree curve.[329,333] Furthermore, large variations were noted in the lengths of the fractured segments.[203,462] This suggests that ductile fractures may originate at points of surface imperfections. Files made of more martensitic alloy tend to work better with longer sweeping or brushing strokes. The notion of "brushing" is not directly related to a paintbrush motion, since this would predispose a file to bending and subsequent fatigue.[346] Rather it should be considered as stroking against the wall away from the danger zone—the inner curvature of a curved root. Specifically, highly martensitic files may benefit from a low-impact gentle brushing with low torque setting.

Box 8.2 Instrument Breakage With Torsional Load (MacSpadden Factor)

For rotary instrument tips, susceptibility to breakage is governed by the quotient of torque needed to fracture divided by working torque. Simply put, the larger the value, the safer the file.

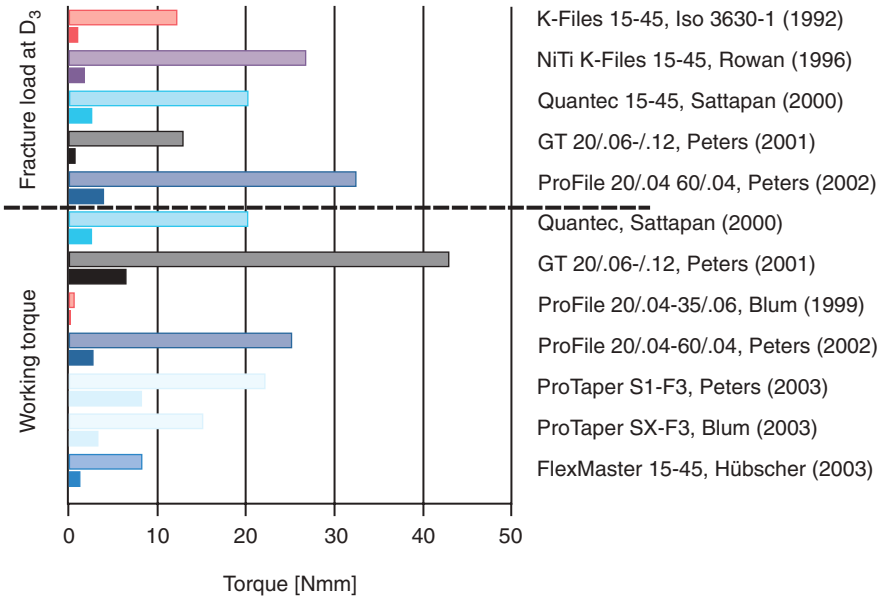


Fig. 8.31 Diagram comparing fracture loads at D_3 *(upper section of graph)* to torques occurring during preparation of root canals *(lower section of graph)*. Filled columns represent the largest file in each set, and open columns show the scores of the most fragile file (see text and Box 8.3 for details).

Rotational speed may influence instrument deformation and fracture. Some studies indicated that ProFile instruments with ISO-norm tip diameters failed more often at higher rotational speed,[117] whereas other studies did not find speed to be a factor.[222] For Vortex, 300 rpm and 400 rpm appeared to reduce the generated torque and force; this higher rpm preset did not result in an increase of instrument fractures in vitro.[35] As mentioned before, there is a trend to using higher rotational speeds and comparatively low torques—for example, for XP-Shaper and BioRace (Brasseler) and for TruNatomy (Dentsply Sirona) instruments.[28]

Clinicians must fully understand the factors that control the forces exerted on continuously rotating NiTi instruments (Box 8.3). To minimize fractures and prevent taper lock, motor-driven rotary instruments should not be forced in an apical direction. Similarly, acute apical curves limit the use of instruments with higher tapers because of the risk of cyclic fatigue. The incidence of instrument fracture can be reduced to an absolute minimum if clinicians use data from well-designed torque and stress studies. Adequate procedural strategies, such as an adequate glide path, a detailed knowledge of anatomic structures, with avoidance of extreme canal configurations, and specific instrumentation sequences may also improve shaping results.

Certain procedures have evolved for removing fractured instruments from root canals (see Fig. 8.25); these are discussed in detail elsewhere in this book (see Chapter 19). Most of those methods require the use of additional equipment, such as a dental operating microscope and ultrasonic units. However, the best way to deal with instrument fracture is prevention. An understanding of the anatomy of the root canal system, together with a clear plan for selecting, sequencing, and using shaping instruments, can certainly help prevent procedural mishaps.

Box 8.3 Factors Governing the Potential for Nickel-Titanium Rotary Instrument Fractures

- Clinician's handling (most important)
- Combination of torsional load, bending, and axial fatigue
- Root canal anatomy
- Manufacturing process and quality

Steps of Cleaning and Shaping

ACCESS—PRINCIPLES

Access cavity preparation is essential for root canal treatment and is described in detail elsewhere in this book (see Chapter 7). However, it should be emphasized at this point that mishaps during access (e.g., perforation) significantly affect the long-term outcome of a root canal–treated tooth. Overenlargement or gouging during access significantly reduces structural strength[354] and may lead to root fracture and nonrestorable conditions.

Rationally, the use of a cylindrical diamond bur, followed by a tapered fluted bur with noncutting tip and perhaps a small round bur is recommended (Fig. 8.32). A recent experiment suggested that more conservative access cavities, in particular in premolars, may increase fracture resistance and permit similar shaping outcomes.[238] The use of ultrasonically powered tips under magnification is aimed at achieving ideal access cavities (Fig. 8.33), including discovery and perhaps relocation of the canal orifices.

CORONAL MODIFICATION

The extension of an access cavity into the coronal-most portion of the root canal has been called "coronal flaring" but

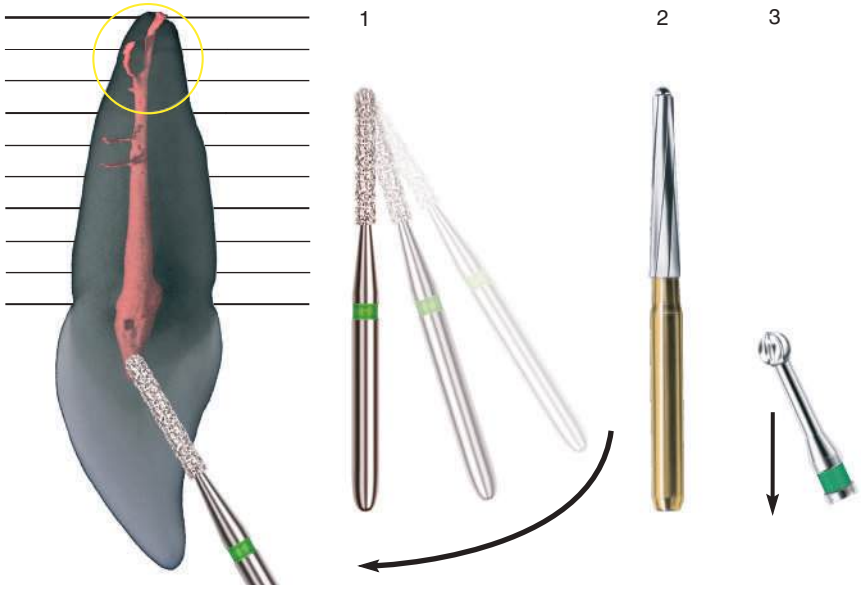


Fig. 8.32 Sequence of instruments used for optimal preparation of an access cavity (e.g., in an incisor). A parallel-sided diamond or steel bur is used to remove overlying enamel in a 90-degree angle toward the enamel surface *(1)*. The bur is then tilted vertically to allow straight-line access to the root canal *(arrow)*. A bur with a noncutting tip (e.g., Endo-Z bur or ball-tipped diamond bur) is then used to refine access *(2)*. Overhangs or pulp horns filled with soft tissue are finally cleared with a round bur used in a brushing or pulling motion *(3)*.

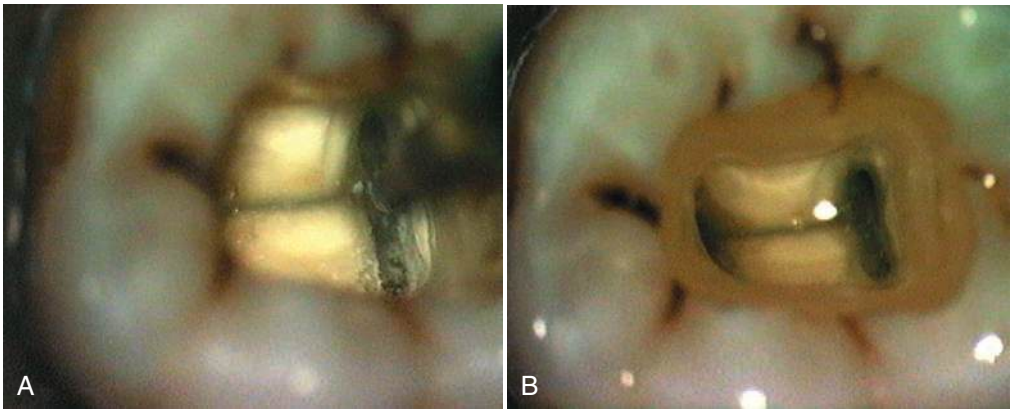


Fig. 8.33 Clinical views of an access cavity in a mandibular molar as seen through an operating microscope (×20). **A,** Modification with an ultrasonically activated tip (**B**).

may be more appropriately described as orifice modification. If a canal is constricted, mineralized, or difficult to access, directed enlargement of the coronal portion prior to any deep entry into the root canal is beneficial. This canal modification should be preceded by a scouting step, in which a small (e.g., size #10) K-file is passively placed several millimeters into the root canal. Tools for preflaring include Gates-Gliddens and dedicated NiTi instruments (Fig. 8.34). Specific orifice shaping rotaries have been developed that are used as a single instrument, rather than in a sequence.

The use of laterally cutting NiTi rotaries allows clinicians to modify the root canal orifice to form a receptacle for subsequent instrumentation. This is related to removal of coronal mineralization and to relocation of the canal pathway by removal of restrictive dentin. For example, a second mesiobuccal canal of a maxillary molar typically exits the pulp chamber toward the mesial under a dentin shelf and then curves towards the distal. It is this curve in a small canal that often leads to formation of a ledge, which in turn prevents shaping to length.

Preenlargement of the coronal portion of a root canal allows files unimpeded access to the apical one third and gives the clinician better tactile control in directing small, adequately precurved negotiating files (Fig. 8.35).

One of the purported benefits of orifice modification is early access of disinfecting irrigation solutions and prevention of pushing contaminated coronal debris apically. This has not been confirmed in experiments. On the other hand, a documented benefit of coronal flaring is mitigation of WL changes during canal preparation.[435]

Comparatively little is known about the efficacy of individual flaring instruments. A recent study compared the

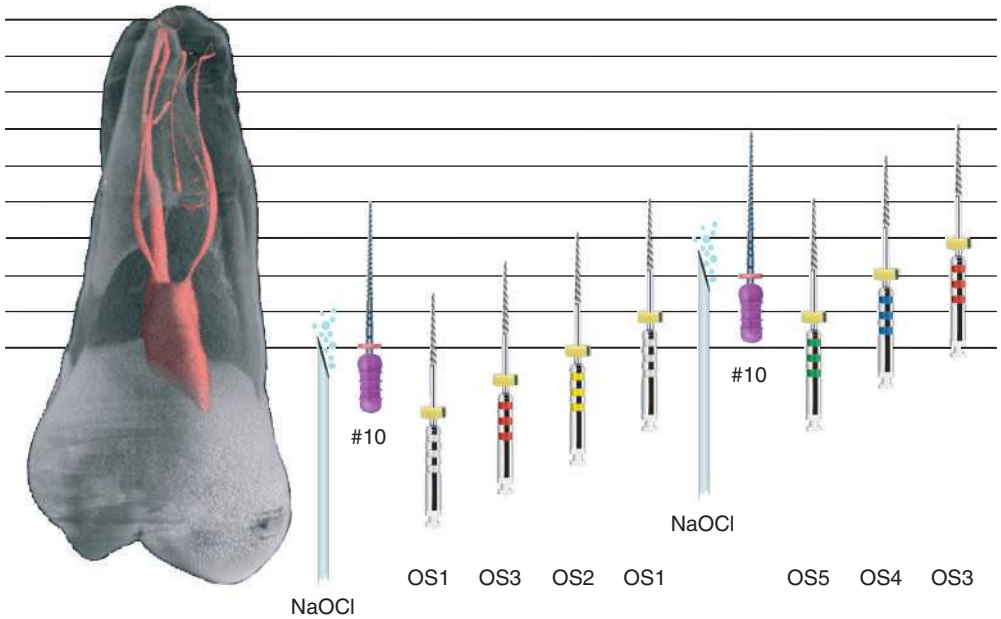


Fig. 8.34 Diagram of coronal enlargement in a more complicated maxillary posterior tooth. This maxillary molar presents several difficulties, including a narrow mesiobuccal canal that exits the pulp cavity at an angle. A possible approach in a case involving difficult entry into the root canal system is to use a small orifice shaper *(OS1)* after ensuring a coronal glide path with a K-file. Use of a sequence of orifice shapers *(OS3 to OS1)* then allows penetration into the middle third of the root canal. Wider canals can accept a second sequence of orifice shapers. Copious irrigation and securing a glide path with a size #10 K-file are prerequisites for use of NiTi rotary instruments.

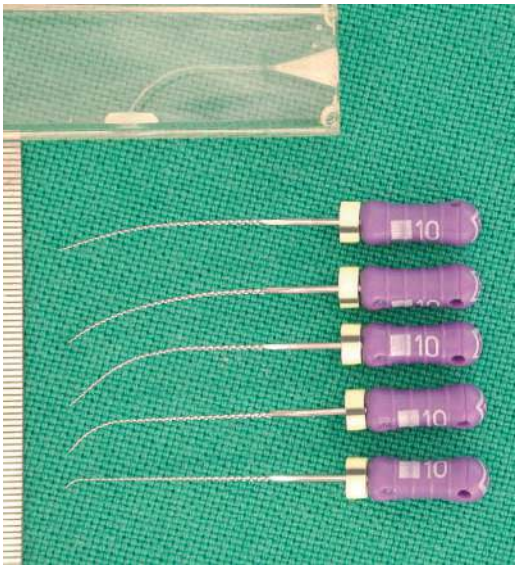


Fig. 8.35 Various precurved, stainless steel hand files for path finding and gauging. Compare the curves in the instruments to the ones in a plastic training block (gradation of ruler is 0.5 mm).

cutting efficacy of radial-landed and triangular cross sections.[292] The latter were shown to be more cutting efficient compared with radial-landed design; somewhat surprisingly, flexible martensitic alloy was cutting more rapidly compared with conventional NiTi alloys.

PATENCY AND GLIDE PATH PREPARATION

A *patency file* is a small K-file (usually a size #10 or #15) that is passively extended just through the apical foramen. Demonstration of apical patency has been suggested as a procedural for most rotary techniques. It is believed to remove accumulated debris, to help maintain WL, and thereby to translate into greater clinical success.[301] One concern with the patency file was that instead of having a cleaning effect, the file would drive contaminated debris through the foramen. However, an in vitro study suggested that the risk of inoculation was minimal when canals were filled with NaOCl.[214] Maintaining patency throughout an endodontic procedure does not lead to an increase in posttreatment symptoms.[21] Only initial clinical evidence exists either favoring the use of a patency file; however, experience suggests that this technique involves relatively little risk and provides some benefit as long as small files are used carefully.

The file that has reached the apical foramen has already formed a so-called glide path. This term has been used in endodontics for more than a decade[337] and relates to securing an open pathway to the canal terminus that subsequent engine-driven instruments can follow. The presence of a glide path is essential for the predictable use of rotary instruments since it minimizes the so-called screw-in or threading-in effect[173] and risk of torsional failure. It has been considered that taper lock may occur if the canal cross section is smaller than the tip of the instrument;[337] in that regard, a glide path reduces the contact area between subsequent shaping instruments and the canal wall, consequently the torsional loads induced by shaping rotary instruments are lower.[52]

It is advocated to explore and shape a root canal at least to a #15 size instrument before the use of a rotary NiTi instrument to full WL[442] to create a glide path for the safe advancement of the rotary shaping instrument tip.[321] The presence of an appropriate glide path is indicated by the fact that a straight size #15 K-file can passively and smoothly travel to WL with long in-and-out movements.

Classically, a glide path was created and secured with K-file sizes #10 and #15 used in watch-winding or balanced-force motion. However, recently NiTi rotary instruments with small tip diameter have been designed specifically to simplify the process after a size #10 K-file has reached WL.

Various small engine-driven instruments were introduced that are aimed at simplifying the glide path process—for example, Pathfiles (Dentsply Sirona),[47] Scout RaCe (FKG),[298] and G-file (MicroMega).[101] Copious irrigation and frequent recapitulations with a smaller file to WL may be required, and in some instances of highly mineralized canals, clinicians must devise strategies using small crown-down and/or step-back sequences.

The creation of a rotary glide path has demonstrated better preservation of the canal anatomy with fewer shaping aberrations[11,47] and has shown a less incidence of postoperative pain.[320] However, some studies have also reported acceptable performance of manual instruments for glide path preparation.[101,303]

WORKING LENGTH DETERMINATION

Devices

Radiographs, tactile sensation, the presence of moisture on paper points, and knowledge of root morphology have been used to determine the length of root canal systems. Following earlier experiments by Custer,[99] subsequently Sunada[439] developed the first commercialized electronic apex locator, suggesting that the apical foramen could be localized using a direct electric current. Currently the electronic apex locator is considered an accurate tool for determining WL.[135] One study reported that the use of electronic apex locators in a dental student clinic resulted in a higher quality of obturation length control and an overall reduction in the number of radiographs taken.[135] However, these devices must not be considered flawless, because several variables are known to affect their accuracy. For example, immature roots can present problems.[208] Once roots mature (i.e., form a narrow apical foramen) and instruments are able to contact the canal walls, an electronic apex locator's accuracy greatly improves. Some investigators have found no statistical difference between roots with vital and necrotic tissue.[157,275] Because apical root resorption is prevalent in necrotic cases with long-standing apical lesions,[473] it may be concluded that apical resorption does not have a significant effect on the accuracy of electronic apex locators.

Recently some clinicians have advocated the use of the electronically determined WL in lieu of WL estimations using the placement of a file in the canal and a radiograph. However, combined use of both radiographic and electronic techniques has been shown to result in greater accuracy.[120] Furthermore, radiographs may add essential anatomic

information that could be missed if electronic apex locators are used exclusively.

The first two generations of electronic apex locators were sensitive to the contents of the canal and irrigants used during treatment. The development of an algorithm called the *ratio measurement method* distinguished the third generation of apex locators.[230] To arrive at this method, the impedance of the canal was measured with two current sources of different frequencies, and a quotient was determined using the electrical potentials proportional to each impedance.[230] This study found that electrolytes did not have a significant effect on the accuracy of the unit. Some third-generation apex locators are the Endex Plus, or Apit (Osada, Los Angeles, CA, USA); the Root ZX (J. Morita, Kyoto, Japan);[135,242] and the Neosono Ultima EZ (Satelec, Mount Laurel, NJ, USA). The Endex Plus device uses 1 and 5 kHz and provides apex location based on subtraction. The Root ZX emits currents at frequencies of 8 and 0.4 kHz and provides apex location based on the resulting quotient. In addition to latest model apex locators, manufacturers have developed smaller models (Fig. 8.36, *B*).

Apex locators are generally safe to use; however, manufacturers' instructions state that they should not be used on patients with pacemakers without consulting the patient's cardiologist. However, when connected directly to cardiac pacemakers in vitro, electronic apex locators did not interfere with the function of the pacemaker[145] and did not interfere with the functioning of any of the cardiac devices tested in a clinical study under ECG monitoring.[487]

Strategies

Anatomic studies and clinical experience suggest that typically teeth are 19 to 25 mm long. Most clinical crowns are approximately 10 mm long, and most roots range from 9 to 15 mm in length. Roots, therefore, can be divided into thirds that are 3 to 5 mm long. An important issue in root canal treatment is the apical endpoint of the prepared shape in relation to the apical anatomy. Traditional treatment has held that canal preparation and subsequent obturation should terminate at the *apical constriction*, the narrowest diameter of the canal (Fig. 8.37). This point is believed to coincide with the cementodentinal junction (CDJ; see Chapter 9) and is based on histologic sections and ground specimens. However, the position and anatomy of the CDJ varies considerably from tooth to tooth, from root to root, and from wall to wall in each canal. Moreover, the CDJ cannot be located precisely on radiographs. For this reason, some have advocated terminating the preparation in necrotic cases at 0.5 to 1 mm short of the radiographic apex and 1 to 2 mm short[187,357,492] in cases involving irreversible pulpitis. Although there is no definitive validation for this strategy at present,[379] well-controlled follow-up studies seem to support it.[297,419]

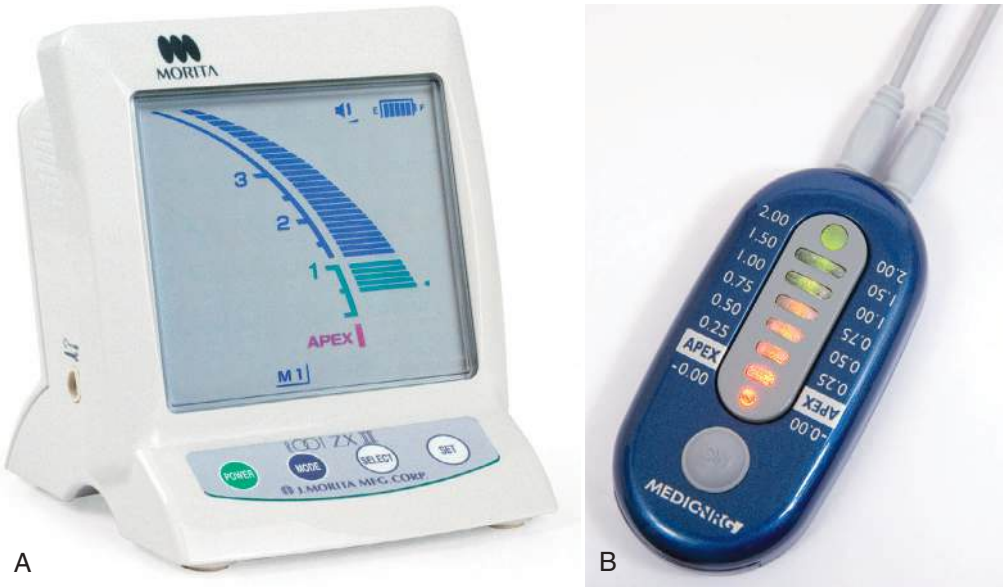
Working to shorter lengths may lead to the accumulation and retention of debris, which in turn may result in apical blockage (Fig. 8.38). If the path to the apex is blocked, working to short lengths may contribute to procedural errors such as apical perforations and fractured instruments. Such obstacles (which consist of collagen fibers, dentin mud, and most importantly, residual microbes) in apical canal areas are a major cause of persistent or recurrent apical periodontitis,[178,296,409] or posttreatment disease (also see Chapters 15 and 16).[138,490]

Using an electronic apex locator has helped clinicians identify the position of apical foramina more accurately and allow safe canal shaping as close as 0.5 mm to the canal terminus.

CANAL ENLARGEMENT/PREPARATION

Rationale

Like the position of the apical constriction, apical diameters are difficult to ascertain clinically.[240] Some clinicians have recommended gauging canal diameters by passing a series of fine files apically until one fits snugly. However, such an



A B

Fig. 8.36 Root ZX apex locator with lip clip and file holder. (**A,** Courtesy J. Morita USA, Irvine, CA, USA). NRG miniature electronic apex locator. (**B,** Courtesy MedicNRG, Kibutz Afikim, Israel.)

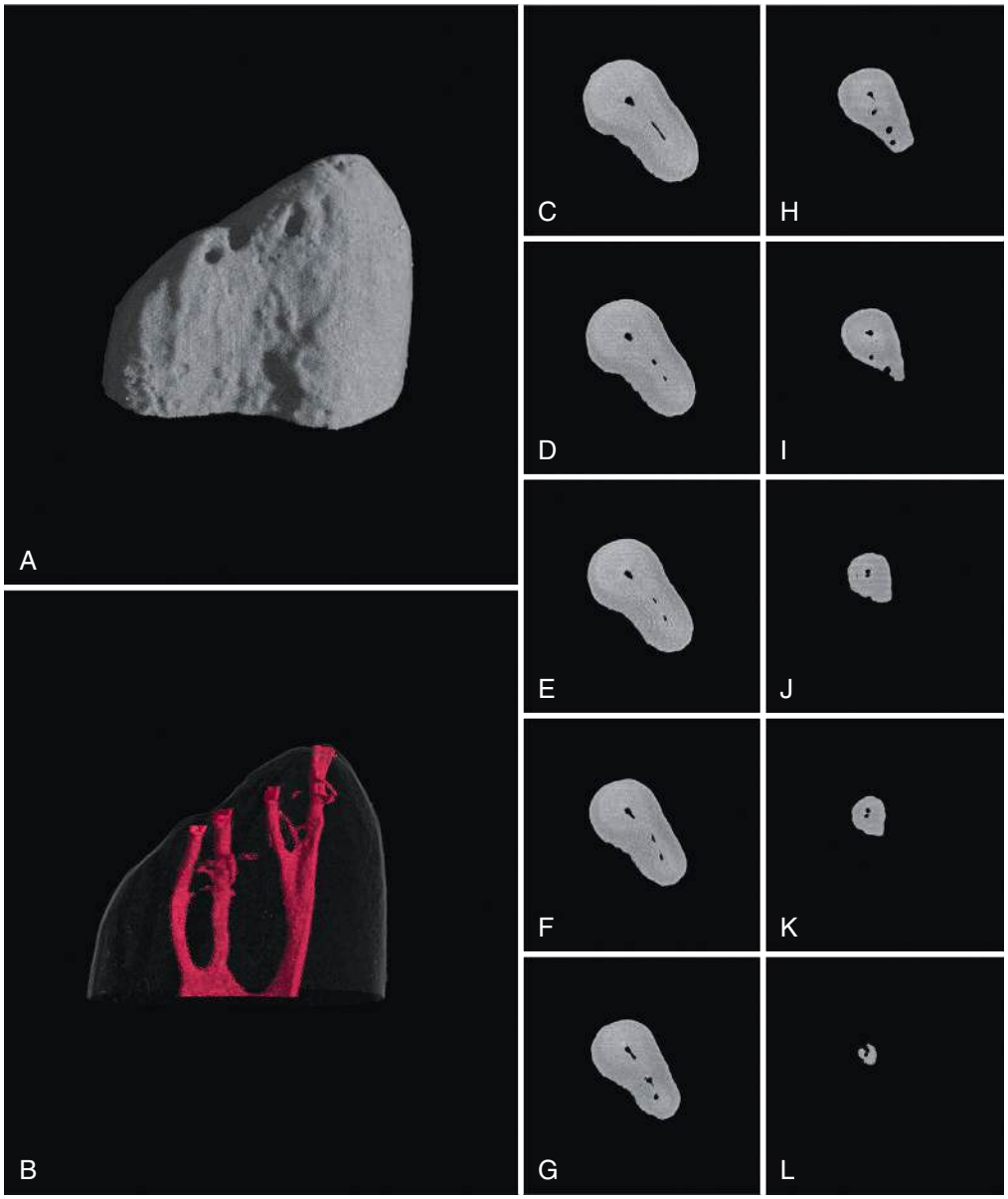


Fig. 8.37 Microcomputed tomographic scan of anatomy of the apical 5 mm of a mesiobuccal root (8 μm resolution). **A–B**, Three-dimensional reconstruction of outer contour and root canal systems. **C-L,** Cross sections 0.5 mm apart.

approach is likely to result in underestimation of the diameter.[486] This is a crucial point because the initial canal size is a major determinant for the desired final apical diameter, with the second relevant dimension being the size that allows deep irrigation needle penetration.

An ongoing debate exists between those who prefer smaller apical preparations combined with tapered shapes and those who favor larger apical preparations for better removal of infected dentin and to allow irrigation fluids access to the apical areas (Table 8.2). Both sides stress the importance of maintaining the original path of the canal during preparation; otherwise, bacteria in the apical one third of the root canal may not be reached by sufficient amounts of an antimicrobial agent.[289] Investigators demonstrated a higher percentage of bacterial elimination in single-root canal systems in situ by using a combination of significant enlargement of the apical third and NaOCl irrigation.[79] However, preparation errors (e.g., zips, canal transportation) may occur in such larger preparations specifically in curved canals, both with stainless steel and NiTi instruments (Fig. 8.39).

Thorough disinfection of the apical part of a root canal is essential, because this area is most likely to harbor residual intraradicular bacteria.[297] Wider apical preparations remove potentially infected dentin, allowing the delivering needle and subsequently antimicrobial irrigants to penetrate the root canal more deeply.[91,130]

A study investigating rotary NiTi files of three tapers (.06, .08, and .10) with file tips in sizes #20, #30, and #40 showed that size #20 instruments left significantly more debris in the apical third compared with size #40 instruments.[464] On the other hand, a study in which half the samples were prepared to a size #25 file and the other half to a size #40 file found no statistically significant difference

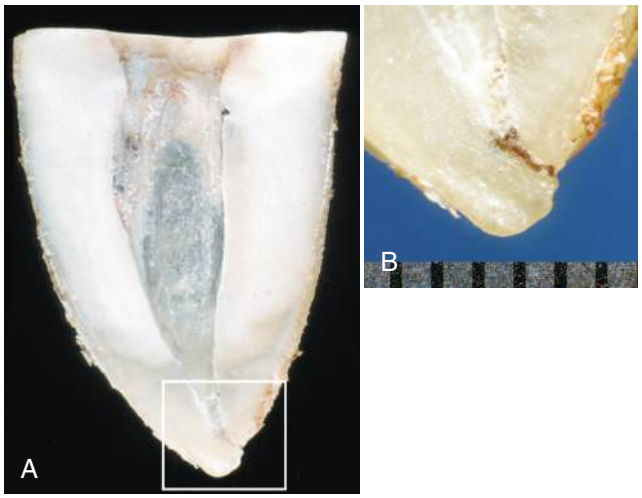


Fig. 8.38 Presence of dentin dust as a possible source of microbial irritation. Tooth #18 underwent root canal therapy. The clinician noted an apical blockage but was unable to bypass it. Unfortunately, intense pain persisted and at the patient's request, the tooth was extracted a week later. **A,** Mesial root of tooth #18; mesial dentin has been removed. **B,** Magnified view (×125) of rectangle in **A** shows an apical block (gradation of ruler is 0.5 mm).

in bacterial growth after instrumentation, with no growth observed after 1 week of treatment with a $Ca(OH)_2$ dressing.[500] Step-down sequences with additional apical enlargement to ISO size #35 and a serial step-back technique with no apical enlargement were compared, using NaOCl and EDTA as irrigants. Here, no significant difference was detected in colony-forming units with or without apical enlargement.[95] These researchers concluded that dentin removal in the apical third might be unnecessary if a suitable coronal taper is achieved.

Despite the disagreement over the appropriate width of a preparation (see Table 8.2), these studies suggest that root canal preparations should be confined to the canal space, should be sufficiently wide, and should incorporate the original root canal cross sections (see Fig. 8.4).

Traditional cleaning and shaping strategies (e.g., the step-back technique) focused immediate preparation of the apical third of the root canal system, followed by various flaring techniques to facilitate obturation.[161,380,453] In an attempt to reach the canal terminus, the clinician first selected a small file, placed an appropriate curve on the instrument, and then tried to work the file to full length. If the terminus could not be reached, the file was removed and,

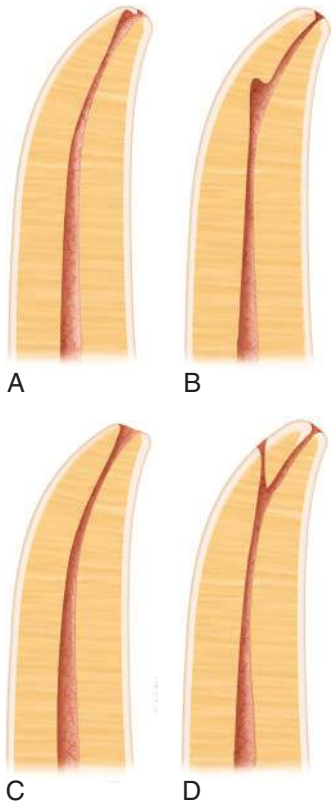


Fig. 8.39 Schematic diagrams showing the most common preparation errors. **A,** Apical zip. **B,** Ledge. **C,** Apical zip with perforation. **D,** Ledge with perforation.

after irrigation, either the same file or a smaller one was inserted. However, not infrequently, full length was not obtained, either because of blockage or because of coronal binding.

Coronal binding is caused by overhangs at the orifice level and when the canal is less tapered than an instrument, making it bind coronally. Moreover, a straight root often contains a curved canal, such as buccal and lingual curvatures that cannot be seen on radiographs.[98,343] In addition, passing a precurved negotiating file through a coronally tight canal will straighten the instrument.[425]

Various instrumentation sequences have been developed for hand and rotary instruments; these are discussed later in this chapter. However, the shape of the access cavity is the prerequisite that must be optimized before any canal preparation can take place (see Chapter 7).

Table 8.2 Characteristics of Wide and Narrow Apical Preparations

Root Canal Preparation	Benefits	Drawbacks
Narrow apical size	Minimal risk of canal transportation and extrusion of irrigants or filling material Can be combined with tapered preparation to counteract some drawbacks Less compaction of hard tissue debris in canal spaces	Little removal of infected dentin Questionable rinsing effect in apical areas during irrigation Possibly compromised disinfection during interappointment medication Not ideal for lateral compaction
Wide apical size	Removal of infected dentin Access of irrigants and medications to apical third of root canal	Risk of preparation errors and extrusion of irrigants and filling material Not ideal for thermoplastic obturation

Basic cleaning and shaping strategies for root canal preparation can be categorized as crown-down, step-back, apical widening, and hybrid techniques. In a crown-down approach, the clinician passively inserts a large instrument into the canal up to a depth that allows easy progress. The next smaller instrument is then used to progress deeper into the canal; the third instrument follows deeper again, and this process continues until the terminus is reached. Both hand and rotary instruments may be used in a crown-down manner. However, instrument sets with various tip diameters and tapers allow the use of either decreasing tapers or decreasing diameters for apical progress. Debate continues as to which of those strategies is superior for avoiding taper lock; currently no compelling evidence favors either of them.

In the step-back approach, WLs decrease in a stepwise manner with increasing instrument size. This prevents less flexible instruments from creating ledges in apical curves while producing a taper for ease of obturation.

As discussed previously, the aim of apical widening is to fully prepare apical canal areas for optimal irrigation efficacy and overall antimicrobial activity. Recently, apical enlargement has been broken down into three phases: preenlargement, apical enlargement, and apical finishing.[476]

Many rotary techniques require a crown-down approach to minimize torsional loads[52] and reduce the risk of instrument fracture. Used sequentially, the crown-down technique can help enlarge canals further. All basic techniques described so far may be combined into a hybrid technique to eliminate or reduce the shortcomings of individual instruments.

Provided adequate tools are used and the access cavity design is appropriate, excessive thinning of radicular structures can be avoided (see Fig. 8.5). Vertical root fractures and perforations are possible outcomes of excessive removal of radicular dentin in zones that have been termed "danger zones."[7] Overenthusiastic filing, for example, may lead to more procedural errors (see Figs. 8.22 and 8.39). On the other hand, ideal preparation forms without any preparation errors and with circular incorporation of the original canal cross sections may be achieved with suitable techniques (see Fig. 8.4).

Techniques

Standardized Technique. The standardized technique adopts the same WL definition for all instruments introduced into a root canal. It therefore relies on the inherent shape of the instruments to impart the final shape to the canal. Negotiation of fine canals is initiated with lubricated fine files in a so-called watch-winding movement. These files are advanced to WL and worked either in the same hand movement or with "quarter-turn-and-pull" until a next larger instrument may be used. Conceptually, the final shape should be predicted by the last instrument used. A single matching gutta-percha point may then be used for root canal filling. In reality, this concept is often violated: curved canals shaped with the standardized technique will be wider than the last used instrument,[16] exacerbated by the pulling portion of the hand movement. Moreover, adequate compaction of gutta-percha in such small a taper (~.02) is difficult or impossible (see Chapter 9).[15]

Step-Back Technique. Realizing the importance of a shape larger than that produced with the standardized approach, Weine et al.[483] suggested the step-back technique, incorporating a stepwise reduction of the WL for larger files, typically in 1 mm or 0.5 mm steps, resulting in flared shapes with .05 and .10 taper, respectively. Incrementally reducing the WL when using larger and stiffer instruments also reduced the incidence of preparation errors, in particular in curved canals. This concept appeared to be clinically very effective.[293]

Although the step-back technique was primarily designed to avoid preparation errors in curved canals, it applies to the preparation of apparently[98,382] straight canals as well. Several modifications of the step-back technique have been described over the years, to explore and provide some enlargement prior to reaching the WL.

Step-Down Technique. Other clinicians[156] described a different approach. They advocated shaping the coronal aspect of a root canal first before apical instrumentation commenced. This technique is intended to minimize or eliminate the amount of necrotic debris that could be extruded through the apical foramen during instrumentation;[129] moreover, by first flaring the coronal two thirds of the canal, apical instruments are unimpeded through most of their length. This in turn may facilitate greater control and less chance of zipping near the apical constriction.[246]

Crown-Down Technique. Multiple modifications of the original step-down technique have been introduced, including the description of the crown-down technique.[133,274,378] The more typical step-down technique includes the use of a stainless steel K-file exploring the apical constriction and establishing WL. In contrast, a crown-down technique relies more on coronal flaring and then determination of the WL later in the procedure.

To ensure penetration during step-down, one may have to enlarge the coronal third of the canal with progressively smaller GG drills or with other rotary instruments. Irrigation should follow the use of each instrument and recapitulation after every other instrument. To properly enlarge the apical third and to round out ovoid shape and lateral canal orifices, a reverse order of instruments may be used starting with a size #20 (for example) and enlarging this region to a size #40 or #50 (for example). The tapered shape can be improved by stepping back up the canal with larger instruments, bearing in mind all the time the importance of irrigation and recapitulation.

The more typical crown-down, or double-flare technique,[133] consisted of an exploratory action with a small file—a crown-down portion with K-files of descending sizes and an apical enlargement to size #40 or similar. The original technique included stepping back in 1-mm increments with ascending file sizes and frequent recapitulations with a small K-file and copious irrigation. It is further emphasized that significant wall contact should be avoided in the crown-down phase to reduce hydrostatic pressure and the possibility of blockage. Several studies[377,378] demonstrated more centered preparations in teeth with curved root canals shaped with a modified double-flare technique and Flex-R files compared with shapes prepared with K-files and

step-back technique. A double-flare technique was also suggested for ProFile rotary instruments.[387]

Balanced Force Technique. Regarding hand movements, a general agreement exists that the so-called balanced force technique creates the least canal aberrations with K-files. This technique has been described as a series of rotational movements for Flex-R files,[362] but it can also be used for K-files and other hand instruments such as GT hand files. Many different explanations have been offered for the obvious and undisputed efficacy of the balanced force approach,[84,241,382] but general agreement exists that it provides excellent canal-centering ability, superior to other techniques with hand instruments.[29,65]

The balanced force technique involves three principle steps.[362] The first step (after passive insertion of an instrument into the canal) is a clockwise rotation of about 90 degrees to engage dentin (Fig. 8.40). In the second step, the instrument is held in the canal with adequate axial force and rotated counterclockwise to break loose the engaged dentin chips from the canal wall; this produces a characteristic clicking sound. Classically, in the third step, the file is removed with a clockwise rotation to be cleaned; however, because files used with the balanced force technique are not precurved, every linear outward stroke essentially is a filing stroke and may lead to some straightening of the canal path. Therefore, in many cases, the clinician may advance farther apically rather than withdrawing the file, depending on the grade of difficulty.

Rotary Instrumentation. NiTi rotary instruments are invaluable devices in the preparation of root canals, although hand instruments may be able to enlarge some canals just as efficiently when used in appropriate sequences. After orifice modification, the access cavity and canals are flooded with irrigant, and a small file is advanced into the canal. Application of a lubricant when using hand instruments can help prevent apical blockage at this early stage. Once the WL has been established (aided by an electronic apex locator and radiographically verified), apical preparation begins to facilitate a glide path for subsequent rotary instrumentation (Fig. 8.41; also see earlier discussion).

Fig. 8.42 illustrates the development of two different shapes in the mesial root canals of a mandibular molar, clearly showing that substantial areas of the root canal surface in either case are not instrumented, even when apical size #50 or 0.09 taper are reached (see *red areas* in Fig. 8.42, *G* and *I*).

Specific Nickel Titanium (NiTi) Instrumentation Techniques

Crown-Down. This was the dominant approach for many years and is still being used, for example for ProFile and other instruments with standard .04 and .06 tapers. It must be noted that the manufacturers' instructions for these systems vary somewhat, and the instructions for GT rotary, RaCe, and the TF vary even more. Clinicians should always consult the manufacturers' instructions for details on working with any endodontic instrument.

Coronal preenlargement is suggested, the WL then is determined as described previously, and an open glide path is secured with K-files up to size #15 or #20, depending on the canal anatomy. If canal size permits, canal preparation begins with .06 taper instruments in descending tip diameters.[52] In more difficult small canals, .06 tapers are followed by .04-tapered instruments, also with descending tip diameters. Apical preparation is performed either with multiple shaping waves, as suggested for GT rotary files,[68] or in a step-back manner.[386] Recapitulation with a small hand file is recommended throughout the preparation.

Single Length. The approach for ProTaper and ProTaper Next instruments differs from that for earlier NiTi rotary files in that no traditional crown-down procedure is performed (see Fig. 8.44, later).

First, size #10 and #15 hand files are passively inserted into the coronal two thirds of a root canal as pathfinding files, which confirm the presence of a smooth, reproducible glide path. This step is essential for ProTaper shaping instruments, because they are mostly side-cutting and have fine, rather fragile tips.

Shaping files S1 and S2 are then passively inserted into the scouted canal spaces, which have been filled with irrigant (preferably NaOCl). If necessary, the SX file can be used at this stage to relocate orifices or remove obstructing dentin. After each shaping file is used, the canals are reirrigated, and a size #10 file is used to recapitulate to break up debris and move it into solution. This process is repeated until the depth of the pathfinding #10 or #15 file is reached.

After irrigation, the apical third is fully negotiated and enlarged to at least a size #15 K-file, and WL is confirmed

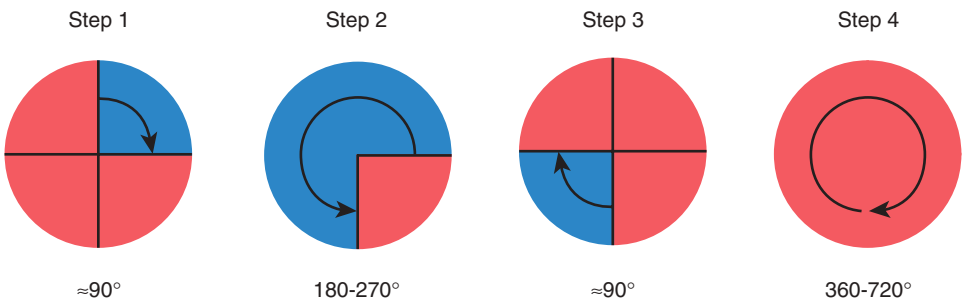


Fig. 8.40 Diagram of handle movements during balanced force hand preparation. STEP 1: After pressureless insertion of a Flex-R or NiTiFlex K-file, the instrument is rotated clockwise 90 degrees, using only light apical pressure. STEP 2: The instrument is rotated counterclockwise 180 to 270 degrees; sufficient apical pressure is used to keep the file at the same insertion depth during this step. Dentin shavings are removed with a characteristic clicking sound. STEP 3: This step is similar to step 1 and advances the instrument more apically. STEP 4: After two or three cycles, the file is loaded with dentin shavings and is removed from the canal with a prolonged clockwise rotation.

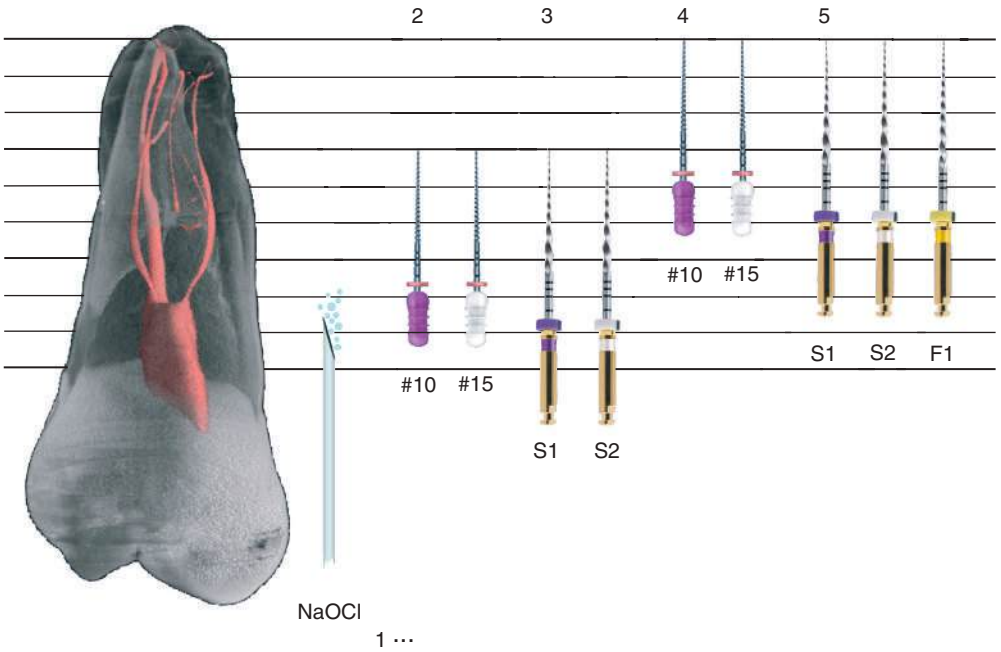


Fig. 8.41 Instrumentation of root canals with the single length technique, in this example using the ProTaper instruments. After irrigation and scouting (*1* and *2*), the coronal thirds are enlarged with shaping files *S1* and *S2*. Hand files then are used to determine the WL and to secure a glide path. Apical preparation is completed with S1 and S2. Finishing files are used to the desired apical width.

(see Fig. 8.44). Depending on the canal anatomy, the rest of the apical preparation can be done with engine-driven ProTaper shaping and finishing hand files. As an alternative, handles may be placed on these instruments' shanks, so that they can be used for a manual balanced force technique.

ProTapers S1 and S2 are then carried to full WL, still in a floating, brushing motion. WL should be confirmed after irrigation and recapitulation with a K-file, aided by an electronic apex locator and/or radiographs. Because of the progressive taper and more actively cutting flutes in the ProTaper design, interferences in the middle and coronal thirds are removed at this stage.

The preparation is finished with one or more of the ProTaper finishing files, used in a nonbrushing manner; because of their decreasing taper, these files will reach the WL passively. Recapitulation and irrigation conclude the procedures (see Fig. 8.41). Many other contemporary engine-driven techniques use the single-length concept—for example, WaveOne Gold, TRUShape, and XP-Shaper. However, the emphasis on Orifice Modification prior to deep instrument penetration allows these techniques to retain many of the benefits of the classic crown-down techniques, such as less change of WL during the procedure, deep and early access of irrigants, and less stress on fragile files.[327]

Hybrid Techniques. For many years, established clinical practice has suggested combining various NiTi preparation systems[79,476] to address certain shortcomings of most current instruments (Box 8.4). Although many combinations are possible, the most popular and useful ones involve coronal preenlargement followed by various apical preparation sequences. However, clinicians must keep in mind that anatomic variations in each canal must be addressed individually with specific instrument sequences. Most important, oval canals extend deep into the apical area,[442,482,491] and apical foramina may in fact be oval in most cases.[64] Naturally, a rotating file can produce a round canal at best (Fig. 8.43); therefore a strategy must be devised for adequately shaping oval canals without overly weakening radicular structure (compare Figs. 8.4 and 8.5).

Histologic slides (see Fig. 8.43) and microcomputed tomography reconstructions show critical areas that were not mechanically prepared despite the use of various individual rotary techniques. The aim of hybridizing NiTi rotary techniques, therefore, is to increase apical size using a fast and safe clinical procedure (Fig. 8.44).

The principle involves the use of a variety of instruments: for example, K-files for establishing patency, ProTaper instruments for body shaping and apical preenlargement, NiTi K-files or LightSpeed instruments for apical widening, and various instruments for final smoothing.[476]

FINAL APICAL ENLARGEMENT

Conceptually, file sizes during canal preparation were termed "initial apical file," "master apical file," and "final file," or IAF, MAF, and FF, respectively. The first file to bind, the IAF, is supposed to give clinicians a guide in determining the final size of their canal preparation. On the other hand, many current instrumentation techniques have, due to their design, predefined final sizes—for example, WaveOne Gold shaping aims at final sizes #25 .07, #35 .06, and #45 .05 taper.

Whichever apical size was conceptually aimed for, after achieving a tapered canal form, the apical size at that point should be verified by apical gauging. K-files or K-FlexOFiles of the same size are often recommended for this step. During gauging, files should be gently moved to WL and the binding point should be noted. Presence of an apical

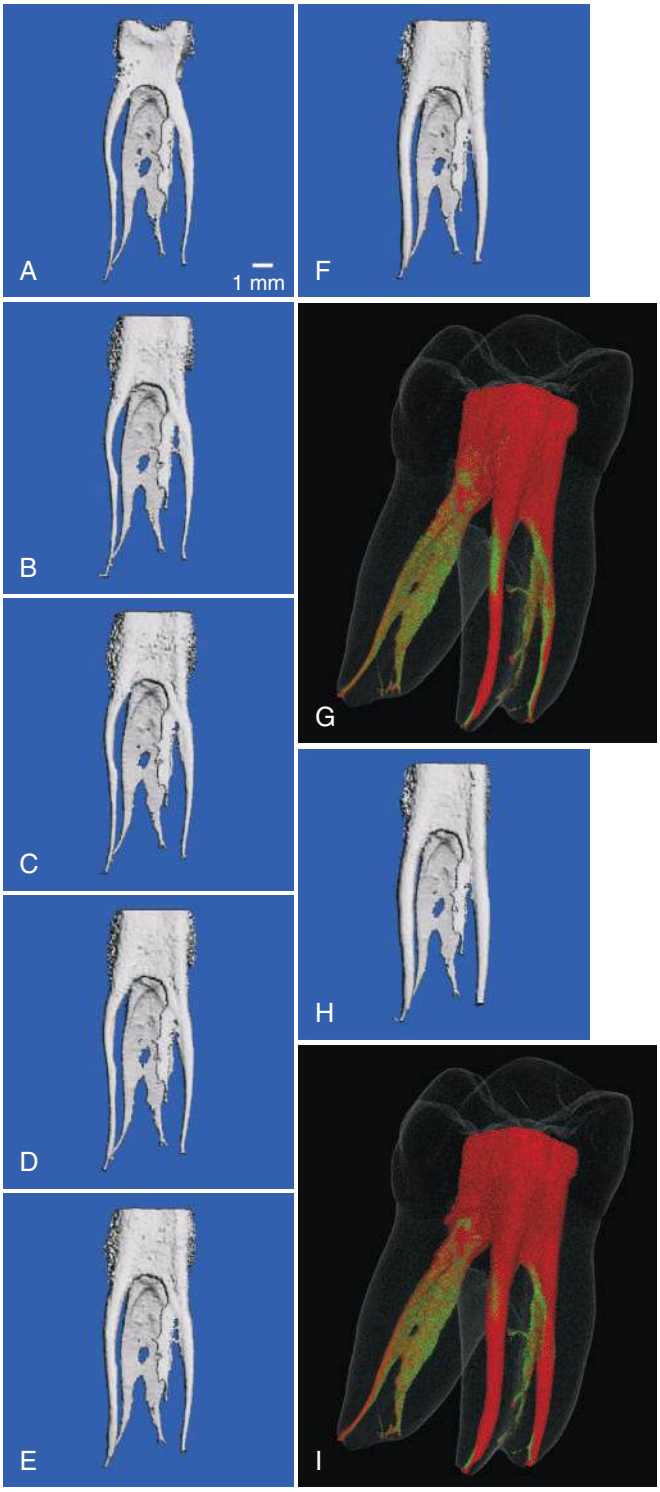


Fig. 8.42 Stepwise enlargement of mesial root canal systems in an extracted mandibular molar demonstrated with micro–computed tomography (μCT) reconstructions. The buccal canal *(left)* was prepared with a LightSpeed (LS) instrument, and the lingual canal *(right)* was shaped with a ProTaper (PT) instrument. **A,** Pretreatment view from the mesial aspect. Note the additional middle canal branching from the lingual canal into the coronal third. **B,** Initial preparation and opening of the orifices, aided by ultrasonically powered instruments. **C,** First step of root canal preparation, up to LightSpeed size #20 and ProTaper shaping file S1. **D,** Further enlargement to LS size #30 and PT shaping file S2. **E,** Apical preparation to LS size #40 and PT finishing file F1. **F,** Additional enlargement to LS size #50 and PT finishing file F2. **G,** Superimposed μCT reconstructions comparing the initial canal geometry *(in green)* with the shape reached after use of the instruments shown in **F**. **H,** Final shape after step-back with LS instruments and PT finishing file F3. **I,** Superimposed μCT reconstructions comparing initial geometry and final shape. Note the slight ledge in the buccal canal after LS preparation and some straightening in the lingual canal after PT preparation.

narrowing means that WL is reached without any shaping effort, with a size-corresponding K-file. However, the next larger K-file size will stay back by a small distance. Finally, maintenance of patency should be verified. If apical gauging stops short of WL, which suggests that a canal is undershaped, additional enlargement should be accomplished with any technique appropriate to the specific anatomy of the root canal. This often necessitates a hybrid technique.

Box 8.4 Benefits o f Using a Combination of Instruments for Endodontic Therapy

- Instruments can be used in a manner that promotes their individual strengths and avoids their weaknesses (most important).
- Hand instruments secure a patent glide path.
- Tapered rotary instruments efficiently enlarge coronal canal areas.
- Less tapered instruments allow additional apical enlargement.

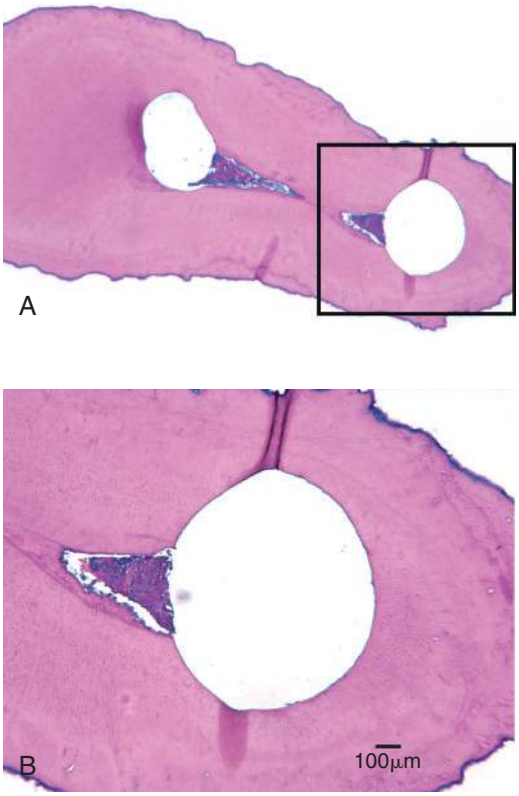


Fig. 8.43 Remaining potentially infected tissue in fins and isthmus configuration after preparation with rotary instruments. **A,** Cross section through a mesial root of a mandibular molar, middle to coronal third of the root. Both canals have been shaped; the left one is transported mesially (×10). **B,** Magnified view of rectangle in **A**. Note the presence of soft tissue in the isthmus area (×63). (Courtesy Professor H. Messer.)

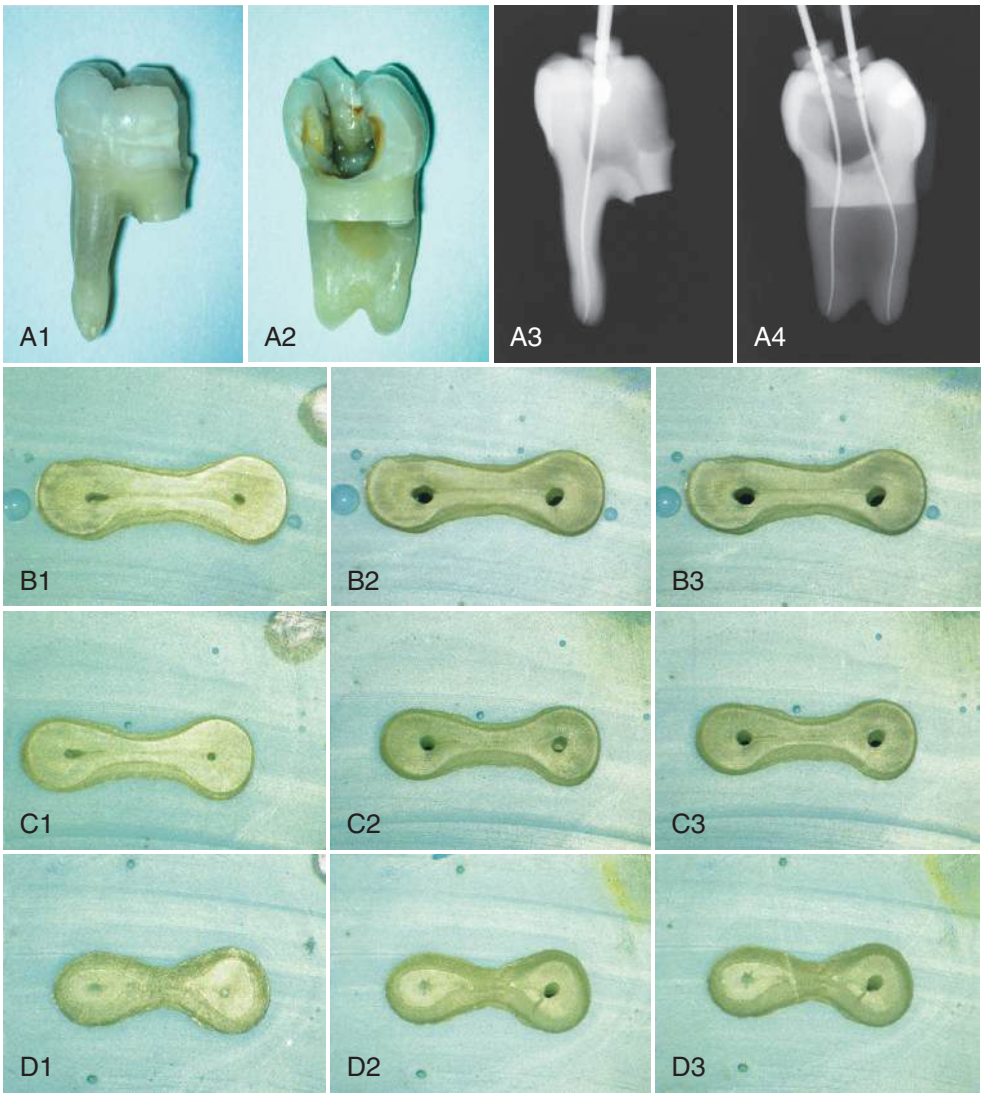


Fig. 8.44 Effect of a hybrid technique on root canal anatomy studied in a Bramante model. **A1–A4,** Both mesial canals of an extracted mandibular molar have been instrumented. Canal cross sections are shown before instrumentation **(B1–D1). B2–D2,** Cross sections after preenlargement with a ProTaper F3 file *(left canal)* and a size #45, .02 taper instrument *(right canal)*. The final apical sizes were LightSpeed (LS) #50 and size #50, .02 taper in the left and the right canal, respectively. (Courtesy Dr. S. Kuttler, Dr. M. Gerala, and Dr. R. Perez.)

Disinfection of the Root Canal System

Irrigation is defined as "to wash out a body cavity or wound with water or a medicated fluid" and aspiration as "the process of removing fluids or gases from the body with a suction device." Disinfectant, meanwhile, is defined as "an agent that destroys or inhibits the activity of microorganisms that cause disease."[1]

The objectives of irrigation in endodontics are mechanical, chemical, and biological. The mechanical and chemical objectives are as follows: (i) Flush out debris, (ii) lubricate the canal, (iii) dissolve organic and inorganic tissue, (iv) prevent the formation of a smear layer during instrumentation or dissolve it once it has formed, and (v) detach and disrupt biofilms.[38] Mechanical effectiveness will depend upon the ability of irrigation to generate optimum streaming forces within the entire root canal system, interact with the canal walls, and reach areas that cannot be prepared by mechanical instruments, including fins, isthmuses, and even lateral canals.[479] Chemical effectiveness depends upon the concentration of the antimicrobial irrigant, the area of contact, and the duration of interaction between irrigant and infected material.[58] Final efficiency of endodontic disinfection depends upon its chemical and mechanical effectiveness.[170]

The biological function of irrigants is related to their antimicrobial effects and their capacity to effectively eradicate or maximally reduce intracanal bacteria.

In principle, irrigants should (i) have high efficacy against anaerobic and facultative microorganisms in their planktonic or biofilm states, (ii) inactivate endotoxin, (iii) be nontoxic when they come in contact with vital tissues, and not cause an anaphylactic reaction.[38]

HYDRODYNAMICS OF IRRIGATION

Irrigation dynamics refers to how irrigants flow, penetrate, and readily exchange within the root canal walls as well as the forces produced by them. A better understanding of the fluid dynamics of different modes of irrigation contributes to achieving predictable disinfection of the root canal system. Hence, in endodontic disinfection, the process of delivery is as essential as the antibacterial characteristics of the irrigants.[58]

The effectiveness of root canal irrigation in terms of debris removal and eradication of bacteria depends on several factors that can be controlled by the operator: penetration depth of the needle, diameter of the root canal, inner and outer diameter of the needle, irrigation pressure, viscosity of the irrigant, velocity of the irrigant at the needle tip, and type and orientation of the needle bevel (Fig. 8.45).

The size and length of the irrigation needle—to root canal dimensions—are of utmost importance for the effectiveness of irrigation. Apical root canal diameter and taper have an

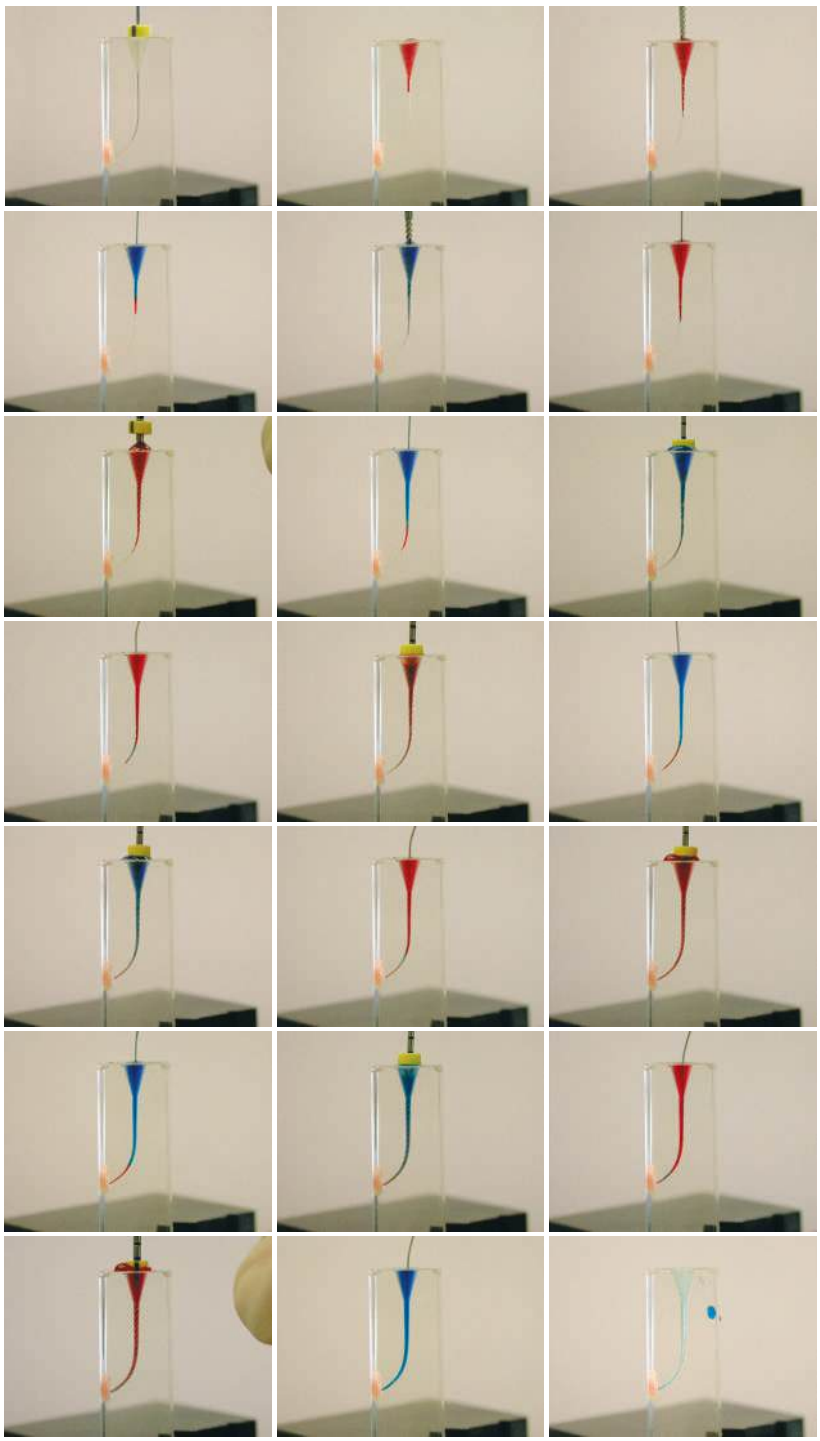


Fig. 8.45 Irrigation and movement of irrigants depends on canal shape. Sequential enlargement of a canal in clear plastic block was performed with a sequence of ProFile instruments in accordance with the manufacturer's recommendations. Alternating irrigation with blue and red fluid was done after each preparation step. Note the apical presence of irrigant after sufficient shape has been provided. Note the distribution of fluid immediately after irrigation with a 30-gauge needle.

impact on needle penetration depth (see earlier in this chapter for details regarding apical preparation size; see Fig. 8.7 and Table 8.2).

The external needle diameter is of relevance for the depth of introduction into the root canal (Fig. 8.46) and rigidity of the needle body, an essential consideration for irrigation of curved canals. Common 27 gauge injection needles have an external diameter of 0.42 mm, but smaller stainless steel irrigation tips with external diameters of 0.32 mm (30 gauge) and even 31 gauge or 0.26 mm (Navitip, Ultradent, South Jordan, UT, USA) are available. The Stropko Flexi-Tip (30 gauge) needle is fabricated from nickel-titanium to improve penetration into curved root canals. More recently, a flexible plastic irrigation tip was introduced (TruNatomy, Dentsply Sirona).

Apical pressure is the force applied perpendicular to the apical wall surface. Internal syringe diameter determines the pressure necessary for moving the syringe plunger. Speed of the plunger determines the velocity with which the irrigant is extruded. Fine needles require more pressure onto the plunger and extrude the irrigant with higher velocity than large needle sizes, which extrude significantly greater amounts of irrigants but for safety reasons cannot be introduced as deeply.

To improve the safety of irrigation and prevent extrusion of the irrigant through the apical foramen, some needle types release the solution via lateral openings and have a closed, safe-ended tip. The orientation of the bevel is crucial to produce a turbulence effect on the dentinal wall of the canal. Wall shear stress has a direct impact on the quality and efficacy of root canal wall débridement[61] and is one of the most critical fluid mechanics parameters. Side-vented and double side-vented needles led to maximum shear stress concentrated on the wall facing the outlet (the proximal outlet for the double side-vented needle).[60]

Flow is defined as the quantity of material passing a given area in a unit time period. The flow depends on many factors. Some of them are pressure: Pressure difference is needed to make flow happen. Flow occurs from higher pressure to lower pressure and the greater the difference, the greater the flow. Flow also changes with the diameter of tube or needle: the larger the needle, the more flow. Finally, the longer the length of the needle, the less the flow. In endodontics, anatomical constrains of the root canal make irrigation flow a challenge.

IRRIGANTS

An optimal irrigant would have all of the characteristics considered beneficial in endodontics but none of the negative or harmful properties. Presently, no solution can be regarded as optimal. However, the combined use of selected irrigation products dramatically contributes to the successful treatment outcomes (Boxes 8.5 and 8.6).

Sodium Hypochlorite

NaOCl is the most commonly used irrigating solution[285] because of its antibacterial capacity and the ability to dissolve necrotic tissue, vital pulp tissue, and organic components of dentin and biofilms in a fast manner.[392]

NaOCl solution is frequently used as a disinfectant or a bleaching agent. It is the irrigant of choice in endodontics, owing to its efficacy against pathogenic organisms and pulp digestion, and satisfies most of the preferred characteristics stated earlier.[285]

History. Hypochlorite was first produced in 1789 in France. Hypochlorite solution was used as a hospital antiseptic that was subsequently sold under the trade names "Eusol" and "Dakin's solution." NaOCl as a buffered 0.5% solution was recommended for the irrigation of wounds during World War I by Dakin.[103] In 1919 Coolidge[97] introduced NaOCl to endodontics as an intracanal irrigation solution.[504]

Mode of Action. When sodium hypochlorite contacts tissue proteins, nitrogen, formaldehyde, and acetaldehyde are formed. Peptide links are fragmented, and proteins disintegrate, permitting hydrogen in the amino groups (-NH-) to be replaced by chlorine (-NCl-) forming chloramines. This plays a vital role in the antimicrobial effectiveness. Necrotic tissue and pus are dissolved, and the antimicrobial agent reaches and cleans the infected areas.

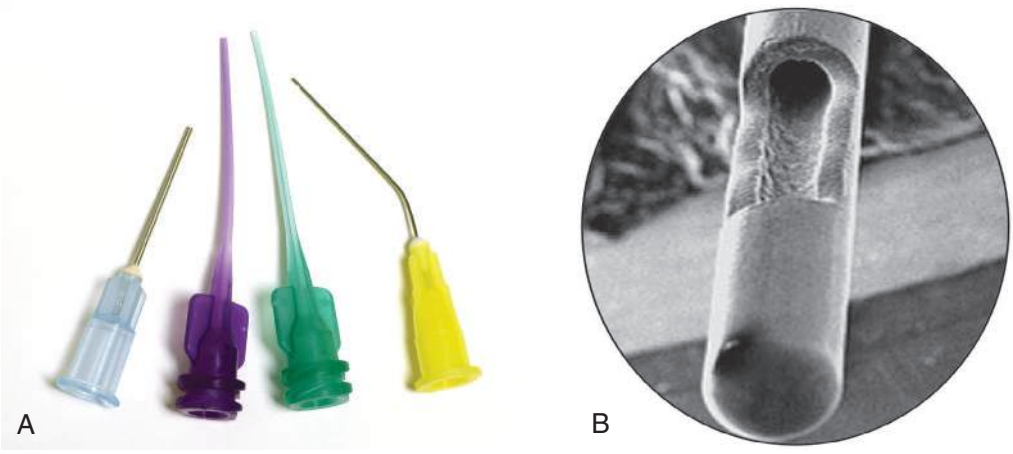


Fig. 8.46. A, Various types of needles for root canal irrigation. Shown are examples with open end and closed end, side vented. These are manufactured from plastic and stainless steel. **B,** SEM image of 30 gauge safety needle. (Courtesy Dr. F. Tay.)

Box 8.5 Benefits of Using Irrigants in Root Canal Treatment

- Removal of particulate debris and wetting of the canal walls
- Destruction of microorganisms
- Dissolution of organic debris
- Opening of dentinal tubules by removal of the smear layer
- Disinfection and cleaning of areas inaccessible to endodontic instruments

Box 8.6 Properties of an Ideal Irrigant for Root Canal Treatment

An ideal irrigant should:

- Be an effective germicide and fungicide.
- Be nonirritating to the periapical tissues.
- Remain stable in solution.
- Have a prolonged antimicrobial effect.
- Be active in the presence of blood, serum, and protein derivates of tissue.
- Have low surface tension.
- Not interfere with repair of periapical tissues.
- Not stain tooth structure.
- Be capable of inactivation in a culture medium.
- Not induce a cell-mediated immune response.
- Be able to completely remove the smear layer, and be able to disinfect the underlying dentin and its tubules.
- Be nonantigenic, nontoxic, and noncarcinogenic to tissue cells surrounding the tooth.
- Have no adverse effects on the physical properties of exposed dentin.
- Have no adverse effects on the sealing ability of filling materials.
- Be convenient in its application.
- Be relatively inexpensive.

Estrela[125] reported in 2002 that sodium hypochlorite exhibits a dynamic balance (Fig. 8.47):

1. Saponification reaction: Sodium hypochlorite acts as an organic and fat solvent that degrades fatty acids and transforms them into fatty acid salts (soap) and glycerol (alcohol), reducing the surface tension of the remaining solution.
2. Neutralization reaction: Sodium hypochlorite neutralizes amino acids by forming water and salt. With the exit of hydroxyl ions, the pH is reduced.
3. Hypochlorous acid formation: When chlorine dissolves in water, and it is in contact with organic matter, it forms hypochlorous acid. Hypochlorous acid is a weak acid with the chemical formula HClO that acts as an oxidizer. Hypochlorous acid ($HOCl^-$) and hypochlorite ions (OCl^-) lead to amino acid degradation and hydrolysis.
4. Solvent action: Sodium hypochlorite also acts as a solvent, releasing chlorine that combines with protein amino groups (NH) to form chloramines (chloramination reaction). Chloramines impede cell metabolism; chlorine is a strong oxidant and inhibits essential bacterial enzymes by irreversible oxidation of SH groups (sulphydryl group).[143]
5. High pH: Sodium hypochlorite is a strong base (pH >11). The antimicrobial effectiveness of sodium hypochlorite, based on its high pH (hydroxyl ions action), is similar to the mechanism of action of calcium hydroxide. The high pH interferes in cytoplasmic membrane integrity due to irreversible enzymatic inhibition, biosynthetic alterations in cellular metabolism, and phospholipid degradation observed in lipidic peroxidation.[125]

Allergic Reactions to Sodium Hypochlorite. Although few reports have been published[127,180] on allergic reactions to NaOCl, real allergies to NaOCl are unlikely to occur, since both Na and Cl are essential elements in the physiology of the human body. It must be remembered that the hypochlorous acid (the active component of the sodium hypochlorite) is a chemical substance that is elaborated by

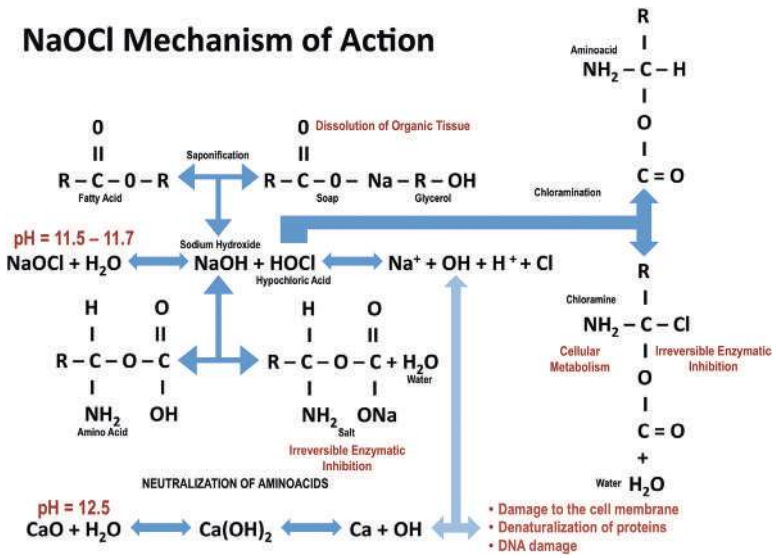


Fig. 8.47 Schematic diagram of the mechanism of action of NaOCl with the main interactions and properties highlighted. This diagram shows the chemical reaction of NaOCl when in contact with organic matter. The solution produces glycerol and soap (saponification reaction) and acts as a fat solvent degrading fatty acids. Moreover, hypochlorous acid is formed with a reduction of the pH and formation of chloramines (chloramination reaction). Damage to the cell membrane, denaturalization of proteins, and DNA damage may be observed. (Courtesy Dr. A. Manzur.)

neutrophils in the process of phagocytosis; it may create local tissue damage when it is produced in excess (liquefaction necrosis: purulent exudate) but does not cause allergic responses. Nevertheless, hypersensitivity and contact dermatitis may occur in rare situations. A recent case report[355] describes a severe chemical burn in an endodontist's eye caused by accidental contact with 3.5% NaOCl used as an irrigant during the root canal.

When hypersensitivity to NaOCl is suspected or confirmed, chlorhexidine (CHX) should not be used either because of its chlorine content. Individuals who genetically have more possibilities than the average population to generate allergies to multiple elements (allergies to food or latex) a skin test for both NaOCl and CHX may be indicated. The use of an alternative irrigant with high antimicrobial efficacy, such as iodine potassium iodide (IKI), should be considered, assuming there is no known allergy to that irrigant. Solutions such as alcohol or tap water are less effective against microorganisms and do not dissolve vital or necrotic tissue; however, $Ca(OH)_2$ could be used as a temporary medicament because it dissolves both vital and necrotic tissue.[17,188]

Temperature. Increasing the temperature of low-concentration NaOCl solutions improves their immediate tissue-dissolution capacity (Fig. 8.48).[504] Furthermore, heated hypochlorite solutions remove organic debris from dentin shavings more efficiently. Bactericidal rates for NaOCl solutions, the capacity of human pulp dissolution, and increased efficacy were detailed in several studies.[418] There are various devices to preheat NaOCl syringes (Fig. 8.49); however, it was demonstrated that as soon as the irrigant touches the root canal system, the temperature reaches the body temperature.[508] Therefore, in situ heating of NaOCl is recommended by some authors. This can be done by activating ultrasonic or sonic tips to the NaOCl inside the root canal for a couple of minutes (see irrigation timings later). Macedo et al.[265] state that the efficacy of NaOCl on dentin is improved by refreshment, ultrasonic activation, and exposure time. In that investigation, a 10°C temperature rise during ultrasonic activation was insufficient to increase the

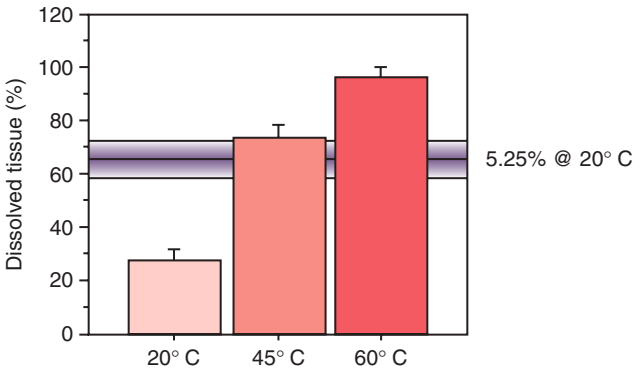


Fig. 8.48 Effect of heating on the ability of 0.5% sodium hypochlorite (NaOCl) to dissolve pulp tissue: NaOCl heated to 113°F (45°C) dissolved pulp tissue as well as the positive control (5.25% NaOCl) did. When the NaOCl was heated to 140°F (60°C), almost complete dissolution of tissue resulted. (Modified from Sirtes G, Waltimo T, Schaetzle M, Zehnder M: The effects of temperature on sodium hypochlorite short-term stability, pulp dissolution capacity, and antimicrobial efficacy, *J Endod* 31:669–671, 2005.)

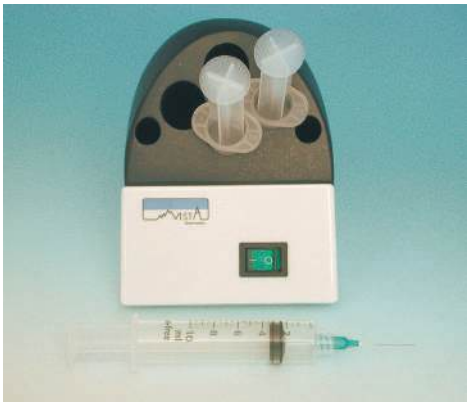


Fig. 8.49 Device for heating syringes filled with irrigation solution (e.g., sodium hypochlorite) before use. (Courtesy Vista Dental Products, Racine, WI.)

reaction rate. However, there are no clinical studies available at this point to support the use of heated NaOCl.[38,94]

Concentration. NaOCl is used in concentrations between 0.5% and 6% for root canal irrigation. Controversy exists over recommended concentrations of sodium hypochlorite during root canal treatment. Several studies have tried to evaluate the influence of different concentrations of NaOCl on various aspects of the treatment such as antimicrobial efficacy, tissue dissolution, penetration into dentin, capacity to remove the smear layer, decalcification of dentin, microhardness, viability of stem cells in the apical region, viscosity, amount of debris extrusion, tooth surface strain, and postoperative pain. The results of these studies have been conflicting in nature. Some in vitro studies have shown that NaOCl in higher concentrations is more effective against *Enterococcus faecalis* and *Candida albicans*.[159,349,478] In contrast, clinical studies have indicated both low and high concentrations to be equally effective in reducing bacteria from the root canal system.[77,100] NaOCl in higher concentrations has a better tissue-dissolving ability.[186] However, in lower concentrations when used in high volumes, it can be equally effective.[290,413] Higher concentrations of NaOCl are more toxic than lower concentrations[430]; however, perhaps due to the anatomy of the root canal system, higher concentrations have successfully been used during root canal treatment, with a low incidence of problems. Higher concentrations cause more cytotoxicity, erosion of dentin, collagen degradation, decalcification, dentin deproteination, and tooth surface strain while reducing microhardness, elastic modulus, and flexural strength of dentin.[468] In the root canal environment, using higher NaOCl concentrations does not necessarily change outcomes, because a high volume of irrigation and frequent exchanges can compensate for the effects of the concentration.[149]

Time. Retention time and the volume of irrigation are essential factors influencing the antibacterial activity of NaOCl.[149] There is conflicting evidence regarding the time course of the antibacterial effect of NaOCl.[38,176] In some articles hypochlorite kills the target microorganism in seconds, even at low concentrations, while other reports have published considerably longer times for the killing of the

same species.[160] Such differences are likely a result of several factors such as irrigant concentration and presence of organic matter (inflammatory exudates, hard and soft tissue remnants and microbial biomass).

Several studies showed[174,291] that the presence of dentin caused marked delays in the killing of *E. faecalis* by 1% NaOCl. Many of the earlier studies were performed in the presence of an unknown amount of organic matter. When such confounding factors are eliminated, it has been shown that NaOCl kills the target microorganisms rapidly, even at low concentrations of less than 0.1%.[176,471] However, in vivo, the presence of organic matter consumes NaOCl and weakens its effect. Therefore continuous replenishing of irrigation solution and allowing sufficient contact time are important factors for the effectiveness of NaOCl.[176]

The chlorine ion, which is responsible for the dissolving and antibacterial capacity of NaOCl, is unstable and consumed rapidly (probably within the first 2 minutes) during the first phase of tissue dissolution,[290] which provides another reason for continuous replenishment. This should especially be considered because rotary root canal preparation techniques have expedited the shaping process. The optimal time that a hypochlorite irrigant at a given concentration needs to remain in the canal system is an issue yet to be resolved.[504] Hypochlorite should be used throughout the instrumentation, and for 1 to 2 minutes after completing the instrumentation.[177]

Toxicity. If NaOCl is used without caution, it is very toxic and destructive to intraoral soft tissues, the periradicular vasculature, and cancellous bone where it can elicit severe inflammatory responses and degradation of organic components. Because of its high pH and strong oxidizing capacity, complete lysis of erythrocytes can occur at low concentration.

Even though the incident of NaOCl accident is reported as low, the actual frequency of such accidents is unknown, as many of them may not be reported, while minor extrusion incidents might even remain undetected due to the absence of severe symptoms.

A survey conducted on 314 diplomates of the American Board of Endodontics indicated that only 132 members reported experiencing a NaOCl accident.[228] Significantly more women experienced sodium hypochlorite accidents compared with men. More maxillary teeth than mandibular teeth and more posterior than anterior teeth were involved. A diagnosis of pulp necrosis with radiographic findings of periradicular radiolucency was positively associated with such accidents. Most respondents reported that patient signs and symptoms entirely resolved within a month. The occurrence of an accident, by itself, did not adversely affect the endodontic prognosis of the involved tooth. Anatomic variations may contribute significantly to the occurrence of a sodium hypochlorite accident.[228]

It is essential to recognize the symptoms and act accordingly. Accidental extrusion of NaOCl into the tissue results in severe pain, edema of neighboring soft tissues, possible extension of edema over injured half side of face/upper lip, profuse bleeding from root canal, profuse interstitial bleeding with hemorrhage of skin and mucosa (ecchymosis), chlorine taste, and irritation of throat after injection into maxillary sinus; secondary infection or even reversible anesthesia or paresthesia is possible.

NaOCl is cytotoxic and a potent dissolver of organic matter. This is the reason why edema may be perceived as a protective tissue response when a hyperosmotic and cytotoxic liquid is extruded into the vicinity of periradicular tissues. The ecchymosis involves the dissolution of blood vessel walls and hemorrhage around the subcutaneous soft tissues. Also, it is a potent oxidant, and the mechanism of injury from undiluted NaOCl is primarily oxidation of proteins. Depending on the concentration employed for root canal irrigation, NaOCl may also be very hypertonic (2800 mOsm/kg for 3% to 6% solutions) concerning tissue fluids; this partially accounts for the rapid appearance of edema during a NaOCl accident.[510]

If NaOCl is inadvertently extruded through the apex, accidents may occur (Fig. 8.50, *A–C*). For the management of these lesions, it is recommended to inform the patient and to control pain with local anesthesia and analgesics and close follow-up in the hours and days after the accident. To gain rapid pain relief, anesthetic infiltration had been attempted. However, if swelling is present, infiltration anesthesia is contraindicated to avoid spreading of any existing infection; a nerve block should be used instead.[167] Very little information was provided regarding the use of

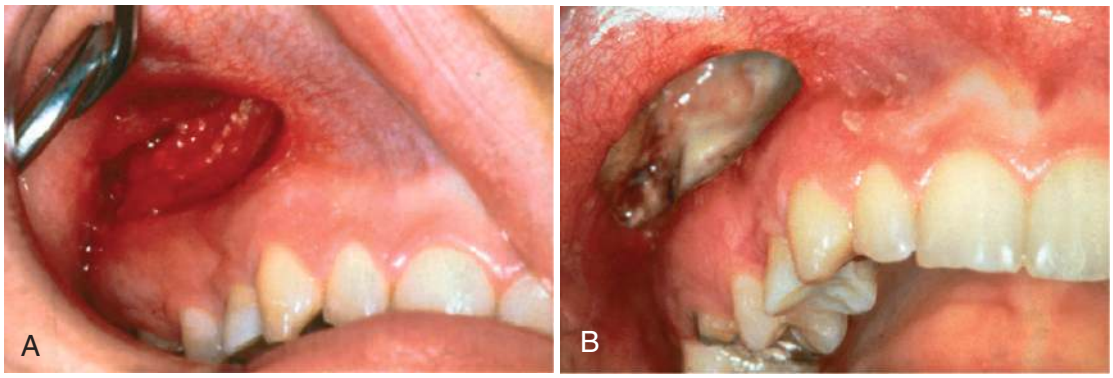


Fig. 8.50 Toxic effect of sodium hypochlorite on periradicular tissues. After root canal treatment of tooth #3, the patient reported pain. **A,** On a return visit, an abscess was diagnosed and incised. **B,** Osteonecrosis was evident after 3 weeks.

vasoconstrictors and the location of the injection attempted for pain relief. Theoretically, vasoconstrictors might limit the diffusion of NaOCl, but this would likely increase the risk of promoting tissue necrosis, especially with highly concentrated solutions promoting local ischemia.[167]

Some reports used saline to dilute the NaOCl or even CHX, but the latest should be avoided to prevent the formation of the precipitate (see later).

In general, postextrusion management is ambulatory using only oral medications. However, it was reported that about one third of patients (18/52) were hospitalized for monitoring and intravenous administration of drugs.[227] The application of extraoral cold compresses for reduction of swelling is also useful. After 1 day, warm compresses and frequent warm mouth rinses for stimulation of local systemic circulation should be used. Patients should be recalled daily to monitor recovery. The use of antibiotics is not obligatory and recommended only in cases of high risk or evidence of secondary infection. The administration of antihistamine is also not obligatory, and the use of corticosteroids is controversial. Further endodontic therapy with sterile saline or CHX as root canal irrigants and referral to hospital in case of worsening symptoms were suggested.[227]

Permanent sequelae could be divided into nerve lesions and scar tissues. Neurologic examination of the trigeminal and facial nerves should systematically be performed once anesthesia has dissipated. Tooth loss has not been reported as a direct result of NaOCl extrusion, but it may be involved. The latter is a real trauma for the patient, and it can lead to subsequent refusal to achieve endodontic treatment.[167]

Chlorhexidine

History. CHX was developed in the United Kingdom and first marketed as an antiseptic cream.[131] It has been used for general disinfection purposes and the treatment of skin, eye, and throat infections in both humans and animals.[131,257] It has been used as an irrigant and medicament in endodontics for more than a decade.[266,286,315]

Molecular Structure. The CHX molecule is a strong base with a pH between 5.5 and 7 that belongs to the polybisguanide group; it consists of two symmetric four-chlorophenyl rings and two bisguanide groups connected by a central hexamethylene chain. CHX digluconate salt is easily soluble in water and very stable (Fig. 8.51).[165]

Mode of Action. CHX, because of its cationic charges, is capable of electrostatically binding to the negatively charged surfaces of bacteria,[105] damaging the outer layers of the cell wall and rendering it permeable.[204] CHX is a wide-spectrum antimicrobial agent, active against gram-positive, gram-negative bacteria and yeasts.[115]

Depending on its concentration, CHX can have both bacteriostatic and bactericidal effects. At high concentrations, CHX acts as a detergent, and exerts its bactericidal effect by damaging the cell membrane and causes precipitation of the cytoplasm. At low concentrations, CHX is bacteriostatic, causing low molecular-weight substances (i.e., potassium and phosphorous) to leak out without the cell being permanently damaged.[39]

Substantivity. Unlike NaOCl, CHX is capable of remaining in the dentin because of the cationic nature of its molecule. Its adsorption to dental hard tissues with gradual and prolonged release at therapeutic levels is usually called the residual effect or substantivity. This effect depends on the concentration of CHX and the contact time with the dentin; moreover, it has been detected in root canal dentin up to 120 days.[427] Komorowski et al.[233] showed that 5-minute application of CHX did not induce substantivity on dentin. Thus a short time inside the root canal can interfere with the adsorption of CHX on dentin.

At low concentrations of 0.005% to 0.01%, only a constant monolayer of CHX is adsorbed on the tooth surface, but at higher concentrations (0.02%) a multilayer of CHX is formed on the surface, providing a reservoir of CHX that can rapidly release the excess into the environment as the concentration of CHX in the surrounding environment decreases.[123] Time and concentration of CHX can influence the antibacterial substantivity, and the conclusions are inconsistent. Some studies demonstrated that 4% CHX has greater antibacterial substantivity than 0.2% after 5 minutes application.[286] Other studies stated that CHX should be left for more than 1 hour in the canal to be adsorbed by the dentin.[1] Komorowski et al.[233] suggested that a 5-minute application of CHX did not induce substantivity, so dentin should be treated with CHX for 7 days. However, when Paquette and Malkhassian[316] in their in vivo study

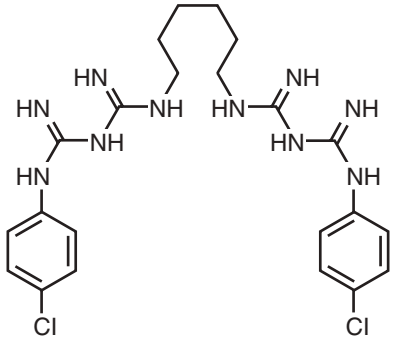


Fig. 8.51 Schematic drawing of the chlorhexidine molecule.

medicated the canals with either liquid or gel forms of CHX for 1 week, neither of them could achieve complete disinfection. Therefore, residual antimicrobial efficacy of CHX in vivo remains to be demonstrated.

Cytotoxicity. In the medical field, CHX is typically used at concentrations between 0.12% and 2.0%. According to Löe,[257] at these concentrations CHX has a low level of tissue toxicity, both locally and systemically. In another report, when 2% CHX was used as a subgingival irrigant, no apparent toxicity was noted on gingival tissues.[258,426] Moreover, CHX rinses have been reported to promote the healing of periodontal wounds.[24] Based on these reports, Jeansonne et al.[215] assumed that the periapical tissues would be as tolerant of CHX as gingival tissues. In two studies when CHX and NaOCl were injected into subcutaneous tissues of guinea pigs and rats, an inflammatory reaction developed; however, the toxic reaction from CHX was less than that of NaOCl.[306] Furthermore, when CHX was applied as a rinse in the extraction sites of third molars on the day of surgery and several days after, it was reported to reduce the incidence of alveolar osteitis.[80] Few reports of allergic and anaphylactic reactions to CHX exist in the literature.[146,305]

Conversely, some studies have reported unfavorable effects of CHX on tissues. Hidalgo and Dominguez[197] demonstrated that CHX is cytotoxic to some lines of cultured human skin fibroblasts. Recently the behavior of osteoblastic human alveolar bone cells in the presence of CHX and povidone-iodine (PI) has been investigated. It has been reported that CHX has a higher cytotoxicity profile than povidone iodine.[78] Faria et al.[132] also demonstrated that CHX injected in the hind paw of mice could induce severe toxic reactions. They reported that CHX induced apoptosis at lower concentrations and necrosis at higher concentrations when added to cultured L929 fibroblast cells.

A byproduct of combining CHX with NaOCl is the formation of toxic products such as parachloroaniline (PCA) that may harm tissues.[42] The toxicity level of CHX on periapical tissues when applied in the root canals, especially with other irrigants, merits further investigation.

Chlorhexidine as an Endodontic Irrigant. CHX has been extensively studied as an endodontic irrigant and intracanal medication, both in vivo[33,255,268,316,494] and in vitro.[40,41,215,233,248,412,443,472] The antibacterial efficacy of CHX as an irrigant is concentration dependent. It has been demonstrated that 2% CHX has better antibacterial efficacy than 0.12% CHX in vitro.[41] When compared with the effectiveness of NaOCl, controversial results can be found. NaOCl has an obvious advantage over CHX with the dissolution capacity of organic matter that CHX lacks; therefore, even though in vitro studies suggest some advantages with the use of CHX, as soon as organic and dental tissue is added, NaOCl is clearly preferable.

When studied in vivo, investigators[359] reported that 2.5% NaOCl was significantly more effective than 0.2% CHX when the infected root canals were irrigated for 30 minutes with either of the solutions. In a controlled and randomized clinical trial, the efficacy of 2% CHX liquid was tested against saline using culture technique. Their results showed better disinfection of the root canals using CHX compared with saline as a final rinse.[503]

In a recent study, the antibacterial efficacy of 2% CHX gel was tested against 2.5% NaOCl in teeth with apical periodontitis, with the bacterial load assessed using real-time quantitative polymerase chain reaction (RTQ-PCR) and colony forming units (CFU). The bacterial reduction in the NaOCl group was significantly higher than the CHX group when measured by RTQ-PCR. Based on culture technique, bacterial growth was detected in 50% of the CHX-group cases compared with 25% in the NaOCl group.[472] On the other hand, another study based on this culture technique revealed no significant difference between the antibacterial efficacy of 2.5% NaOCl and 0.12% CHX liquid when used as irrigants during the treatment of infected canals.[414]

In a systematic review, Ng and colleagues[301] demonstrated that abstaining from using 2% CHX as an adjunct irrigant to NaOCl was associated with superior periapical healing. In summary, CHX lacks a tissue-dissolving property; therefore, NaOCl is still considered the primary irrigating solution in endodontics.

Interaction Between CHX, NaOCl, and EDTA. NaOCl and CHX when in contact, produce a change of color and a precipitate (Fig. 8.52). The reaction is dependent on the concentration of NaOCl. The higher the concentration of NaOCl, the larger the precipitate is if 2% CHX is used.[42]

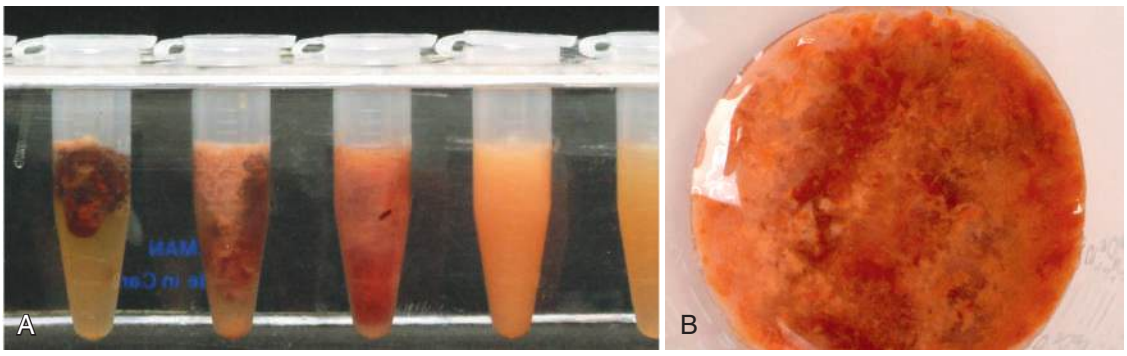


Fig. 8.52 Red precipitate forming after contact between NaOCl and chlorhexidine. **A,** When 2% chlorhexidine (CHX) is mixed with different concentration of NaOCl, a change of color and precipitate occurs. The higher the concentration of NaOCl, the larger is the precipitate formation. **B,** Detail of the interaction between 2% CHX and 5% NaOCl.

Furthermore, concerns have been raised that the color change may have some clinical relevance because of staining, and the resulting precipitate might interfere with the seal of the root obturation. Basrani et al.[42] evaluated the chemical nature of this precipitate and reported the formation of 4-chloroaniline (PCA). Furthermore, a recent study[232] using a specific mass spectrometry analysis (TOF-SIMS), demonstrated the penetration of PCA inside dentinal tubules. PCA is toxic in humans with short-term exposure, resulting in cyanosis, which is a manifestation of methemoglobin formation. The combination of NaOCl and CHX causes color changes and formation of a possibly toxic insoluble precipitate that may interfere with the seal of the root obturation. Alternatively, the canal can be dried using paper points before the final CHX rinse.[504]

The combination of CHX and EDTA produces a white precipitate, so a group of investigators[351] did a study to determine whether the precipitate involves the chemical degradation of CHX. The precipitate was produced and redissolved in a known amount of dilute trifluoroacetic acid. Based on the results, CHX was found to form a salt with EDTA rather than undergoing a chemical reaction.

Allergic Reactions to Chlorhexidine. Allergic responses to CHX are rare and there are no reports of reactions following root canal irrigation with CHX.[209] The sensitization rate has been reported in several studies to be approximately 2%.[236] However, some allergic reactions such as anaphylaxis, contact dermatitis, and urticaria have been reported following direct contact to mucosal tissue or open wounds.[342,391,472] Patients with history of atypical reaction to any irrigant should undergo a basic immunologic evaluation before CHX or NaOCl are used. If any of these tests is positive, alternative substance should be use.[104]

Decalcifying Agents. Debris is defined as dentin chips or residual vital or necrotic pulp tissue attached to the root canal wall. Smear layer was defined by the American Association of Endodontists in 2003 as a surface film of debris retained on dentin or other surfaces after instrumentation with either rotary instruments or endodontic files; it consists of dentin particles, remnants of the vital or necrotic pulp tissue, bacterial components, and retained irrigants. While it has been viewed as an impediment to irrigant penetration into dentinal tubules (Fig. 8.53), there is still a controversy about the influence of smear layer on the outcome of endodontic treatment. Some researchers emphasize the importance of removing the smear layer to allow irrigants, medications, and sealers to penetrate dentinal tubules and improve disinfection. On the other hand, other researchers focused on keeping the smear layer as protection for bacterial invasion, apical and coronal microleakage, bacterial penetration of the tubules, and the adaptation of root canal materials. The majority of the conclusions on the smear layer are based on in vitro studies. A recent clinical study by Ng et al.[301] found that the use of EDTA significantly increased the odds of success of retreatment.

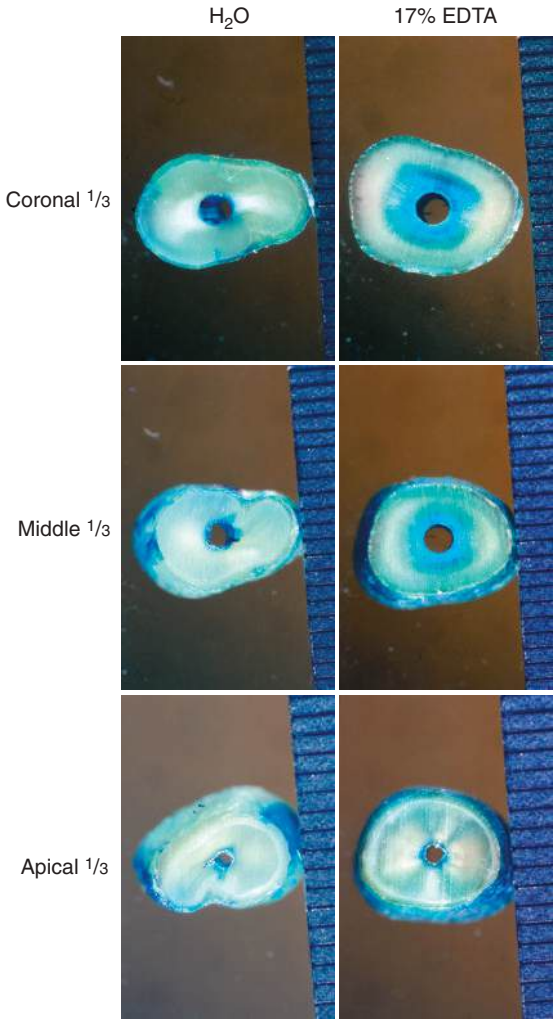


Fig. 8.53 Penetration of irrigants into dentinal tubules after root canal preparation with different dentin pretreatments. *Left column,* Irrigation with tap water and then with blue dye. *Right column,* Smear layer is removed with 17% EDTA, applied in high volume and with a 30-gauge needle, followed by irrigation with blue dye. Note the comparable diffusion of dye in the apical sections, whereas dye penetrated deeper into the dentin in the two coronal sections.

Ethylenediaminetetraacetic Acid

Ethylenediaminetetraacetic acid, widely abbreviated as EDTA, is a polycarboxylic amino acid, and a colorless, water-soluble solid EDTA is often suggested as an irrigation solution because it can chelate and remove the mineralized portion of the smear layer (Fig. 8.54). It is a polyaminocarboxylic acid with the formula $[CH_2N(CH_2CO_2H)_2]_2$. Its prominence as a chelating agent arises from its ability to sequester di- and tricationic metal ions such as Ca^{2+} and Fe^{3+}. After being bound by EDTA, metal ions remain in solution but exhibit diminished reactivity.[433]

History. The compound was first described in 1935 by Ferdinand Munz, who prepared it from ethylenediamine and chloroacetic acid. Chelating agents were introduced into endodontics as an aid for the preparation of narrow and calcified root canals in 1957 by Nygaard-Østby.[206] Today, EDTA is mainly synthesized from ethylenediamine (1,2-diaminoethane), formaldehyde (methanal), and sodium cyanide.[206]

Mode of Action. On direct exposure for an extended time, EDTA extracts bacterial surface proteins by combining with

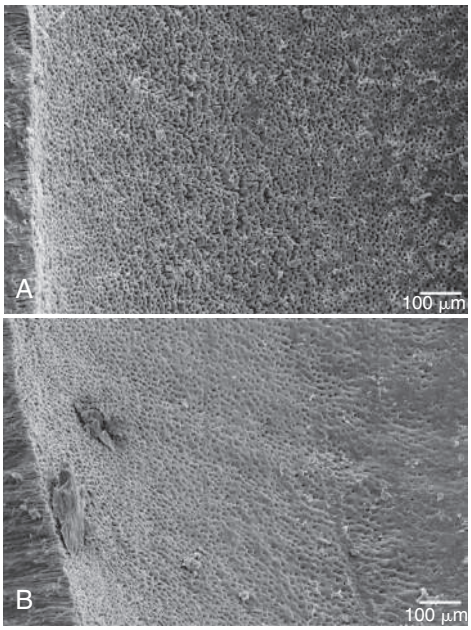


Fig. 8.54 Example of canals with minimal smear layer. **A,** Middle third after irrigation with 17% ethylenediaminetetraacetic acid (EDTA) and 2.5% sodium hypochlorite (NaOCl). **B,** Apical third with some particulate debris.

metal ions from the cell envelope, which can eventually lead to bacterial death.[206] Chelators such as EDTA form a stable complex with calcium. When all available ions have been bound, equilibrium is formed and no further dissolution takes place; therefore, EDTA is self-limiting.[206]

Applications in Endodontics. EDTA alone usually cannot remove the smear layer effectively; a proteolytic component, such as NaOCl, must be added to remove the organic components of the smear layer.[158] For root canal preparation, EDTA has limited value alone as an irrigation fluid.[158] EDTA is generally used in a concentration of 17% and can remove the smear layers when in direct contact with the root canal wall for less than 1 minute. Even though EDTA has self-limited action, if it is left in the canal for longer or NaOCl is used after EDTA, erosion of dentin has been demonstrated (Fig. 8.55).

Although citric acid appears to be slightly more potent at similar concentration than EDTA, both agents show high efficiency in removing the smear layer. In addition to their cleaning ability, chelators may detach biofilms adhering to root canal walls.[168] This may explain why an EDTA irrigant proved to be highly superior to saline in reducing intracanal microbiota, even though its antiseptic capacity is relatively limited.[168] Antiseptics such as quaternary ammonium compounds (EDTAC) or tetracycline antibiotics (MTAD; see irrigants combination later) have been added to EDTA and citric acid irrigants, respectively, to increase their antimicrobial capacity. The clinical value of this, however, is questionable. EDTAC shows similar smear-removing efficacy as EDTA, but it is more caustic.

The effect of chelators in negotiating narrow, tortuous, calcified canals to establish patency depends on both canal width and the amount of active substance available since the demineralization process continues until all chelators

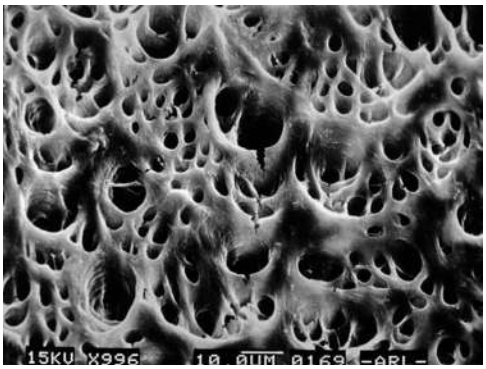


Fig. 8.55 Scanning electron microscope (SEM) image of root canal dentin exposed to 2 minutes of EDTA.

have formed complexes with calcium.[206,506] Therefore, studies should be read with caution because one study showed demineralization up to a depth of 50 µm into dentin,[210] but other reports demonstrated significant erosion after irrigation with EDTA.[44]

A comparison of bacterial growth inhibition showed that the antibacterial effects of EDTA were stronger than citric acid and 0.5% NaOCl but weaker than 2.5% NaOCl and 0.2% CHX.[410] EDTA had a significantly better antimicrobial effect than saline solution. It exerts its strongest effect when used synergistically with NaOCl, although no disinfecting effect on colonized dentin could be demonstrated.[192]

Interaction of EDTA and NaOCl. Investigators[164] studied the interactions of EDTA with NaOCl. They concluded that EDTA retained its calcium-complex ability when mixed with NaOCl, but EDTA caused NaOCl to lose its tissue-dissolving capacity, with virtually no free chlorine detected in the combinations. Clinically, this suggests that EDTA and NaOCl should be used separately. In an alternating irrigating regimen, copious amounts of NaOCl should be administered to wash out remnants of the EDTA. In modern endodontics, EDTA is used once the cleaning and shaping are completed for around 1 minute. It can be activated with ultrasonic activation for better penetration in dentinal tubules. It should be taken into consideration that raising the temperature of EDTA is not desirable, as chelators have a temperature range that they work their best. When EDTA is heated from 20 to 90 degrees, the calcium binding capacity decreases.[505]

HBPT and Other Phosphate Bases Tensides

HEBP (1-hydroxyethylidene-1, 1-bisphosphonate; also called etidronic acid) is a weak chelator.[504] It is a potential alternative to EDTA because it has no short-term reactivity with NaOCl. It can be used in combination with NaOCl without affecting its proteolytic or antimicrobial properties.[504] It is nontoxic[26,111] and it is used in medicine to treat bone diseases.[314]

Combination Irrigants. Irrigants with low surface tension are assumed to penetrate better in dentinal tubules and anatomic irregularities. Detergents (e.g., Tween 80) are added to irrigants to lower their surface tension. However, Boutsioukis and Kishen[58] did not find support

for this rationale, perhaps because the effect of surface tension is important only at the interface between two immiscible fluids (e.g., between irrigant and an air bubble, but not between irrigant and dentinal fluid). Recent studies have also confirmed that surfactants do not enhance the ability of NaOCl to dissolve pulp tissue[221] or the efficacy of common chelators to remove calcium or smear layer.[58,111,506]

BioPure MTAD and Tetraclean. Two other irrigants, based on a mixture of antibiotics, citric acid, and a detergent, are commercially available. They are capable of removing both the smear layer and organic tissue from the infected the root canal system.[454] MTAD (Fig. 8.56), introduced by Torabinejad and Johnson[455] at Loma Linda University in 2003, is an aqueous solution of 3% doxycycline, a broad-spectrum antibiotic; 4.25% citric acid, a demineralizing agent; and 0.5% polysorbate 80 detergent (Tween 80).[455] It is mixed from a liquid and powder before use. MTAD has been recommended in clinical practice as a final rinse after completion of conventional chemomechanical preparation.[44]

Tetraclean (Ogna Laboratori Farmaceutici, Muggio, Italy) is a combination product similar to MTAD. The two irrigants differ in the concentration of antibiotics (doxycycline 150 mg/5 mL for MTAD and 50 mg/5 mL for Tetraclean) and the kind of detergent (Tween 80 for MTAD, polypropylene glycol for Tetraclean).

Mode of Action. All of the tetracyclines are derivatives of the four-ringed nucleus that differ structurally regarding the chemical groups at 2, 5, 6, 7 positions. These derivatives exhibit different characteristics, such as absorption, protein binding, metabolism, excretion, and degree of activity against the susceptible organism.[191] Tetracyclines inhibit protein synthesis by reversibly binding to the 30S subunit of bacterial ribosome in susceptible bacteria. It is effective against *Aggregatibacter actinomycetemcomitans* (A.a), *Porphyromonas gingivalis*, and *Prevotella intermedia* and affects both gram-positive and gram-negative bacteria, with a more pronounced gram-negative effect. Tetracycline is a bacteriostatic antibiotic, but in high concentrations tetracycline may also be bactericidal. Doxycycline, citric acid, and Tween 80 together may have a synergistic effect on the disruption of the bacterial cell wall and the cytoplasmic membrane.



Fig. 8.56 Container with BioPure MTAD. (Courtesy Dentsply Tulsa Dental Specialties, Tulsa, OK, USA.)

Smear Layer Removal. In two studies, the efficacy of MTAD or EDTA in the removal of the smear layer was confirmed, but no significant difference between these two solutions was reported.[444,445]

Antibacterial Efficacy. Earlier independent in vitro research on MTAD showed its antimicrobial efficacy over conventional irrigants.[106,235] Torabinejad et al.[455] found that MTAD was effective in killing *E. faecalis* up to a 200× dilution. Shabahang and Torabinejad[396] showed that the combination of 1.3% NaOCl as a root canal irrigant and MTAD as a final rinse was significantly more effective against *E. faecalis* than the other regimens.[397] A study using extracted human teeth contaminated with saliva showed that MTAD was more effective than 5.25% NaOCl in disinfection of the teeth. In contrast to the previously mentioned studies, later research suggested less than the optimal antimicrobial activity of MTAD.[152,219,225] Krause et al.,[235] using bovine tooth sections, showed that 5.25% NaOCl was more effective than MTAD in disinfection of dentin discs inoculated with *E. faecalis*.[396,411]

Clinical Trials. Malkhassian et al.[267] in a controlled clinical trial of 30 patients reported that the final rinse with MTAD did not reduce the bacterial counts in infected canals beyond levels achieved by chemomechanical preparation using NaOCl alone.

Protocol for Use. MTAD was developed as a final rinse to disinfect the root canal system and remove the smear layer. The effectiveness of MTAD to altogether remove the smear layer is enhanced when a low concentration of NaOCl (1.3%) is used as an intracanal irrigant before placing 1 mL of MTAD in a canal for 5 minutes and rinsing it with an additional 4 mL of MTAD as the final rinse.[396]

Combinations of Irrigants

QMiX was introduced in 2011 as a combination product introduced for root canal irrigation (Fig. 8.57). It is recommended to be used at the end of instrumentation, after NaOCl irrigation. According to the patent,[175] QMix contains a CHX-analog, Triclosan (N-cetyl-N,N, N-trimethylammonium bromide), and EDTA as a decalcifying agent; it



Fig. 8.57 QMiX 2in1. Combination of CXH, EDTA, and detergent. QMix irrigating solution is a single solution used as a final rinse after bleach for one-step smear layer removal and disinfection (Courtesy Dentsply Tulsa Dental Specialties, Tulsa, OK, USA).

is intended as an antimicrobial irrigant as well as the removal of canal wall smear layers and debris.

Protocol. QMiX is suggested as a final rinse. If sodium hypochlorite was used throughout the cleaning and shaping, saline could rinse out NaOCl remnants to prevent the formation of any precipitate.

Smear Layer Removal. Stojic et al.[438] investigated the effectiveness of smear layer removal by QMiX using scanning electron microscopy. QMiX removed smear layer equally well as EDTA. Dai et al.[102] examined the ability of two pH versions of QMiX on the removal of canal wall smear layers and debris using an open canal design. Within the limitations of an open canal design, the two experimental QMiX versions are as useful as 17% EDTA in removing canal wall smear layers after the use of 5.25% NaOCl as the initial rinse.

Antibacterial Efficacy and Effect on Biofilms. Stojic et al.[438] also assessed the in vitro efficacy of QMiX against *E. faecalis* and mixed plaque bacteria, both in the planktonic phase and biofilms. QMiX and 1% NaOCl killed all planktonic *E. faecalis* and plaque bacteria in 5 seconds. QMiX and 2% NaOCl killed up to 12 times more biofilm bacteria than 1% NaOCl ($P < .01$) or 2% CHX ($P < .05$; $P < .001$). Wang et al.[480] compared the antibacterial effects of different disinfecting solutions on young and old *E. faecalis* biofilms in dentin canals using a novel dentin infection model and confocal laser scanning microscopy. Six percent NaOCl and QMiX were the most effective disinfecting solutions against the young biofilm, whereas, against the 3-week-old biofilm, 6% NaOCl was the most effective, followed by QMiX. Both were more effective than 2% NaOCl and 2% CHX. Morgental et al.[291] showed that QMiX was less effective than 6% NaOCl and similar to 1% NaOCl in bactericidal action. According to their in vitro study, it appears that the presence of dentin slurry has the potential to inhibit most current antimicrobials in the root canals system.

Moreover, Ordinola et al.[307] found that several endodontic irrigants containing antimicrobial compounds such as chlorhexidine (QMiX), cetrimide, maleic acid, iodine compounds, or antibiotics (MTAD) lacked an effective antibiofilm activity when the dentin was infected intraorally. The irrigant solutions 4% peracetic acid and 2.5% to 5.25% sodium hypochlorite decreased the number of live bacteria significantly in biofilms, providing also cleaner dentin surfaces ($P < .05$). They concluded that several chelating agents containing antimicrobials could not remove nor kill significantly biofilms developed on intraorally infected dentin, except for sodium hypochlorite and 4% peracetic acid. Dissolution ability is mandatory for an appropriate eradication of biofilms attached to dentin.

Clinical Trials. The efficacy and biocompatibility of QMiX were as yet only demonstrated via nonclinical in vitro and ex vivo studies. Further clinical research from independent investigators is needed to corroborate the findings.

Iodine Potassium Iodide. IKI is a root canal disinfectant that is used in concentrations ranging from 2% to 5%. IKI kills a broad spectrum of microorganisms found in root canals but shows relatively low toxicity in experiments using tissue cultures.[430] Iodine acts as an oxidizing agent by reacting with free sulfhydryl groups of bacterial enzymes, cleaving disulphide bonds. *E. faecalis* often is associated with therapy-resistant periapical infections (see Chapter 16), and combinations of IKI and CHX may be able to kill $Ca(OH)_2$-resistant bacteria more efficiently. One study[416] evaluated the antibacterial activity of a combination of $Ca(OH)_2$ with IKI or CHX in infected bovine dentin blocks. Although $Ca(OH)_2$ alone was unable to destroy *E. faecalis* inside dentinal tubules, $Ca(OH)_2$ mixed with either IKI or CHX effectively disinfected dentin. Others[31] demonstrated that IKI was able to eliminate *E. faecalis* from bovine root dentin when used with 15-minute contact time. A distinct disadvantage of iodine is a possible allergic reaction in some patients. Although iodine is not generally considered an allergen, some patients are hypersensitive to this compound and may be considered to have an iodine "allergy."

INTRACANAL MEDICATION

Intracanal medication is used when treatment cannot be completed in one appointment (see Chapters 5 and 15). The surviving intracanal bacteria after cleaning and shaping may increase between appointments.[76,477] The main objectives for the intracanal medication are restrict bacterial regrowth, supply continued disinfection, and create a physical barrier.

Calcium Hydroxide

Calcium hydroxide is the most popular intracanal medication use. It was introduced by Hermann[194] in 1920. Although its use was well documented for its time, evidence of its efficacy in clinical endodontics is controversial. A series of articles[76,77] promoted the antibacterial efficacy of $Ca(OH)_2$ in human root canals. Subsequent studies substantiated these reports,[308,420] and the routine use of $Ca(OH)_2$ as an interappointment intracanal medicament became widespread.[373]

However, newer clinical studies and systematic reviews failed to show a clear benefit of $Ca(OH)_2$ to further eliminate bacteria from the root canal.[325,373] $Ca(OH)_2$ mostly is used as a slurry of $Ca(OH)_2$ in a water base; at body temperature, less than 0.2% of the $Ca(OH)_2$ is dissolved into $Ca++$ and OH ions. Because $Ca(OH)_2$ needs water to dissolve, water should be used as the vehicle for the $Ca(OH)_2$ paste. In contact with air, $Ca(OH)_2$ forms calcium carbonate ($CaCO_3$). However, this is a lengthy process and of little clinical significance. $Ca(OH)_2$ paste, with a significant amount of calcium carbonate, feels granular because the carbonate has a very low solubility.[367,420]

$Ca(OH)_2$ is a slow-acting antiseptic; direct-contact experiments in vitro show that a 24-hour contact period is required for complete killing of enterococci.[367,420] Another study of 42 patients found that NaOCl canal irrigation reduced the bacteria level by only 61.9%, but use of $Ca(OH)_2$ in the canals for 1 week resulted in a 92.5% reduction.[403] These researchers concluded that $Ca(OH)_2$ should be used in infected cases to more predictably obtain débridement and decontamination.

In addition to killing bacteria, $Ca(OH)_2$ has the useful ability to hydrolyze the lipid moiety of bacterial lipopolysaccharides (LPS), thereby inactivating the biologic activity of

the lipopolysaccharide and reducing its effect.[368] This is a desirable effect because dead cell wall material remains after the bacteria have been killed and can continue to stimulate inflammatory responses in the periradicular tissue.

The main characteristics of $Ca(OH)_2$ include limited solubility, high pH, broad-spectrum antimicrobial agent, and antimicrobial action sustained for long periods.

Other Uses of $Ca(OH)_2$. Long-term calcium hydroxide treatment can be used to induce apexification of the immature tooth with pulpal necrosis before placing an obturation material such as gutta-percha in the root canal system.[137] Also, in cases where revascularization/guided pulpal repair is desired, $Ca(OH)_2$ may be used instead of antibiotic pastes (for more details, see later and Chapter 12).

Clinical Protocol. Calcium hydroxide should be in contact with the tissue to act. $Ca(OH)_2$ powder may be mixed with sterile water or saline and placed into the canal using a lentulo paste filler.[324] Alternatively, the mix may be applied from sterile, single-dose packages (e.g., Calasept [J.S. Dental, Ridgefield, CT, USA]; Calcijet [Centrix, Shelton, CT, USA]; and DT Temporary Dressing [Global Dental Products, North Bellmore, NY, USA]; Fig. 8.58). The mixture should be thick to carry as many $Ca(OH)_2$ particles as possible; however, it should not be overdried to retain enough moisture and to promote continued dissociation with a resulting high pH. For maximum effectiveness, the root canal should be filled homogeneously to the WL with a slurry, and a radiograph should be taken to confirm this.

Accidents with $Ca(OH)_2$ placement have been reported and the dentist should be aware of them, and because it is a strong base that may cause irreversible injury to vital tissue that comes into contact with this substance, specifically nerve damage may occur. Therefore, care must be taken not to extrude $Ca(OH)_2$ through the apex, as this carries the risk of reaching the maxillary sinus or the mandibular canal. Calcium hydroxide can also cause blindness when it comes into contact with the eye. Clinicians should take adequate precautions to prevent this serious complication.[256]

Limitations of Calcium Hydroxide. There are some concerns regarding the use of $Ca(OH)_2$. The handling and proper placement of $Ca(OH)_2$ present a challenge to the average clinician.[417] Moreover, the removal of $Ca(OH)_2$ is frequently incomplete, resulting in a residue covering 20% to 45% of the canal wall surfaces, even after copious irrigation with saline, NaOCl, or EDTA.[242] Residual $Ca(OH)_2$ can shorten the setting time of zinc oxide eugenol–based endodontic sealers.[271] Notably, it may interfere with the seal of the root filling and compromise the quality of treatment. An additional concern is that $Ca(OH)_2$ is not effective against several endodontic pathogens, including *E. faecalis* and *C. albicans*.[478] In fact, in vitro studies have shown that dentin can inactivate the antibacterial activity of $Ca(OH)_2$,[174,347] and one clinical study[325] demonstrated that the number of bacteria-positive canals increased after $Ca(OH)_2$ medication. Other studies have further indicated that $Ca(OH)_2$ could not predictably eliminate bacteria or that cultures changed from negative to positive after $Ca(OH)_2$ placement.[325,477]

When different studies report inconsistent results, a systematic review and meta-analysis technique can clarify conflicting research data and the current state of knowledge regarding specific issues. Therefore, based on the current best available evidence, $Ca(OH)_2$ has limited effectiveness in eliminating bacteria from human root canals when assessed by culture techniques. The quest for better antibacterial protocols and sampling techniques must continue to ensure that bacteria can be reliably eradicated before obturation.

Chlorhexidine

In vitro, CHX used as intracanal medicament has at least as good or even better antimicrobial efficacy than $Ca(OH)_2$.[411] Notably, 2% CHX was very effective in eliminating a biofilm of *E. faecalis*.[253] In vivo, it inhibits experimentally induced inflammatory external root resorption when applied for 4 weeks.[255] In infected root canals, it reduces bacteria as effectively as $Ca(OH)_2$ when applied for 1 week.[33] Unlike $Ca(OH)_2$, CHX has a substantive antimicrobial activity that, if imparted onto the root dentin, has the potential to prevent bacterial colonization of root canal walls for prolonged periods.[215,233] This effect depends on the concentration of CHX, but not on its mode of application, which may be as a liquid, gel, or controlled-release device.[40]

CHX as intracanal medication has been the focus of many in vitro[36,114,233,247,411] and in vivo studies.[33,255,268,316,509] The results of in vitro experiments were mostly in

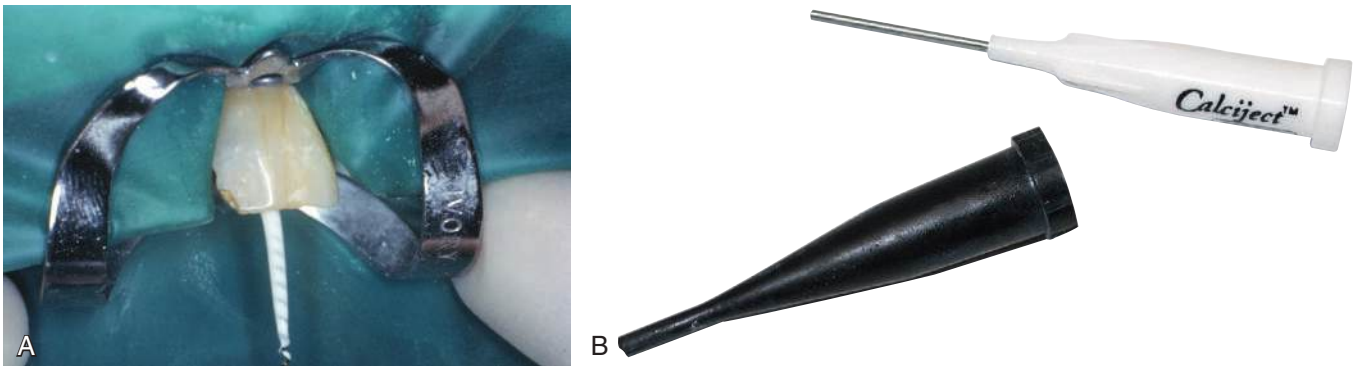


Fig. 8.58 A, Application of calcium hydroxide paste in the canal with a lentulo spiral. **B,** Calcijet is a calcium hydroxide prefilled, easy-to-use, single-dose syringe system. Centrix NeedleTube cartridges can be used for direct syringe injection into the root canal. (*A,* Courtesy Dr. S. Friedman. *B,* Courtesy Centrix, Shelton, CT, USA.)

favor of CHX regardless of its mode of application. In vivo, variable results were reported; one of the reasons for this is that researchers[174] developed an experimental model using dentin powder particles to investigate the possible inactivation of some antibacterial medicaments when they come in contact with dentin. They showed that dentin powder had inhibitory effects on all medicaments tested. The effect was dependent on the concentration of the medicament and the duration of contact. The effect of $Ca(OH)_2$ was abolished by the presence of dentin powder. The effect of 0.05% CHX and 1% NaOCl was reduced but not eliminated by the presence of dentin.

In vitro and animal studies suggest that CHX has the potential to replace $Ca(OH)_2$ as an intracanal medicament, but the listed limitations of the in vitro experiments make the conclusions obtained challenging to extrapolate to actual clinical situations. In vivo, human studies are therefore required to test the efficacy of CHX as an intracanal medication.

Compared with the number of in vitro studies, very few clinical studies have been conducted to assess the effectiveness of CHX as an intracanal medication. An in vivo investigation[33] assessed the antibacterial efficacy of three different intracanal medications: camphorated paramonochlorophenol, $Ca(OH)_2$, and 0.12% CHX liquid by applying them for 1 week in single-rooted teeth of patients. Using a culture method, it was reported that the proportions of positive cultures were not significantly different among the tested medications, but they were slightly lower in teeth medicated with CHX (0.12%) liquid than those medicated with camphorated paramonochlorophenol or $Ca(OH)_2$.[33]

Another in vivo study evaluated the antibacterial effectiveness of 2% CHX liquid as an intracanal medication in teeth with apical periodontitis.[316] The results showed a moderate increase in bacterial counts during a medication period of 7 to 14 days that was similar to outcomes seen and reported for $Ca(OH)_2$ by Peters et al.[325] It was speculated that the CHX liquid may have partially escaped from the apical foramen and that the higher-viscosity gel form might have been better suited as an intracanal medication.[316]

However, a different study[268] demonstrated that intracanal medication with $Ca(OH)_2$, 2% CHX gel, or a mixture of $Ca(OH)_2$/CHX applied for 7 days did not reduce the bacterial concentration beyond what was achieved after chemomechanical preparation using 1% NaOCl. The results were not significantly different among the medication groups. Similar results were found by other investigators,[267] when after a randomized controlled trial of 30 patients, they concluded that a final rinse with MTAD (a *m*ixture of *t*etracycline, *a*cid, and *d*etergent) and intracanal application of 2% CHX gel did not reduce bacterial counts beyond levels achieved by chemomechanical preparation using NaOCl.

Chlorhexidine Mixed With Calcium Hydroxide. During the last few years, researchers have studied the combination of $Ca(OH)_2$ and CHX, with the concept that their antimicrobial properties interact in a synergistic fashion that enhances their efficacy. The high pH of $Ca(OH)_2$ was unaffected when combined with CHX.[42] However, the results have not been conclusive. Some in vitro studies had reported an improved antibacterial action when both agents were combined,[41,128,504] while other studies reported contradictory results.[181]

Recent animal studies have evaluated the tissue reactions to the mix of $Ca(OH)_2$/CHX, showing that the combination exerts excellent antimicrobial properties[421] and improves healing of the periapical tissues. In vivo studies have shown that the mix is at least as good as both agents applied separately in necrotic teeth with apical periodontitis,[268] as well as in previously treated cases with persistent apical periodontitis. A more recent study utilizing a CHX-based protocol of 0.12% CHX as an irrigant followed by a 7-day intracanal medication of $Ca(OH)_2$/0.12% CHX has shown promising results. The authors concluded that chemomechanical canal shaping using 0.12% CHX solution as an irrigant significantly reduced the number of intracanal bacteria, but it failed to render the canals bacteria free. Further intracanal medication with a $Ca(OH)_2$/CHX paste significantly improved the results by reducing the number of bacteria.[408] Therefore, it seems that the usefulness of mixing $Ca(OH)_2$ with CHX remains controversial.

Phenolic Preparations

Phenol (C_6H_5OH) and phenolic preparation used to be very commonly used intracanal medicament in endodontics. It was thought that because of their volatile properties, it could penetrate dentinal tubules and anatomical irregularities. However, it was later demonstrated that these compounds have a short life and their volatility can diffuse through the temporary fillings and also through the periapical tissue causing toxicity. Despite the severe toxicity of phenolic preparations, derivatives of phenol, such as paramonochlorophenol (C_6H_4OHCl), thymol ($C_6H_3OHCH_3C_3H_7$), and cresol ($C_6H_4OHCH_3$), remain available. Currently, $Ca(OH)_2$ or no medication is preferred.[147] Phenol is a nonspecific protoplasm poison that has an optimal antibacterial effect at 1% to 2%. Many dental preparations use much too high a concentration of phenol (e.g., in the range of 30%).[147] At such a concentration, the antimicrobial effect in vivo is lower than optimal and of short duration.[279] Derivatives of phenol are stronger antiseptics and more toxic than phenol. Phenolic compounds are available as camphorated solutions.[430] Camphoration results in a less toxic phenolic compound because it slows the release of toxins to the surrounding tissues. Studies in vitro have shown that phenol and phenol derivatives are highly toxic to mammalian cells, and their antimicrobial effectiveness does not sufficiently balance their toxicity.[430] Phenols are ineffective antiseptics under clinical conditions.[76]

Formaldehyde

Formaldehyde, used as formocresol, is highly toxic, mutagenic, and carcinogenic; however, historically it has been used extensively in endodontic therapy.[249] These formulations are still being recommended in pediatric dentistry when treating deciduous teeth. The formaldehyde component of formocresol may vary substantially between 19% and 37%. Tricresol formalin, another formaldehyde preparation, contains 10% tricresol and 90% formaldehyde.[249] All of these preparations have formaldehyde content well above the 10% generally used for fixation of pathologic specimens. Formaldehyde is volatile and releases antimicrobial vapors when applied to a cotton pellet for pulp chamber disinfection. All formaldehyde preparations are potent toxins with antimicrobial effectiveness much lower than their

toxicity.[431,432] There is no clinical reason to use formocresol as an antimicrobial agent for endodontic treatment, based on what is known at this time. The alternatives are better antiseptics with significantly lower toxicity.[431,432]

Halogens

Chlorinated solutions have been used for many years to irrigate root canals. They are also used as intracanal dressings in the form of chloramine-T, an N-chloro-tosylamide sodium salt. Iodine, in the form of IKI, is a very effective antiseptic solution with low tissue toxicity. IKI is an effective disinfectant for infected dentin and can kill bacteria in infected dentin in 5 minutes in vitro.[367] IKI releases vapors with a durable antimicrobial effect. The solution can be prepared by mixing 2 g of iodine in 4 g of potassium iodide; this mixture then is dissolved in 94 mL of distilled water. Tincture of iodine (5%) has proved to be one of the few reliable agents for disinfection of rubber dam and tooth surfaces during the preparation of an aseptic endodontic work-field.[288]

Steroids

Steroids have been used locally, within the root canal system to reduce pain and inflammation. Ledermix (Lederle Pharmaceuticals, Wolfratshausen, Germany) is a commercially available product that was developed about 1960 by Prof. André Schroeder.[390] The active ingredients are the potent anti-inflammatory corticoid triamcinolone acetonide in combination with the broad-spectrum antibiotic demeclocycline. It is an intracanal medicament paste popularly used in some countries. Ledermix paste has been advocated as an initial dressing, mainly if the patient presents with endodontic symptoms.[390] Ledermix paste contains triamcinolone acetonide as an anti-inflammatory agent, at a concentration of 1%.[229] The clinical effect is a rapid relief of pain associated with acute inflammatory conditions of the pulp and periodontium.

Ledermix paste is a nonsetting, water-soluble paste material for use as root canal medicament or as a direct or indirect pulp-capping agent. The mechanisms of action of this substance are based on inhibition of the ribosomal protein synthesis in the bacteria. The release and dentin diffusion characteristics of triamcinolone from Ledermix paste, when used as a root canal medicament, have been investigated under different conditions.[5,6] Collectively, these studies show that triamcinolone is released from Ledermix paste in the root canal and can reach the systemic circulation via diffusion through dentinal tubules, lateral canals, and the apical foramen. Also, because of its root resorption inhibition property, it was tested for replanted teeth in dogs. The results showed that the groups treated with Ledermix, triamcinolone, and demeclocycline had significantly more favorable healing and more remaining root structure than the group filled with gutta-percha and sealer (positive control).[85] The use of the steroid triamcinolone in combination with a zinc oxide eugenol paste has recently been popularized with the introduction of Odontopaste (Australian Dental Manufacturing, Brisbane, Australia); however, data regarding the clinical efficacy of this product are limited.

Triple-Antibiotic Paste

Antibiotic combinations have been studied over the past few years as a regimen during regenerative endodontic strategies. The literature is inconsistent on the effectiveness of double and triple antibiotic pastes (DAP and TAP), respectively, against mono- and multispecies biofilms. The triple-antibiotic regimen, composed of metronidazole, ciprofloxacin, and minocycline, was first tested[374] for its effectiveness against *Escherichia coli*–infected dentin in vitro. The same research group also tested its bactericidal efficacy against microbes from carious dentin and infected pulp. They found that the mixture of antibiotics is sufficiently potent to eradicate the bacteria.[448] The clinical effectiveness of the triple-antibiotic paste in the disinfection of immature teeth with apical periodontitis has been reported.[488] One potential concern of using an intracanal antibiotic paste is that it may cause bacterial resistance. In addition, intracanal use of minocycline can cause tooth discoloration, creating potential cosmetic complications. For this reason, a dual paste (metronidazole, ciprofloxacin) and, alternatively, abandonment of this protocol in favor of $Ca(OH)_2$ have been considered.[245] Another reason for such a change could be the reported high toxicity to stem cells of paste prepared from antibiotic powder,[364] and possible allergic reactions in patients, remain important concerns with this category of medicaments. With the current available evidence, it remains unclear if intracanal medicaments are effective against multispecies biofilms.[299]

Lubricants

In root canal treatment, lubricants are mostly used to emulsify and keep in suspension debris produced by mechanical instrumentation. Although irrigation solutions serve as lubricants for hand instrumentation, particular gel-type substances are also marketed. Two of these are wax-based RC-Prep, which contains EDTA and urea peroxide, and glycol-based Glyde. Another purported function of lubricants is to facilitate the mechanical action of endodontic hand or rotary files. A study evaluating the effects of lubrication on cutting efficiency found that tap water and 2.5% NaOCl solutions increased cutting efficiency compared with dry conditions.[501] The authors of this study cited the ability of an irrigant to remove debris as the factor for the increased efficiency. Similarly, a reduction of torque scores was found when canals in normed dentin disks were prepared with ProFile and ProTaper instruments under irrigation, but the use of a gel-type lubricant did not, resulting in similar torques as in dry, nonlubricated canals.[55,330]

In summary, irrigation is an essential step in root canal treatment to ensure disinfection. NaOCl is the irrigant of choice because of its tissue-dissolving and disinfecting properties. EDTA or other chelators should be used at the end of a procedure to remove the smear layer, followed by a final flush with NaOCl for 1 minute for maximum cleaning efficiency and to minimize dentin erosion. This strategy also minimizes the inactivation of NaOCl by chemical interactions.

DISINFECTION DEVICES AND TECHNIQUES

Syringe Delivery

Application of an irrigant into a canal using a syringe and needle allows exact placement, replenishing of existing fluid, rinsing out of larger debris particles, as well as

allowing direct contact to microorganisms in areas close to the needle tip. In passive syringe irrigation, the actual exchange of irrigant is restricted to 1 to 1.5 mm apical to the needle tip, with fluid dynamics taking place near the needle outlet.[59,504] Volume and speed of fluid flow are proportional to the cleansing efficiency inside a root canal.[59] Therefore, both the diameter and position of the needle outlet determine successful chemomechanical débridement; placement close to WL is required to guarantee fluid exchange at the apical portion of the canal, but close control of insertion depth is required to avoid extrusion.[59,201] Therefore, the choice of an appropriate irrigating needle is important. Although larger-gauge needles allow a quicker and larger amount of fluid exchange, the wider diameter does not allow cleaning of the apical and narrower areas of the root canal system (see the discussion of irrigation dynamics earlier in this chapter; see also Fig. 8.7). Excess pressure or binding of needles into canals during irrigation with no possibility of backflow of the irrigant should be avoided under all circumstances[205] to prevent extrusion into periapical spaces. In immature teeth with wide, apical foramina or when the apical constriction no longer exists, special care must be taken to prevent irrigation extrusion and potential accidents.[100]

There are different sizes and types of irrigation needles. The size of the irrigation needle[91] should be chosen depending on the canal size and taper.[79,281] Most root canals that have not been instrumented are too narrow to be reached effectively by disinfectants, even when appropriate irrigation needles are used (see Figs. 8.7 and 8.45). Therefore, effective cleaning of the root canal must include intermittent agitation of the canal content with a small endodontic instrument[276,466] to prevent debris from accumulating in general (see Fig. 8.43) and specifically at the apical portion of the root canal.

Preparation size[281] and to some extent taper[95] determine how closely a needle can be placed to the final apical millimeters of a root canal. Open-ended needles are recommended over the end open needles to prevent extrusion of the irrigant. Some needles and suction tips may be attached to the air/water syringe to increase both the speed of irrigant flow and the volume of irrigant. Examples include the Stropko Irrigator (Vista Dental Products), an adapter that connects to the air/water syringe and accepts standard Luer-lock needle tips for irrigant removal and application as well as air-drying.

Manually Activated Irrigation

Liquid placed inside the root canal more effectively reaches crevices and mechanically untouched areas if it is agitated inside the root canal. Corono-apical movements of the irrigation needle,[205] stirring movements with small endodontic instruments,[276,467] and manual push-pull movements using a fitted master gutta-percha cone have been recommended.[202]

Other than conventional irrigation, additional techniques for endodontic disinfection have been proposed and tested, including laser systems and gaseous ozone. Specific devices for endodontic irrigation and/or disinfection have been introduced, among which are the EndoActivator System (Dentsply Sirona), passive ultrasonic irrigation (PUI), the EndoVac (Discus, Culver City, CA, USA), the Safety-Irrigator (Vista Dental Products, Racine, WI, USA), the Self-adjusting File (see earlier), photoactivation disinfection, ozone generators, and others. These devices use pressure, vacuum, oscillation, and a combination of suction.

Sonically Activated Irrigation

The EndoActivator System uses safe, noncutting polymer tips in an easy-to-use subsonic handpiece to quickly and vigorously agitate irrigant solutions during endodontic therapy (Fig. 8.59).

In a study,[116] the safety of various intracanal irrigation systems was analyzed by measuring the apical extrusion of irrigant. The authors concluded that EndoActivator had a minimal statistically insignificant amount of irrigant extruded out of the apex in comparison with manual, ultrasonic, and Rinsendo (Dürr Dental, Bietigheim-Bissingen, Germany) groups.[116]

When cleanliness of the root canal walls was analyzed,[366] investigators suggested that both passive sonic or ultrasonic irrigation rendered root canals significantly cleaner than manual preparation in comparison with manual syringe irrigation.[74,465] When comparing sonic with ultrasonic irrigation, the results can be controversial. The majority of the studies favors ultrasonic irrigation.[217,366] The difference lies in the oscillating movements: sonic devices range between 1500 and 6000 Hz, and ultrasonic equipment requires vibrations greater than 20,000 Hz.[217,262] If the sonic device is left in the canal for extended periods, cleaning is improved. Sonic or ultrasonic irrigation may be carried out with activated smooth wires or plastic inserts, endodontic instruments, or activated irrigation needles. Examples include EndoSonor (Dentsply Maillefer) and EndoSoft ESI (EMS, Nyon, Switzerland) inserts, IrriSafe (Satelec, Acteon, Merignac, France), the EndoActivator System, and the Vibringe sonic syringe (Cavex, Haarlem, Netherlands). Inadvertent cavitation of root canal walls has not been observed with sonic activation of instruments.[276]

Ultrasonic-Assisted Irrigation

Ultrasonic devices were first introduced in endodontics by Richman.[356] The files are driven to oscillate at ultrasonic frequencies of 25 to 30 kHz to mechanically prepare the



Fig. 8.59 The EndoActivator, a sonic frequency system. (Courtesy Dentsply Tulsa Dental Specialties, Tulsa, OK, USA.)

root canal walls.[475] Recently, it has been shown that ultrasonically driven files are useful to activate the irrigation liquids inside the root canal system by inducing acoustic streaming and cavitation.

Two types of ultrasonic irrigation have been described in the literature[10,485]: one in which irrigation is combined with simultaneous ultrasonic instrumentation (UI) and another without simultaneous instrumentation, called *passive ultrasonic instrumentation* (PUI). During UI, the file is intentionally brought into contact with the root canal wall. However, because of the complex canal anatomy, UI never contacts the entire walls, and it may result in the uncontrolled cutting of the root canal walls without effective disinfection.[10] In a study by Macedo[265] instrument oscillation frequency, ultrasonic power, and file taper determined occurrence and extent of cavitation. Some degree of cavitation occurred between the file and canal surface and reached lateral canals and isthmuses.

PUI was first described by Weller et al.[485] The term "passive" is related to the noncutting action of the ultrasonically activated file.[10] PUI relies on the transmission of acoustic energy from an oscillating file or smooth wire to an irrigant in the root canal.

PUI should be introduced in the canal once the root canal system has a final apical size and taper. A fresh solution of irrigant should be introduced, and a small file or smooth wire (e.g., size #15) is ultrasonically activated. Since the root canal has already been shaped, the file or wire can move more freely,[467] and the irrigant can penetrate the apical part of the root canal system,[237] with the cleaning effect being more significant.[261] Using this noncutting approach, the potential to create aberrant shapes within the root canal is reduced to a minimum. A file larger than a #15 or #20 requires a wide root canal to reduce oscillation dampening by wall contact.

Ultrasonic activation of the irrigant seems to improve débridement of the root canal system in vitro; the results in vivo present some controversy. Therefore objective guidelines regarding their risks and benefits have not been ascertained.[251] Ultrasonic activation of the irrigant can be intermittent or continue. The ProUltra PiezoFlow (Dentsply Sirona) has been introduced to irrigate and activate the liquids at the same time. The device consists mainly of an ultrasonically energized needle connected to a reservoir of sodium hypochlorite (NaOCl). This continuous ultrasonic irrigation (CUI) system allows simultaneous continuous irrigant delivery and ultrasonic activation; unlike PUI, it does not require the intermittent replenishment of irrigant between ultrasonic file activations. Research shows better elimination of debris and better penetration of irrigant into dentinal tubules.[82,218]

Negative Apical Pressure Irrigation

Another approach to afford better access to irrigation solution to the apical portion of the canal is the so-called negative-pressure irrigation (Fig. 8.60). Here, irrigant is delivered into the access chamber, and a very fine needle connected to the dental unit's suction device is placed into the root canal. Excess irrigant from the access cavity is then transported apically and ultimately removed via suction. First, a macrocannula, equivalent to an ISO size #55, .02 taper instrument, removes coronal debris. Subsequently, a microcannula, equivalent to a size #32, .02 taper, removes particles lodged close to WL. Such a system is commercially available (EndoVac, Discus Dental) and may prove a valuable adjunct in canal disinfection.[302] One of the main characteristics of the system is safety. Several studies demonstrated that EndoVac use is unlikely to extrude irrigation solution through the apex. On the other hand, because the irrigation is deposited in the coronal area, the irrigant flow in the apical last millimeters of the canal is very passive, and some concerns were expressed that flow is laminar and passive in the apical region. In a recent study, the apical negative pressure mode of irrigation generated the lowest wall shear stress.[86]

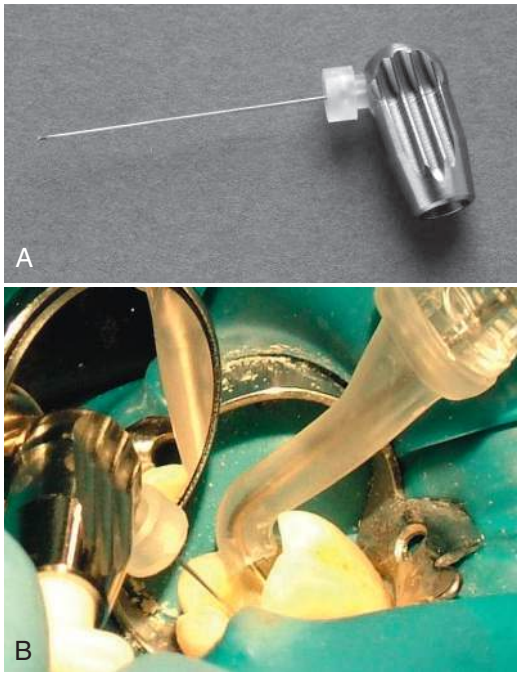


Fig. 8.60 The EndoVac system. **A,** Magnification of the closed-ended microcannula (**A**). **B,** Clinical view of the EndoVac system combined with the Safety-Irrigator. (*A,* Courtesy Discus Dental LLC, Culver City, CA. *B,* Courtesy Dr. A. Azarpazhooh.)

Another device that makes use of pressure-suction technology is the RinsEndo system (Dürr Dental, Bietigheim-Bissingen, Germany). It aspirates the delivered rinsing solution into an irrigation needle that is placed close to WL and at the same time activates the needle with oscillations of 1.6 Hz amplitude.[63]

Safety-Irrigator

The Safety-Irrigator (Vista Dental Products) is an irrigation/evacuation system that apically delivers the irrigant under positive pressure through a thin needle containing a lateral opening and evacuates the solution through a large needle at the root canal orifice (Fig. 8.61). The Safety-Irrigator features a large coronal evacuation tube, enabling the user to irrigate and evacuate simultaneously safely. It fits any standard Luer-lock syringe. Designed to limit the risk of NaOCl accidents, this "negative-pressure" irrigation device comes fully assembled and fitted with side-vented irrigating needle for added safety. This system was tested in vitro to evaluate the removal of dentin debris from artificially made grooves in standardized root canals and showed



Fig. 8.61 The Safety-Irrigator. (Courtesy Vista Dental Products, Racine, WI, USA.)

that there was no significant difference between the manual dynamic activation (MDA) with a nontapered gutta-percha cone, the Safety Irrigator, and the apical negative pressure irrigation. These techniques produced better cleaning efficacy than syringe irrigation ($P < .005$) but significantly worse than MDA with a tapered cone ($P < .05$). CUI was significantly better than all the other techniques tested in this study ($P < .001$).[218]

GentleWave System

The GentleWave (GW) system (Sonendo Inc, Laguna Hills, CA) was introduced in the United States in 2016. The device is composed of a console, a so-called procedure instrument, which is a single-use tip attached to a handpiece (for either molars or anterior and premolar treatment), and a central unit that contains three individual irrigation solution containers, one waste canister, a degassing system, and a pressure generator (Fig. 8.62). The system delivers an energized flow of irrigation solutions from the central unit to the procedural instrument. According to the manufacturer, the fluid stream entering the tooth creates a shear force, which in turn causes hydrodynamic cavitation. The implosion of micro-bubbles then creates an acoustic field of broadband frequencies, which travels through the fluid into the root canal system. The programmed irrigation regimen begins with 3% NaOCl followed by 8% EDTA, with a rinse of distilled water in between and at completion. The fluid within the root canal space is continuously collected and removed from the chamber through a five-point vented suction system built in the sealing lid of the procedural instrument.[179] A sealed environment is needed between the tooth and the GW Procedure Instrument to allow for constant refreshing of degassed procedure fluids and simultaneous evacuation of debris. The system operates within an air-free, sealed root canal system, filled only with procedure fluid and appears to induce a vertical flow within the root canal system with a slight negative pressure at the apices, which reduces the potential for extrusion. The system applies advanced fluid dynamics, acoustics, and tissue dissolution chemistry to remove tissue and debris from the entire root canal system simultaneously.[179]



Fig. 8.62 The GentleWave system (Sonendo, Irvine, CA, USA). The system uses multisonic energy to develop a broad section of waves within the irrigation solution to clean inside the roots canal system. It has two main components: a hand piece and a console (shown in figure).

The studies investigating cleaning and disinfection of the root canal system concluded that GW irrigation results in a faster dissolution rate than other systems[179] and that can clean canals efficiently with minimal instrumentation and can reduce residual debris[287] and promote better penetration into dentinal tubules compared with needle irrigation.[83]

The latter study,[83] using MicroCT, further concluded that GW irrigation was effective in the removal of accumulated dentin from the isthmus of mesial roots in mandibular molars. GentleWave irrigation was superior to needle irrigation but statistically similar to continuous ultrasonically powered irrigation (CI), with debris reductions for canals and isthmuses, respectively, of 96.4% and 97.9% for GW and 91.2% and 93.5% for CI.

As of the writing of this chapter, only three in vivo studies are published, all by the same group. The first two reports[405,406] evaluated the 6- and 12-month healing outcome after endodontic treatment using GW. The cumulative success was 97% at 12 months, with 92% and 5% of the necrotic pulp cases classified as healed and healing, respectively. However, in these studies, only 23% of the total sample had preoperative signs of asymptomatic apical periodontitis, which is a well-established outcome predictor.

The subsequent report[404] addressed this concern by assessing the 12-month healing outcome in teeth with a score greater than 3 in the periapical index (PAI). A reduction in PAI scores occurred in 98% of the teeth that received treatment with GW, with 82% and 16% of the teeth classified as healed and healing, respectively. However, despite initial encouraging data, more independent studies on GW are essential to fully describe the clinical applicability of this technique.

Laser-Activated Irrigation

Lasers are widely used in dentistry and include a diode, Nd:YAG, erbium and CO_2, which produces radiation in both the near and far infrared electromagnet spectrum.[166] Laser devices have been proposed to improve the efficacy of irrigants.[166] Lasers have been studied for their ability to clean and effectively disinfect root canals. The Er:YAG laser wavelength (2940 nm) has the highest absorption in water and

high affinity to hydroxyapatite, which makes it suitable for use in root canal treatment.[96]

Laser energy may be used to activate irrigant solutions in different ways, for example at a molecular level, as in photoactivated disinfection (PAD) or at a bulk flow level as in laser-activated irrigation (LAI). Several studies in vivo and ex vivo indicated that laser activated irrigation is promising in removing smear layer[150] and dentin debris[108] in less time than PUI. The mechanism of action[51] is based on the generation of a secondary cavitation effect with expansion and the successive implosion of fluids.[51]

These results are in agreement with data related to a new erbium laser technique that used photon-induced photoacoustic streaming (PIPS) of irrigants. In that technique, the laser tip is placed into the coronal access opening of the pulp chamber only and is kept stationary without advancing into the orifice of the canal.[118] The use of a tapered and stripped tip with specific minimally ablative laser settings is required: low energy (20 mJ), pulse repetition rate (15 Hz), and short pulse duration (50 μs). The difference in laser penetration and bacterial killing is attributed to the difference in the degree of absorption of different wavelengths of light within the dentin. Bergmans et al.[45] concluded in their in vivo study that the Nd:YAG laser irradiation is not an alternative, but a possible supplement to existing protocols for canal disinfection as the properties of laser light may allow a bactericidal effect beyond 1 mm of dentin. Endodontic pathogens that grow as biofilms, however, are challenging to eradicate even upon direct laser exposure.[58]

Photoactivated Disinfection

Photodynamic therapy (PDT) or light-activated therapy (LAT) may have endodontic applications because of its antimicrobial effectiveness.[185] In principle, antimicrobial photodynamic therapy (APDT) is a two-step procedure that involves the introduction of a photosensitizer (step 1: photosensitization of the infected tissue) followed by light illumination (step 2: irradiation of the photosensitized tissue) of the sensitized tissue, which would generate a toxic photochemistry on the target cell, leading to cell lysis. Each of these elements used independently will not have any action, but together they have a synergism effect of producing antibacterial action. Indeed, in vitro experiments showed promising results as an adjunct disinfected device. Shrestha and Kishen[400] concluded that the tissue inhibitors existing within the root canal affected the antibacterial activity of PDT at varying degrees and further research is required to enhance their antimicrobial efficacy in an endodontic environment. Photosensitizers combined with nanoparticles such as Rose Bengal may have a promising antibacterial efficacy.[401]

Antibacterial Nanoparticles

The concept of nanotechnology was first discussed in 1959 by Richard Feynman; the topic has recently been reviewed by Shrestha and Kishen.[402] Nanotechnology is the science conducted at the nanoscale, using particles about 1 to 100 nm, and it can be used in chemistry, biology, physics, materials science, and engineering. The particles can be classified depending on their shape as atom clusters, grains, fibers, films, nanoholes, and composites from these combinations. Nanomaterial properties vary majorly from other materials due to the increase in surface area per unit mass.

In dentistry, nanotechnology is currently investigated in the following fields: local nanoanaesthesia, hypersensitivity cure, tooth repositioning, nanorobotic dentifrice, dental durability, cosmetics, nanodiagnostics, and gene therapy.[49] More specifically, the research on nanomaterials is focusing on nanocomposites, nanosolution, nanoadhesives, and impression materials.

Nanoparticles in endodontics are being developed for their efficient disinfection of root canals, due to the broad-spectrum antibacterial activity. The nanoparticles evaluated in endodontics include Chitosan (CS-np), zinc oxide (ZnO-np), and silver (Ag-np) nanoparticles.[273]

CS-np have been widely studied and appear to be effective against monospecies *(E. faecalis)* and multispecies biofilms, even in the presence of tissue inhibitors, and it has been attributed to their ability to disrupt the cell wall.[402] Silver nanoparticles are also being evaluated for use as root canal disinfecting agents. In gel and liquid form, such nanoparticles have been shown to be able to kill and disrupt *E. faecalis* biofilm.

Biactive glass (SiO_2-Na_2O-CaO-P_2O_5) is another revolutionary introduction in the field of nanotechnology, and it has been suggested for root canal disinfection. The antimicrobial effect of bioactive glass is due to its ability to maintain an alkaline environment over a period of time.[273] When used in root canals, bioactive glass was found to kill bacteria, but the mechanism of action was not pH related and dentin did not seem to alter its effect.[507]

It is recently being researched for its use for endodontic regeneration in which diseased or necrotic pulp tissues are removed and replaced with healthy pulp tissues to revitalize teeth. It was reported the first use of nanostructured and functionalized multilayered films was as a new active biomaterial for endodontic regeneration.

HealOzone. Ozone, in the gaseous or aqueous phase, is a powerful and reliable antimicrobial agent against bacteria, fungi, protozoa, and viruses. The mechanism of action is based on the destruction of cell walls and cytoplasmic membranes of bacteria and fungi by the oxidant potential of the ozone. During this process, ozone attacks glycoproteins, glycolipids, and other amino acids and inhibits the enzymatic control system of the cell with an increase of the membrane permeability leading to immediate functional cessation. The ozone molecules then readily enter the cell and effectively kills microorganisms.[272]

It is applied to oral tissues in the following forms: ozonated water, ozonated olive oil, and oxygen/ozone gas. In vitro studies showed promising results on their antibacterial effect, clinical results demonstrated the opposite.[126,193,294,295] Ozone gas is currently used clinically for endodontic treatment; however, the results from in vivo studies are inconclusive.[193,294,295] This inconsistency is attributed to the lack of information about the maximum duration of application and concentration that should be used.[27,226] The hazards of ozone, when used in endodontics, have not been investigated. Care should be taken concerning the patient's and the dentist's exposure to the gas.

Water Preparations. Super-oxidized water,[171] also called electrochemically activated water[270,422] or potential oxidative water,[189,394] is effectively saline that has been

electrolyzed to form superoxidized water, hypochlorous acid, and free chlorine radicals. It is commercially available as Sterilox (Sterilox Technologies, Radnor, PA, USA). This solution is nontoxic to biologic tissues, yet able to kill microorganisms. The solution is generated by electrolyzing saline solution—a process no different from that used in the commercial production of NaOCl.[136] The difference, however, is that the solution accumulating at the anode is harvested as the anolyte and at the cathode as the catholyte. These solutions display properties that are dependent upon the strength of the first saline solution, the applied potential difference, and the rate of generation. The technology that allows harvesting of the respective solutions resides in the design of the anode and the cathode and originates either in Russia (electrochemically activated water) or Japan (potential oxidative water).[269] Although the solutions bear different names, the principles in the manufacturing process appear to be similar.

The use of superoxidized water is sparsely described in the endodontic literature but shows early promise. The solutions from both technologies have been tested for their ability to débride root canals,[189] remove the smear layer,[394,422] and kill bacteria[200] and bacterial spores.[259] Results are generally favorable, while also showing biocompatibility with vital systems.[211]

Anolyte and catholyte solutions generated from one such technology (Radical Waters Halfway Johannesburg, South Africa) have shown promise as antibacterial agents against laboratory-grown, single-species biofilm models.[151] Such solutions have been recommended as suitable for removing biofilms in dental unit water lines[269] and have even been marketed for this purpose. Cautious clinicians may prefer to wait for more studies to demonstrate safety and efficacy under normal clinical setting conditions before adopting newer, less tested irrigating solutions.

Criteria to Evaluate Cleaning and Shaping[4]

WELL-SHAPED CANALS

The main aims of canal shaping are to directly remove tissues and microbial irritants and to provide sufficient geometric space for subsequent obturation (see Table 8.2). In order to achieve these goals, the prepared canal should include the original canal (see *red areas* in Fig. 8.4); there should be an apical narrowing and the canal should be tapered. These concepts have been popularized by Schilder[386] and are still maintained today.[3,124]

Therefore, a well-shaped canal is defined more specifically by the absence of procedural errors (see later) and the achievement of disinfection; more recently, another element was added to this equation—the retention of as much tooth structure as feasible.[154]

The fact that a canal is adequately shaped can often be determined by a clinician from radiographs and clinical experience—for example, when fitting a cone. At this time, the feel would be of tug-back, a slight resistance to pull; a radiograph should demonstrate a symmetrical canal shape lateral of the cone, the presence of an intact apical narrowing, and no thinned out radicular wall sections.

Using magnification, clinicians should inspect the canal orifice and the coronal third of each shaped canal for clean canal walls.[319] Immediately after irrigation with sodium hypochlorite, an absence of visible turbidity and effervescence should be noted. If present, these phenomena along with visible deposits on the canal walls are indicative of organic matter still in suspension or adherent to the radicular walls.

SIGNS OF MISHAPS

Instrument Fracture

Most reports suggest that manual endodontic file fracture or rotary instrument fracture occur at a rate of approximately 1% to 6% and 0.4% to 5%, respectively.[434,463] Such fractures are untoward events and perceived as such by clinicians.[50] Evidently retained instrument fragments limit access of disinfecting irrigants to the root canal system, possibly impeding sufficient elimination of microorganisms.[178] However, the current clinical evidence does not suggest that the presence of a retained instrument must result in a significant higher rate of failing root canal treatments when done by specialists.[434]

In general, instruments used in rotary motion break in two distinct modes—torsional and flexural.[329,375,462] Torsional fracture occurs when an instrument tip is locked in a canal while the shank continues to rotate, thereby exerting enough torque to fracture the tip. This also may occur when instrument rotation is sufficiently slowed in relation to the cross-sectional diameter. In contrast, flexural fracture occurs when the cyclic loading leads to metal fatigue. This problem precludes the manufacture of continuously rotating stainless steel endodontic instruments, because steel develops fatal fatigue after only a few cycles.[393] NiTi instruments can withstand several hundred flexural cycles before they fracture,[250,348,462] but they still can fracture in the endodontic setting after a low (i.e., below 10,000) number of cycles.[87]

Repeated loading and cyclic fatigue tests for endodontic instruments are not described in pertinent norms. Initially, rotary instruments such as GG burs and Peeso reamers were tested with a superimposed bending deflection.[62] In another study, stainless steel and NiTi hand files were rotated to failure in steel tubes with an acute 90-degree bend and an unspecified radius.[393] Under these conditions, size #40 stainless steel instruments fractured after fewer than 20 rotations, whereas various NiTi files of the same size withstood up to 450 rotations.

Cyclic fatigue was also evaluated for ProFile .06 taper instruments using a similar device.[498] The number of rotations to failure for unused control instruments ranged from 1260 (size #15 files) to 900 (size #40 files). These scores did not change when the instruments were tested under simulated clinical conditions such as repeated sterilization and contact with 2.5% NaOCl. Subsequently, control instruments were compared with a group of instruments used in the clinical setting in five molar cases;[497] again, no significant differences were found in resistance to cyclic fatigue.

One study[183] used a different testing method involving tempered metal cylinders with radii of 5 and 10 mm that produced a 90-degree curve. They reported fatigue fractures

for size #15, .04 taper ProFile instruments after about 2800 cycles with the 10 mm cylinders. In size #40, .04 taper ProFile instruments, fractures occurred after about 500 cycles with the 5-mm cylinders. In comparison, size #15, .06 taper ProFile instruments also failed after about 2800 revolutions with the 10-mm cylinders, but failure occurred in size #40, .06 taper ProFile specimens after only 223 cycles with the 5-mm cylinders.

Rotary NiTi instruments with larger tapers and sizes consistently fractured after fewer rotations,[345] and although the radius of the curves was halved, fatigue-life was reduced by 400%. Another investigation[183] reported similar results for selected HERO instruments, and their findings were confirmed by other tests on GT rotary instruments. Size #20, .06 taper GT files failed after 530 rotations in a 90-degree curve with a 5-mm radius; size #20, .12 taper GT files failed after 56 rotations under the same conditions.[334]

Reuse of rotary instruments depends on safety, specifically on assessment of fatigue and also the potential to properly clean NiTi surfaces.[34,304,311,398,424,428,458] Specific instruments perform differently in this regard, since fatigue depends more on the amount of metal in cross section at the point of stress concentration[163,457] than on the specifics of instrument design.[88]

On the other hand, manufacturers constantly claim that their instrument has been equipped with design elements that render it more fatigue resistant. For example, LightSpeed LSX is manufactured without a milling process. However, no data have been published regarding its fatigue resistance. GTX is manufactured from a novel NiTi alloy, M-Wire, to increase its fatigue resistance.[220] However, investigators[234] could not confirm these findings. Similarly, another study[243] did not find the TF, which is not milled and hence believed to be fatigue resistant,[143] to perform better than conventionally manufactured ProFile rotaries. Another feature, electropolishing (as discussed earlier), does not appear to confer a significantly increased fatigue resistance to EndoSequence[243,353] and RaCe.[457,458,495] One possible reason for these variable outcomes are the different testing environments used in vitro[90]; clinically, even greater variability is to be expected.

Recently, cyclic fatigue tests are sometimes conducted under more directly clinically applicable conditions, such as varying ambient temperature or introducing dynamic movement. For example, while the latest martensitic alloys we showed to have significantly increased fatigue resistance at room temperature, this benefit was greatly reduced at body temperature.[22,109]

Attempts have been made to use tests according to norms and specifications described for stainless steel hand instruments such as K-files and Hedström files,[2] since no comparable norms exist for instruments used in continuous rotary motion. Consequently, a number of models have been devised to assess specific properties of NiTi rotary instruments, including torque at failure, resistance against cyclic fatigue, and others (Fig. 8.63). These systems may simultaneously assess torque at failure, working torque axial force, and cyclic fatigue.

According to the norms mentioned previously, torque at failure is recorded with the apical 3 mm of the instrument firmly held in the testing device while the instrument's handle is rotated. A wide variety of rotary NiTi endodontic instruments have been tested in this way. For example, ProFile NiTi rotary files in ISO sizes #25, #30, and #35 (.04 taper) fractured at 0.78, 1.06, and 1.47 Ncm, respectively.[441]

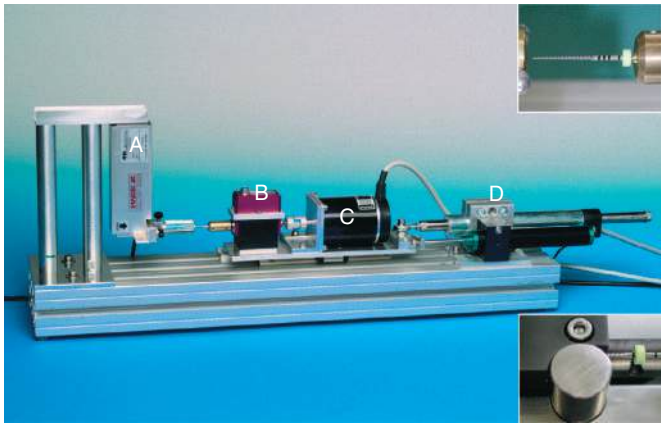


Fig. 8.63 Testing platform for analysis of various factors during simulated canal preparation with rotary endodontic instruments. Labeled components are a force transducer *(A)*, a torque sensor *(B)*, a direct-drive motor *(C)*, and an automated feed device *(D)*. For specific tests, a cyclic fatigue phantom or a brass mount compliant with ISO No. 3630-1 *(inserts)* may be attached.

When analyzing clinical factors involved in instrument fracture, one must consider both torsional load and cyclic fatigue.[375] However, these are not separate entities, especially in curved canals.[57] Working an instrument with high torque may lower resistance to cyclic fatigue.[141] Conversely, cyclic prestressing has been shown to reduce the torsional resistance of ProTaper finishing files,[462] as well as K3[30] and MTwo.[345] Also, cyclic fatigue occurs not only in the lateral aspect when an instrument rotates in a curved canal but also axially when an instrument is bound and released by canal irregularities.[48]

The torque generated during canal preparation depends on a variety of factors, and an important one is the contact area.[52] The size of the surface area contacted by an endodontic instrument is influenced by the instrumentation sequence or by the use of instruments with different tapers.[388] A crown-down approach is recommended to reduce torsional loads (and thus the risk of fracture) by preventing a large portion of the tapered rotating instrument from engaging root dentin (known as *taper lock*).[52,499]

The clinician can further modify torque by varying axial pressure, because these two factors are related (see Fig. 8.20).[388] In fact, a fairly light touch is recommended for all current NiTi instruments to avoid forcing the instrument into taper lock. The same effect might occur in certain anatomic situations, such as when canals merge, dilacerate, and divide.

The torsional behavior of NiTi rotary endodontic instruments cannot be described properly without advanced measurement systems and a new set of norms. However, the clinician must be able to interpret correctly the stress-strain curves for all rotary NiTi instruments used in the clinical setting to be able to choose an appropriate working torque and axial force.

Therefore a careful evaluation should be performed before the attempt is made to remove any retained fragment

(see Fig. 8.25). In fact, Ward and others[481] suggest any attempt of fragment removal only when the fragment is located coronal of a significant root canal curve and thus visible with the aid of magnification.

There are sophisticated means and strategies to remove retained fragments, which are described in detail in Chapter 20 of this book.

Of note, assessment of physical parameters governing rotary root canal preparation was considered crucial because NiTi rotary in vitro had increased risk of fracture compared with K-files. Some clinicians also describe instrument fracture as a main issue for concern.[50]

In a study using plastic blocks, as many as 52 ProFile Series 29 instruments became permanently deformed.[450] Three fractures were reported in a subsequent study on ISO-norm ProFile size .04 instruments, and three other instruments were distorted.[66] An even higher fracture incidence was shown in a study on rotary instruments used in plastic blocks in a specially designed testing machine.[446] These findings were supported by two studies, in which high fracture incidences were reported for LightSpeed and Quantec rotary instruments used in a clinical setting.[32,375]

On the other hand, as stated earlier, a retrospective clinical study suggests similar outcomes with and without retained instrument fragments[434]; moreover, others' experience suggests that the number of rotary instrument fractures is lower than previously estimated.[112,489] Removal of such fragments is possible in many situations, but there is also the potential for further damage (e.g., perforation) rather than successful removal.[440,489] Consequently, a benefit-versus-risk analysis should be carried out prior to attempts to remove NiTi instrument fragments, addressing the reasons and the clinical consequences of instrument fracture.

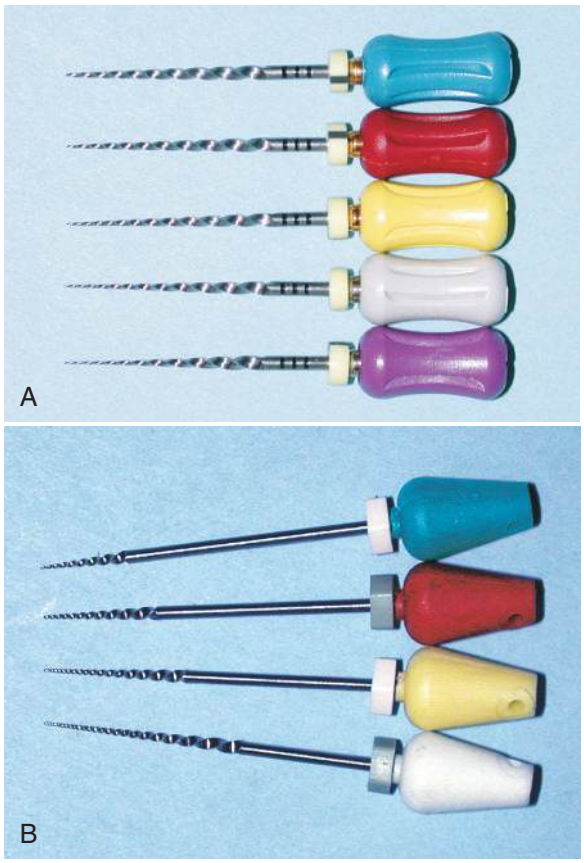


Fig. 8.64 Instruments with increased taper that can be used by hand. **A,** ProTaper instruments with special handles attached to rotary instrument shanks. **B,** GT hand instruments.

Canal Transportation

Perhaps the most frequent adverse outcomes during canal shaping are aberrations from the original canal path. Much has been written on the appearance of such aberrations using labels such as zip and elbow formation, ledging, perforation, stripping, and others.[484]

Canal transportation is at the root of all these clinical problems and may be defined as "the removal of canal wall structure on the outside curve in the apical half of the canal due to the tendency of files to restore themselves to their original linear shape during canal preparation."[3]

As files tend to straighten in the canal, this typically occurs toward the inner (or convex) radicular wall at midroot as well as to the outer curvature apically. Such a shift of the canal axis during shaping results in excessive loss of dentin and may ultimately result in strip perforation, while apical transportation may lead to zipping or overt apical perforation (see Fig. 8.39).

Conceptually, any canal preparation will shift the canal axis, which is often determined as the center of gravity in cross sections. It has been held that a transportation of about 100 to 150 μm may be clinically acceptable.[207,326]

If canal transportation has led to ledge formation, subsequent instruments will bypass the area of the ledge only when adequately precurved. In case of rotary instrumentation, hand-operated instruments of comparable size are then recommended (Fig. 8.64).

Perforation

As indicated, perforation may be the ultimate result of canal transportation when it occurs within the root canal system. Other perforations are those in the access cavity; a discussion of access cavity preparation may be found in Chapter 7. Obviously, preparing mineralized canals requires advanced operator skills and is facilitated by magnification (Fig. 8.65).

Three types of perforations can be defined—strip perforations that occur toward the furcation in multirooted teeth (also known as "danger zones"[7]), perforations associated with canal curvatures, and perforations though the apical foramen.

Blockage

A canal may become impassable during the process of cleaning and shaping due to two distinct, but often connected, occurrences. A ledge is a dentinal shelf that is created by shaping instruments that straighten and dig into the convex side of the canal wall. In less severe cases, ledges can be corrected and smoothed out with precurved instruments. This condition may lead to false paths and impede optimal obturation, when WL cannot be reached with master cones.

A blockage refers mainly to a root canal area that is filled with densely compacted debris or collagenous pulp remnants (see Fig. 8.38). It may also be caused by other obstacles such

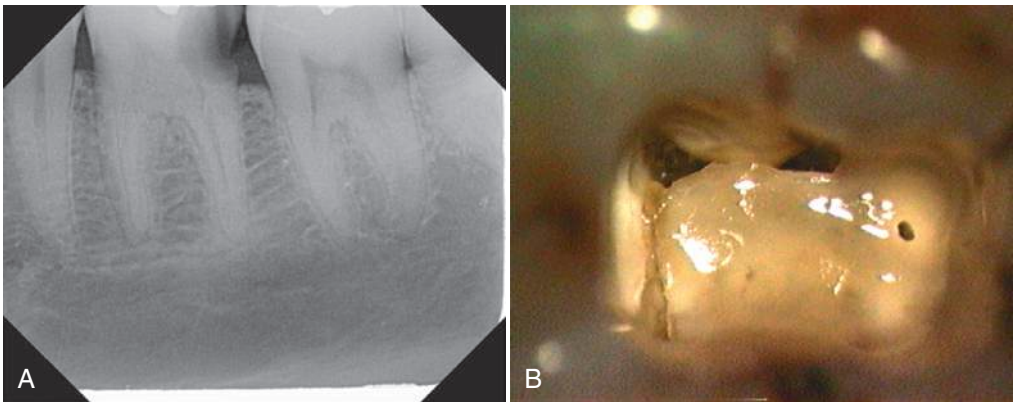


Fig. 8.65 Evidence of coronal hard-tissue deposition. **A,** Periapical radiograph of tooth #19 shows evidence of reduced coronal and radicular pulp space. **B,** Intraoral photograph, taken through an operating microscope (×25), of access cavity of the tooth shown in **A**; note the calcific metamorphosis.

as fractured files or remnants of a preexisting coronal or radicular filling materials. Clinically the presence of ledge or block is signaled by the inability for a straight flexible instrument to penetrate deeper into the root canal; however, this needs to be differentiated from a small or mineralized canal, which causes longer portions of the instrument's cutting flutes to bind.

Obviously, such a blockage prevents the apical canal portion from being disinfected. For details of strategies to deal with ledge and blockage, refer to Chapter 10.

Another reason for a perceived blockage may be a very abrupt canal curvature.

Sample Protocol for Contemporary Cleaning and Shaping Procedures

- Using well-angulated preoperative radiographs, analyze case difficulty using well-established parameters.
- Place rubber dam and estimate WL.
- Prepare a conservative access cavity sufficient enough to reveal all root canal orifices.
- Scout canals with a #10 K-file in the presence of a lubricant.
- If the selected series of rotary instruments advances easily to the estimated WL, confirm patency and determine WL using an electronic apex locator.
- If the instruments meet resistance and the file does not progress gently to WL, use a dedicated NiTi instrument. It is prudent to modify the orifice to create a coronal receptacle for the subsequent rotaries. Negotiate, confirm patency, and determine WL.
- Create a reproducible glide path to WL with appropriate instruments.
- Irrigate with sodium hypochlorite throughout the shaping procedure.
- Advance the selected series of rotary instruments (based on canal anatomy) passively in the presence of sodium hypochlorite to shape the middle third. When shaping canals that have a larger buccal-lingual dimension, consider shaping as two canals.
- Clean cutting flutes routinely upon removal and remove debris with an alcohol-moistened gauze. If the selected rotary does not progress easily, remove irrigate, recapitulate with a #10 K-file, and choose a different, often smaller, instrument.
- Use copious irrigation, reverify canal patency and working length throughout and upon completion of shaping. Gauge the size of the foramen with an appropriate hand file.
- Protocol for irrigation:
  - Irrigate using copious amounts of sodium hypochlorite.
  - Activate the irrigant.
  - Select irrigation solution for smear layer management.
  - Perform final irrigation.
- Dry the canal thoroughly and obdurate with a technique that promotes a three-dimensional fill.
- Restore the endodontically treated tooth in a timely manner.

Summary

Cleaning and shaping are important, interdependent steps in root canal treatment. Cleaning, as demonstrated by an intracanal surface free of smear layer, can be done only after root canals have been sufficiently enlarged to accommodate adequate irrigation needles. Canal preparation is optimized when mechanical aims are fulfilled and enlargement is acceptable; such aims include avoiding both significant preparation errors and weakening of the radicular structure, which can result in fractures.

Taken together and performed to a high standard, the procedures described in this chapter lay the foundation for biologic success in both straightforward (Fig. 8.66) and more complicated (Fig. 8.67) clinical cases. Recall radiographs confirm favorable outcomes or biologic success (i.e., prevention or healing of periradicular periodontitis) over the years. Similarly, adherence to the principles discussed leads to predictable outcomes for root canal treatments.

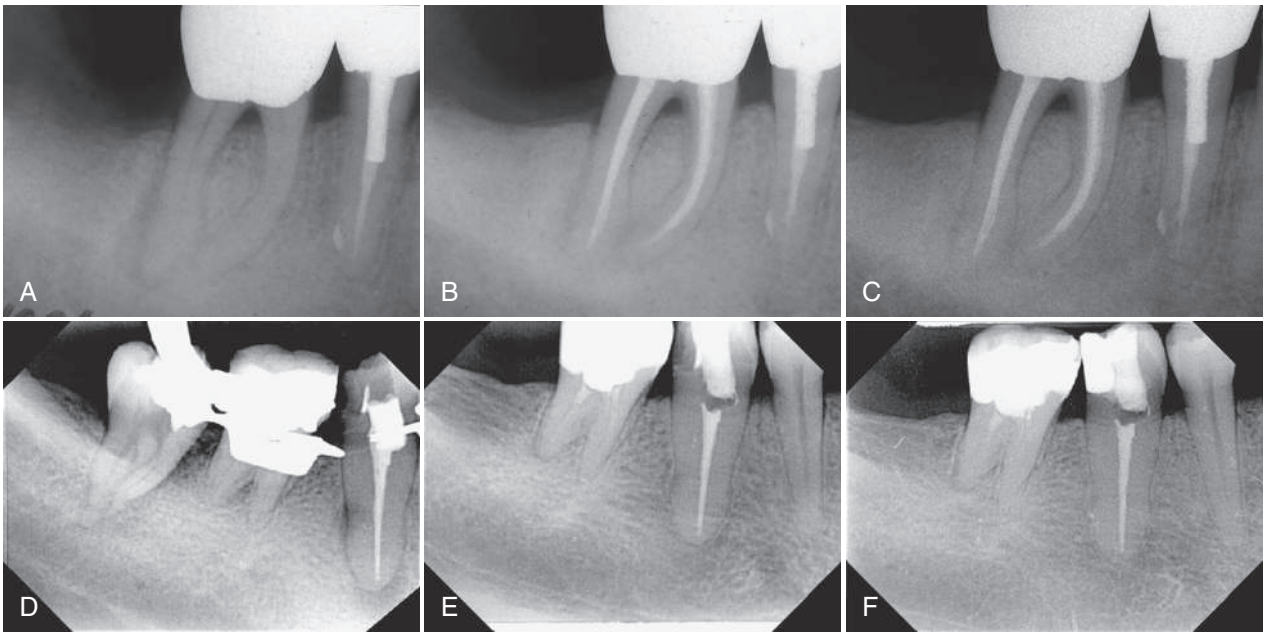


Fig. 8.66 Clinical cases treated according to the principles detailed in this chapter. **A,** Pretreatment radiograph of tooth #30 with a periradicular lesion. **B,** Postobturation radiograph. **C,** Two-year follow-up radiograph shows osseous healing. **D,** Immediate postobturation radiograph of tooth #29 shows both a periapical and a lateral osseous lesion. **E–F,** One-year and 3-year follow-up radiographs show progressing osseous healing. Note the imperfect obturation of tooth #30.

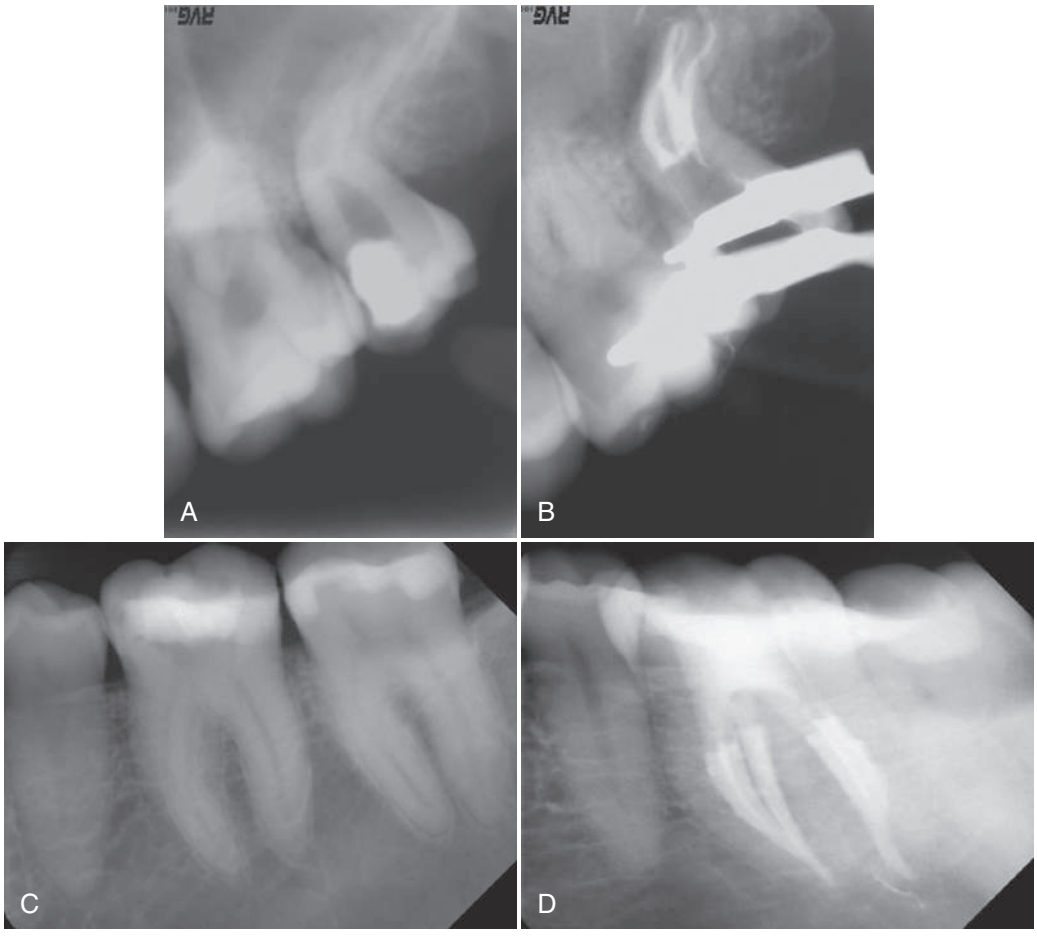


Fig. 8.67 Complicated clinical cases treated with hybrid techniques. **A,** Pretreatment radiograph of tooth #16 indicates laceration and significant curvature of all roots. **B,** Posttreatment radiograph shows multiple planes of curvature. **C,** Pretreatment radiograph of tooth #19, which was diagnosed with irreversible pulpitis. **D,** Angulated posttreatment radiograph shows three canals in the mesiobuccal root canal system, all of which were prepared to apical size #50. (*A–B*, Courtesy T. Clauder; ***C–D***, Courtesy Dr. H. Walsch.)

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